Statistics and Econometrics with R

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Contents

Welcome

This book is the result of several years of teaching introductory statistics and econometrics at Azim Premji University. It was shaped not only by lectures and classroom discussions, but also by countless questions from students who wanted to understand *why* statistical methods work, rather than simply *how* to apply them.

Those questions made me rethink the way statistics is usually introduced.

Many introductory courses place considerable emphasis on mathematical derivations, definitions and problem-solving techniques. These are important, and this book does not seek to diminish their value. Yet, in my experience, many students complete such courses having learned the mechanics of statistical procedures without fully appreciating the logic that makes them work. They know the steps involved in carrying out a hypothesis test or estimating a regression model, but the underlying ideas — the beauty of reasoning from a sample to an entire population — often remain hidden beneath the mathematics.

This book begins from a different premise. Before introducing notation, formulas or proofs, it asks a simple question: **what problem are we trying to solve?** Once that question is clear, intuition follows naturally. The mathematics is then introduced not as something to be memorised, but as a precise language for expressing ideas that the reader has already encountered through examples, simulation and reasoning.

The philosophy of the book can be summarised in five words:

> **Question** $\rightarrow$ **Intuition** $\rightarrow$ **Rigour** $\rightarrow$ **Computation** $\rightarrow$ **Interpretation**

Every chapter follows this progression. We begin with a real question, usually drawn from economics or public policy. We build intuition through examples and simulation. We then develop the mathematical foundations carefully and without unnecessary shortcuts. Finally, every idea is implemented in R using real data so that the transition from theory to empirical analysis becomes seamless.

This integration of intuition, mathematical reasoning and programming reflects the way statistics is practised today. There is no separate chapter devoted to software. Programming is not an optional add-on; it is part of the process of learning statistical thinking. Every table, figure and numerical result in this book is generated directly from R, allowing readers to reproduce every analysis for themselves.

The examples throughout the book are grounded, wherever possible, in the Indian economy. Rather than relying on unfamiliar datasets and examples, we study questions involving wages, education, poverty, agriculture, inflation, employment, health and public policy. My hope is that students will not only learn statistical methods but also develop a deeper understanding of the economic and social issues that shape contemporary India.

This emphasis is particularly important because empirical economics has changed dramatically over the past two decades. Modern applied research increasingly seeks to answer causal questions rather than merely document statistical relationships. Although this book develops the foundations of frequentist statistics, estimation and regression analysis, it does so with this broader objective in mind. The methods introduced here are intended not simply as computational tools, but as the foundations upon which modern empirical economics is built.

This book also draws extensively on the outstanding textbooks that have shaped the teaching of statistics and econometrics over many decades. I have learned enormously from those works, and this book should be viewed as a companion to that tradition rather than a replacement for it. Its contribution lies not in presenting new statistical methods, but in presenting familiar ideas through a pedagogical approach that places intuition, conceptual understanding and empirical application at its centre.

The book is written primarily for undergraduate students in India who are studying quantitative social science — economics, but equally public policy, sociology, political science, development studies and related disciplines.

I want to be explicit about one group of readers in particular. Many students arrive in these programmes without a mathematics training in school, and find statistics and econometrics genuinely difficult as a result. In my experience the difficulty is almost never a lack of ability. It is that most textbooks are written for readers who already have the background, and a student without it is left trying to learn the ideas and the notation simultaneously, from a book that explains only the second.

This book assumes no such background. Every idea is introduced before the symbols for it, mathematical steps are shown rather than left to the reader, and nothing is described as obvious. Motivated readers should be able to work through it independently.

My hope is that readers finish the book not only knowing how to perform statistical analyses in R, but also understanding the logic of frequentist inference and developing the habits of mind needed to think critically about empirical evidence.

I owe a considerable debt to my colleagues in economics at Azim Premji University. Good pedagogy is treated here as a serious professional undertaking rather than an afterthought, and the care that colleagues take — day after day, and largely unremarked — to become better teachers and better practitioners has been invaluable. That environment has done a great deal to bring this project to light.

Finally, this book would never have been written without my students. Every question asked after class, every thoughtful challenge to an explanation, every mistake that forced me to rethink the way I taught, and every conversation about data and economics has left its mark on these pages. I owe a sincere debt of gratitude to all the students who have taken my courses in introductory statistics and econometrics over the years. Your curiosity, patience and willingness to ask difficult questions have continually pushed me to become a better teacher. This book is the result of that shared journey, and I hope it helps future students discover the same excitement for statistics that you brought into my classroom.

How to use this book

Statistics is not a subject that rewards passive reading. Understanding comes from asking questions, working through examples, making mistakes and revisiting ideas. This book is written with that philosophy in mind, and you will get the most out of it if you engage with it actively.

Read the book in order

The first few units are designed as a sequence rather than a collection of independent topics. Ideas introduced early — such as variation, repeated sampling and statistical inference — form the foundation for everything that follows. While later chapters can be used as references, I strongly recommend reading the book from beginning to end the first time.

Work through the mathematics

Do not skip the derivations because they look unfamiliar. Every mathematical result is developed gradually, with intermediate steps shown. Try to work through each derivation yourself before reading ahead. The objective is not to memorise proofs, but to understand where formulas come from and why they work.

Think in words before thinking in mathematics

One piece of advice has stayed with me throughout my academic career. During my PhD, my advisor once told me:

> If you can explain the problem clearly in simple English, you have already solved half of it. The mathematics is just a matter of practice.

Over the years I have found this to be remarkably true, and nowhere more so than in statistics and econometrics.

Students often assume that success in these subjects depends primarily on being good at mathematics. Mathematical reasoning is undoubtedly important, but most difficulties arise much earlier than the algebra. They arise because we lose sight of the question we are trying to answer. Before writing down a formula, ask yourself: what am I trying to learn? What information do I have? Why should this method work at all? If you can answer those questions clearly in words, the mathematics usually becomes far easier to follow.

Throughout this book I encourage you to explain ideas to yourself in plain English before expressing them in symbols. Statistical notation is a powerful language, but it is still a language. Its purpose is to communicate ideas precisely, not to replace logical reasoning.

Whenever you find yourself stuck in a derivation or confused by a formula, pause and return to the underlying question. More often than not, the mathematics becomes much less intimidating once the logic is clear.

Write the code yourself

Every analysis in this book is accompanied by R code. Resist the temptation to copy and paste it. Typing the code yourself forces you to pay attention to every line and every argument. Once the code works, experiment with it. Change a parameter, modify the data, or deliberately introduce an error. Some of the most valuable lessons come from understanding why something fails rather than why it succeeds.

Think before calculating

Whenever a new method is introduced, pause before looking at the computation. Ask yourself what you expect the answer to be, and why. Developing statistical intuition is just as important as obtaining the correct numerical result.

Attempt the exercises first

The exercises are an essential part of the book, not an optional supplement. Attempt every question before consulting the solutions, even if you are unsure how to proceed. Struggling with a problem is often where the deepest learning takes place. The solutions are intended to explain your reasoning, not replace it.

Use the book as a reference

Once you have worked through the material, the book can also serve as a reference. Use the search function at the top of the page to revisit concepts, definitions, examples or R commands. You are not expected to remember every formula or every line of code. What matters is understanding the underlying ideas and knowing where to find them when needed.

Learn by connecting ideas

Throughout the book, try to ask not only how a method works, but why it was introduced in the first place. Every chapter begins with a question and develops the statistical tools needed to answer it. If you understand the question, the mathematics that follows will feel much more natural.

Above all, remember that statistics is a way of thinking rather than a collection of techniques. The goal of this book is not simply to help you perform statistical analyses in R, but to help you reason carefully about evidence, uncertainty and data. If you finish the book asking better questions than when you began, then it will have achieved its purpose.

A note on the data

Every dataset used in this book is either built into R, supplied with the book, or generated directly by code that you can read and run yourself. This means that every table, figure and numerical result can be reproduced by running the accompanying script from beginning to end.

Reproducibility is a fundamental principle of statistical work. If an analysis cannot be recreated from the original data and the accompanying code, its results cannot be independently verified. Good statistical practice therefore means leaving behind a complete record of every step that produced the final result.

> **REMEMBER THIS**
>
> A result that cannot be reproduced by running a script from top to bottom is not yet a result.

One dataset deserves a special mention. The class survey used throughout the book was collected from students enrolled in this course. Before it was included here, all identifying information was removed and each student was assigned an anonymous identification number. Protecting the privacy of participants is one of the first responsibilities of anyone working with data about people.

Later in the book we return to questions of sampling, data collection and research ethics. For now it is enough to remember that every dataset represents real individuals, and good statistical practice begins with treating their information responsibly.

What you need

This book requires two free programs, which should be installed in the following order.

1. **R** — the statistical programming language, available from cran.r-project.org.
2. **RStudio Desktop** — an integrated development environment that provides a more convenient interface for working with R, available from posit.co/download/rstudio-desktop.

Several chapters use additional R packages. These need to be installed only once, by running the following command in the RStudio console:

```r
install.packages(c("tidyverse", "BSDA", "wooldridge", "AER", "stargazer"))
```

No other software is required, and everything used in this book is freely available.

Mathematical background

This book assumes a basic acquaintance with introductory probability and descriptive statistics. Specifically, it will help to have met:

- basic probability rules and notation;

- random variables and their probability distributions;
- probability mass functions, probability density functions and cumulative distribution functions.

Everything else the book relies on is built up within it. Measures of central tendency and dispersion are reviewed in Unit 1 before they are used, and random variables, expected values and variances are developed there from first principles rather than assumed. If those terms are unfamiliar, that is where they are introduced.

The focus throughout is statistical inference and data analysis rather than probability theory itself. Where an idea from probability is needed, it is recalled briefly at the point of use.

A word to readers who are uncertain whether they have enough mathematics for this. Almost everything in the list above is notation and vocabulary rather than technique, and the difficulty of statistics lies elsewhere — in understanding what a question is asking and what the data can answer. If you can follow an argument carefully and are willing to work through the derivations rather than skip them, you have what this book requires. Begin at Unit 1 and consult a probability text only if something there genuinely does not yield.

Licence and citation

If you spot an error, the source of every chapter is on GitHub and corrections are welcome as pull requests.

Corrections from students are genuinely welcome; several of the sharper points in this book exist because someone asked why.

To cite:

> Kulkarni, K. (2026). *Statistics and Econometrics with R.* Azim Premji University. https://kedarkulkarni1991.github.io/stats-econometrics-r/ Licensed under CC BY 4.0.

Chapter 1

Towards Inferential Statistics

You already know how to calculate the average of a sample.

In this unit, you will learn why that average is almost certainly wrong.

At first glance, this seems like a strange claim. After all, there is only one way to calculate a sample mean, and if the arithmetic is correct, the answer must be correct as well. Yet the number you obtain from a sample is almost never exactly equal to the quantity you ultimately wish to know — the mean of the population from which that sample was drawn.

Suppose you wish to estimate the average income of households in India. Since surveying every household is impossible, you select a sample of 2,000 households and calculate its mean income. The calculation is straightforward, but the result immediately raises a question. How close is this sample mean to the true population mean?

The difficulty is that the sample you have drawn is only one of many that could have been observed. Had a different 2,000 households been selected, the sample mean would almost certainly have been different. Draw another sample, and the answer changes again. Every statistic calculated from a sample therefore reflects two things at once: the characteristics of the underlying population, and the randomness introduced by the sampling process.

This simple observation marks the boundary between descriptive and inferential statistics.

Until now, our objective has been to describe the data that we observe. Measures of central tendency, such as the mean and the median, summarise where observations are concentrated. Measures of dispersion, such as the variance and standard deviation, describe how widely they are spread. Histograms, bar charts and density plots reveal the overall shape of a distribution and help us identify skewness, outliers and other important features. Together, these tools provide a concise description of a dataset and are the natural starting point of every empirical investigation.

Most empirical research, however, is concerned not with the sample itself but with the population from which it was drawn. Economists survey a few thousand households to understand millions. Polling agencies interview a sample of voters to estimate national voting intentions. Medical researchers evaluate new treatments on a limited number of patients before recommending them to the wider population. In every case, the sample serves as evidence about a population that cannot be observed directly.

The central problem of inferential statistics is therefore straightforward to state: **how can we use the information contained in a sample to learn about the population from which it came?**

Throughout this course we adopt the **frequentist** approach to answering this question. Rather than treating the observed sample as unique, we imagine drawing many samples from the same population and calculating the same statistic repeatedly. Some sample means will be larger than others, some smaller. Understanding how statistics vary across these hypothetical samples allows us to assess how much confidence we should place in the estimate computed from the one sample we actually observe.

Before we can understand why, we need to understand where the uncertainty comes from. The answer lies in randomness.

1.1 Randomness and variation

> *Where does the uncertainty come from?*

The uncertainty that surrounds every statistical estimate has a single source: randomness.

If we were to repeat the same study with a different sample, we would almost certainly obtain a different average. Inferential statistics is concerned with understanding this variability. Without randomness, every sample would be identical, every estimate would equal the population value, and there would be no uncertainty to quantify.

The idea is easiest to see with a die.

Suppose we roll a fair die once.

```r
set.seed(1)
sample(1:6, size = 1)
```

```
#> [1] 1
```

The outcome is a one. Beyond that, there is little to say. A single observation tells us almost nothing about the process that generated it. Was the die fair? Would the next roll also be a one? Is one an unusually common outcome? One observation cannot answer these questions.

Now imagine repeating the experiment ten thousand times.

```
rolls <- sample(1:6, size = 10000, replace = TRUE)

round(prop.table(table(rolls)), 3)
```

```
#> rolls
#>     1     2     3     4     5     6
#> 0.169 0.162 0.160 0.170 0.173 0.166
```

Each individual roll remains unpredictable, but something remarkable happens collectively. The outcomes no longer appear as isolated events. They form a stable pattern. Every face occurs with almost the same frequency, close to one-sixth of the time, exactly as we would expect from a fair die.

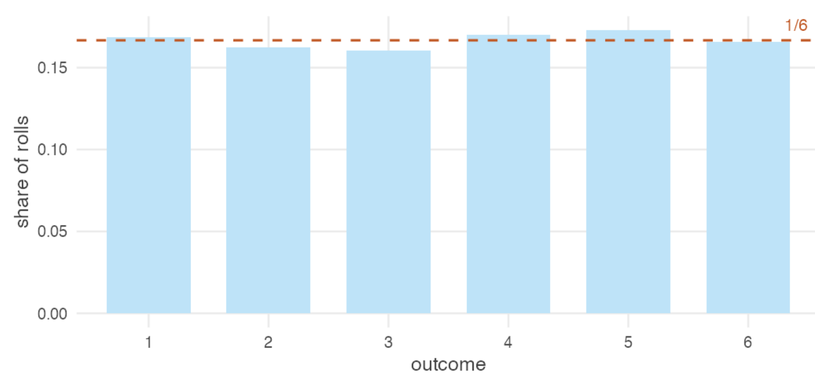

Figure 1.1: Ten thousand rolls of a fair die. No single roll was predictable. The pattern across all of them is.

The important point is not that the die turned out to be fair. It is that a pattern exists at all.

To see why that matters, imagine removing the randomness altogether. Suppose every face of the die is labelled three.

```
loaded <- rep(3, 10000)

table(loaded)
```

```
#> loaded
#>     3
#> 10000
```

```r
sd(loaded)
```

```
#> [1] 0
```

Every roll now produces exactly the same outcome. There is no variation, no uncertainty and no pattern waiting to emerge through repetition. The average is always three. The standard deviation is zero. Repeating the experiment a million times teaches us nothing that we did not already know after the first roll.

The contrast between the two dice illustrates a fundamental idea. Randomness makes individual outcomes unpredictable, but repeated observations reveal stable patterns. Statistics is built on this coexistence of uncertainty and regularity. If outcomes never varied, there would be no need for inference; if they varied without any underlying regularity, inference would be impossible.

Dice provide a simple illustration, but the same principle governs every empirical investigation. Household surveys interview different families. Election polls reach different voters. Clinical trials recruit different patients. Agricultural experiments observe different harvests. Every study could have produced a different dataset. The central challenge of inferential statistics is to understand what can be learned from **one realisation of a process that could have produced many others**.

1.2 Describing variation

> *Once variation exists, how do we describe it?*

Variation is what makes statistics necessary. The next question is equally natural: how do we describe it?

A die has only six possible outcomes, so its behaviour is easy to summarise. Real datasets are rather different. Consider the annual earnings of twenty thousand Indian workers.

```r
workers <- read.csv("data/wages-india-synthetic.csv")

nrow(workers)
```

```
#> [1] 20000
```

```r
head(workers$annual_earnings)
```

```
#> [1]  17773  60847 115739  34834  30784   2068
```

This dataset is simulated, but it closely resembles the earnings distribution observed in large household surveys such as the India Human Development Survey (IHDS). Throughout this book, we will use it to illustrate statistical ideas and methods.

Twenty thousand numbers are too many for anyone to understand by inspection. We need ways of reducing the data without losing its essential features. The first step is simply to look at it.

```r
ggplot(workers, aes(annual_earnings / 1000)) +
  geom_histogram(bins = 60, fill = book_palette$fill, colour = "white",
                 linewidth = 0.15) +
  scale_x_continuous(labels = scales::comma) +
  labs(x = "Annual earnings (₹ thousand)",
       y = "Number of workers") +
  theme_book()
```

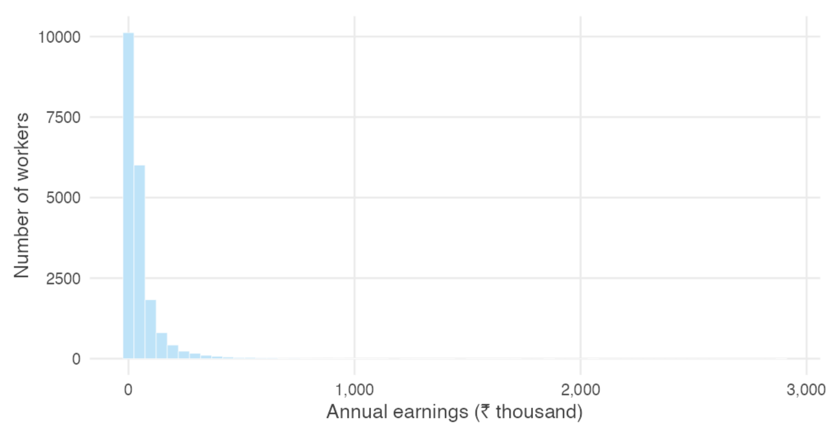

Figure 1.2: Annual earnings of 20,000 Indian workers.

The histogram immediately reveals two important features. Most workers earn relatively modest incomes, while a much smaller number earn substantially more. The distribution is therefore concentrated on the left with a long tail extending to the right — a pattern known as **right skewness**.

One picture already tells us a great deal, but we can describe the same distribution more precisely using a few numerical summaries.

```
c(mean   = mean(workers$annual_earnings),
  median = median(workers$annual_earnings))
```

```
#>   mean median
#>  54181  24220
```

The mean annual earnings are more than twice the median. This difference reflects the long right tail visible in the histogram. A relatively small number of high earners pull the mean upwards, whereas the median simply divides the distribution into two equal halves and is much less affected by extreme values.

Variation also concerns how widely observations are dispersed.

```
c(sd  = sd(workers$annual_earnings),
  iqr = IQR(workers$annual_earnings))
```

```
#>     sd    iqr
#> 100141  46756
```

The **standard deviation** measures the typical distance of observations from the mean, while the **interquartile range** measures the spread of the middle fifty per cent of the distribution. Together they indicate that earnings vary considerably across workers.

Sometimes we are interested not only in the overall distribution but also in how it differs across groups. A density curve provides a smooth representation of the same data and makes such comparisons easier.

```
ggplot(workers, aes(annual_earnings / 1000, colour = residence)) +
  geom_density(linewidth = 0.8) +
  scale_colour_manual(values = c(Rural = book_palette$ink,
                                 Urban = book_palette$accent),
                      name = NULL) +
  coord_cartesian(xlim = c(0, 400)) +
  labs(x = "Annual earnings (₹ thousand)", y = NULL) +
  theme_book() +
  theme(axis.text.y = element_blank(),
        legend.position = "top")
```

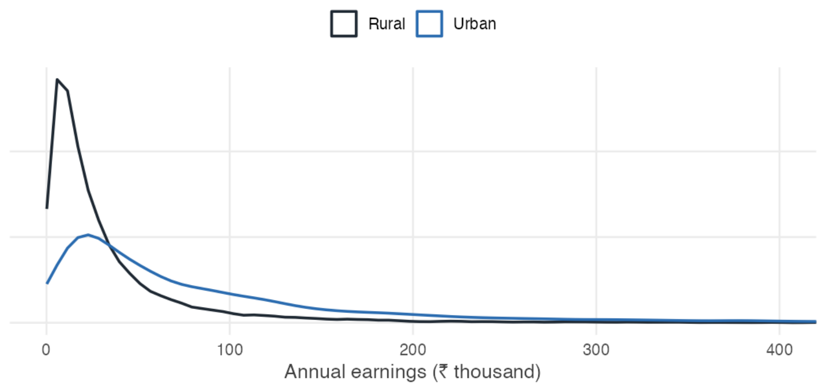

Figure 1.3: Density curves for rural and urban workers.

The entire urban distribution lies to the right of the rural distribution, indicating that urban workers generally earn more than rural workers. The density curve makes this comparison immediately visible without requiring us to examine thousands of individual observations.

By this point, twenty thousand observations have been reduced to a handful of informative summaries: a picture of the distribution, measures of its centre and spread, and a comparison across groups. This is the purpose of **descriptive statistics**. Rather than examining every observation individually, we summarise the data in ways that preserve its most important features.

Descriptive statistics is always the starting point of a statistical analysis. Before asking whether a result is statistically significant or whether it can be generalised beyond the data at hand, we must first understand what the observed data look like. Only then can we ask the central question of this unit: if we had collected a different sample, would these summaries have been different?

1.3 The limits of descriptive statistics

> *How far can description take us?*

Descriptive statistics tells us everything we could want to know about the data we have. The difficulty is that the data we have are rarely the data we ultimately care about.

The twenty thousand workers in our dataset were not selected individually because they were special. They are one sample drawn from a much larger population of Indian workers. Had a different set of workers been surveyed, the dataset would have been different — and so too would the summaries we calculated in the previous section.

To see this, let us treat our dataset as the population and repeatedly draw samples of two thousand workers.

```
set.seed(1)

s1 <- workers$annual_earnings[sample(nrow(workers), 2000)]
s2 <- workers$annual_earnings[sample(nrow(workers), 2000)]
s3 <- workers$annual_earnings[sample(nrow(workers), 2000)]
```

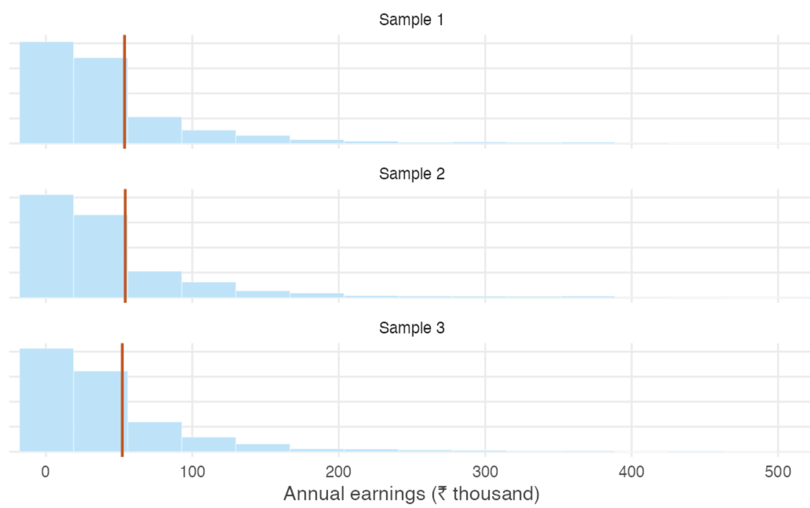

Figure 1.4: Three samples of two thousand workers drawn from the same population.

The three histograms look remarkably similar. Each has the same broad shape: most workers earn relatively little, while a small number earn much more. Nothing suggests that one sample is “better” than another.

Yet the numerical summaries are not identical.

```
c(sample1 = mean(s1),
  sample2 = mean(s2),
  sample3 = mean(s3))
```

```
#> sample1 sample2 sample3
#>   53727   54185   52176
```

Each sample produces a different mean.

The same is true of the median.

```
c(sample1 = median(s1),
  sample2 = median(s2),
  sample3 = median(s3))
```

```
#> sample1 sample2 sample3
#>   24974   25238   24599
```

And of the standard deviation.

```
c(sample1 = sd(s1),
  sample2 = sd(s2),
  sample3 = sd(s3))
```

```
#> sample1 sample2 sample3
#>   99376   94857   86327
```

Every statistic changes because the sample itself has changed. None of the calculations is incorrect, nor has anything gone wrong with the survey. The differences arise simply because different individuals happened to be included.

The exercise above highlights a distinction that lies at the heart of statistical inference. The **population** is the complete collection of units about which we wish to draw conclusions, while the **sample** is the subset that we happen to observe. Because different samples contain different observations, the statistics computed from them also differ. This inherent variability is known as **sampling variation**.

Sampling variation is neither a mistake nor evidence of poor data collection. It is an unavoidable consequence of drawing inferences about a population from a sample. Even when a survey is carefully designed and correctly implemented, a different sample would generally produce different numerical summaries.

This observation also marks the boundary between descriptive and inferential statistics. Descriptive statistics summarises the sample accurately: the sample mean is the mean of the observed data, the sample median is its median, and so on. What descriptive statistics cannot tell us is how closely these sample summaries reflect the corresponding characteristics of the population, or how much they would differ had another sample been drawn.

Inferential statistics addresses precisely these questions. By recognising that the observed sample is only one of many that could have been selected, it provides a framework for assessing the uncertainty associated with sample-based conclusions and for drawing conclusions about the underlying population.

1.4 What if we repeated the survey?

> *If our conclusions depend on the particular sample we happened to observe, how can we learn anything about the population?*

The previous section ended with a simple but unsettling observation. A different sample produces a different mean. The same is true of the median, the standard deviation and every other statistic we compute.

One possibility is to collect another sample.

Imagine repeating the survey. A new team visits a different set of households, following exactly the same sampling design. They compute the average annual earnings and obtain a value that differs slightly from the first survey.

Now imagine repeating the exercise again.

And again.

Every new survey produces a new sample, and every sample produces a new mean. None of these means is "correct" or "incorrect". Each is simply the result of observing a different subset of the population.

Of course, conducting thousands of nationwide surveys would be prohibitively expensive. Fortunately, we do not need to. Since we are treating our dataset as the population for the purposes of this chapter, we can recreate this thought experiment on a computer.

Suppose we repeatedly draw random samples of 2,000 workers from our population. For each sample we calculate the mean annual earnings and store the result. After repeating this process one thousand times, we have one thousand sample means.

```r
set.seed(1)

sample_means <- replicate(
  1000,
  mean(sample(workers$annual_earnings,
              size = 2000,
              replace = TRUE))
)
```

What do these one thousand means look like?

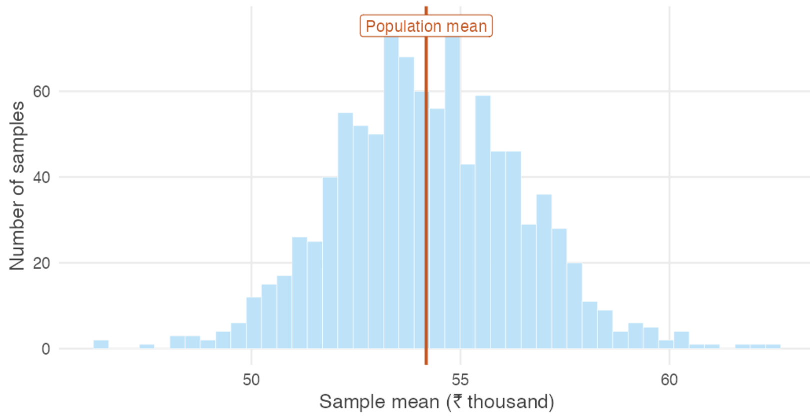

Figure 1.5: Distribution of the sample mean based on 1,000 samples of 2,000 workers.

Several features stand out immediately.

First, the sample means are not all identical. Even though every sample was drawn from the same population using the same procedure, each survey produced a slightly different average.

Second, the variation is not arbitrary. Most of the sample means cluster within a relatively narrow range, while only a few lie far from the centre. Extreme values are possible, but they are uncommon.

Finally, the distribution itself has a remarkably regular shape. Although the earnings of individual workers are highly skewed, the distribution of the sample means is much more symmetric and concentrated.

The thousand means are not just a collection of numbers. They form a distribution of their own. Notice what this distribution describes. It is not the distribution of workers' earnings — that was the distribution we began with. Instead, it is the distribution of the sample mean obtained by repeating the survey many times. The object of interest has changed. We are no longer studying individual workers; we are studying the behaviour of the sample mean.

This distinction lies at the heart of inferential statistics. The original distribution describes the variability in earnings across workers. The new distribution describes the variability in the sample mean across repeated samples. It is this second source of variation that determines how much confidence we should place in the average computed from a single survey.

The distribution formed by a statistic over repeated samples is called its **sampling distribution**. Every statistic has a sampling distribution: the mean, the median, the standard deviation, and many others. In this chapter we focus on the sampling distribution of the sample mean because it provides the foundation for much of statistical inference.

Two features of Figure 1.5 are too striking to ignore. First, the sample means are tightly concentrated around the population mean. Second, despite the strong skewness of individual earnings, their sampling distribution is almost bell-shaped. Neither feature is a coincidence. Both are consequences of two fundamental results in probability theory: the Law of Large Numbers and the Central Limit Theorem. Together, these results explain why a single sample can provide reliable information about an entire population.

1.5 Why do larger samples give more reliable answers?

> *What determines how much they vary?*

Figure 1.5 answered one important question. Repeating the survey many times does not produce wildly different sample means. Instead, the sample means cluster around the population mean.

A natural question follows. What determines how much they vary?

One possibility is the size of the survey itself. Intuition suggests that a survey of twenty workers should produce averages that vary much more from one survey to the next than a survey of two thousand workers. But how much difference does sample size really make?

To find out, let us repeat the experiment from the previous section several times. Instead of fixing the sample size at 2,000 workers, we consider surveys of different sizes. For each sample size we draw one thousand independent samples and compute the average annual earnings.

```
set.seed(3)

earn  <- workers$annual_earnings
sizes <- c(20, 100, 500, 2000)

by_size <- do.call(rbind, lapply(sizes, function(n) {
  data.frame(n = factor(paste("n =", n), levels = paste("n =", sizes)),
             m = replicate(4000, mean(sample(earn, n, replace = TRUE))))
}))
```

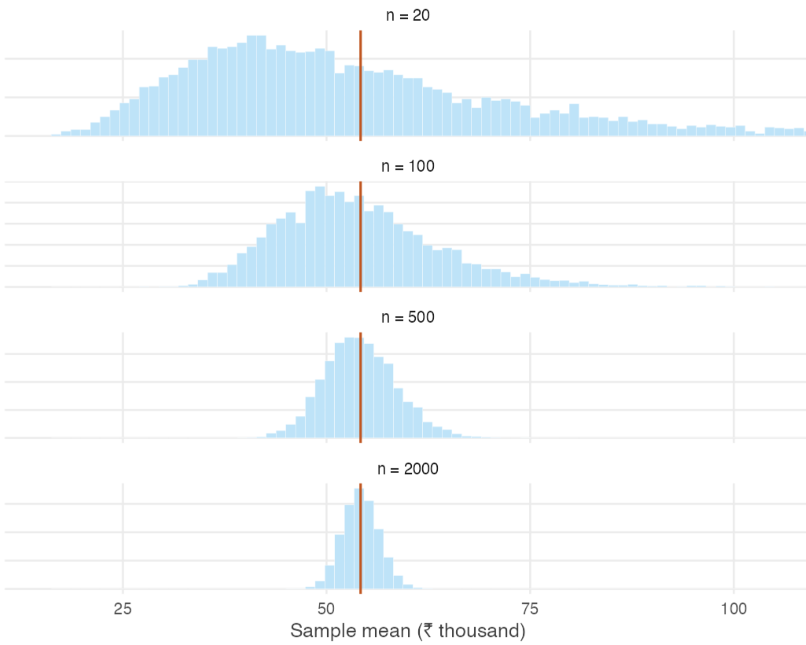

Figure 1.6: Sampling distributions of the sample mean for different sample sizes.

The pattern is unmistakable.

When the sample contains only a small number of workers, the sample means are widely dispersed. Some surveys produce averages well above the population mean, while others fall well below it. The averages vary substantially from one survey to the next.

As the sample size increases, the sampling distribution becomes progressively narrower. The sample means remain centred at the population mean, but they fluctuate much less. Large departures from the population mean become increasingly uncommon, while averages close to the population mean become increasingly likely.

```r
round(tapply(by_size$m, by_size$n, sd))
```

```
#>   n = 20  n = 100  n = 500 n = 2000
#>    22998     9973     4433     2252
```

This behaviour reflects a simple idea. Every sample contains chance variation. Some samples happen to include more high-income workers than average, while others include more low-income workers. In a small sample these random differences have a noticeable effect on the average. As the sample grows, unusually high and unusually low observations increasingly offset one another, and the sample mean settles closer to the population mean.

The behaviour we have just observed is not merely an empirical regularity. It is one of the most fundamental results in probability theory and forms the foundation of statistical inference. It is known as the **Law of Large Numbers**.

In informal terms, the Law of Large Numbers states that as the sample size increases, the sample mean gets closer and closer to the population mean. Large samples do not eliminate randomness, but they reduce its influence. Although no single survey is guaranteed to produce the correct answer, larger surveys are increasingly likely to do so.

The Law of Large Numbers explains why surveys, opinion polls, clinical trials and economic censuses can provide reliable information about populations without observing every individual. The key is not that they observe everyone, but that they observe enough.

EVIDENCE FROM THE REAL WORLD

Real demonstrations of the Law of Large Numbers are surprisingly hard to find. The law concerns what happens when the same random process is repeated a great many times, and few processes in economic life are repeated identically enough, often enough, for the convergence to be visible.
The cricket toss is an exception. Before every international match a coin is tossed, and over nearly a century and a half some teams have accumulated thousands of these tosses while others have played only a handful.

```
toss <- read.csv("data/cricket-toss.csv")
keep <- c("team", "span", "matches", "pct_toss_won")

# the most and least fortunate teams, and the most experienced
rbind(head(toss[order(-toss$pct_toss_won), keep], 2),
      head(toss[order( toss$pct_toss_won), keep], 2),
      head(toss[keep], 3))
```

```
#>            team      span matches pct_toss_won
#> 90       Greece 2019-2024      15        80.00
#> 88 Cook Islands 2022-2025      17        70.59
#> 94    Africa XI 2005-2007       6        16.67
#> 91        China 2023-2024      11        27.27
#> 1       England 1877-2025    2124        49.25
#> 2     Australia 1879-2025    2116        51.23
#> 3         India 1932-2025    1926        49.58
```

Greece has won 80% of its tosses and Africa XI barely one in six, while

England, after 2,124 matches, sits at 49.2%. Nobody is luckier than anybody else. The teams with extreme records are simply the teams that have played few matches.
That is the Law of Large Numbers, and the agreement with theory is close. If the toss is fair, a team with n matches should have a toss-win percentage scattered around 50 with a standard deviation of $50/\sqrt{n}$ percentage points. Restricting attention to teams above a given number of matches:

```r
lln_toss <- t(sapply(c(0, 50, 100, 200, 500), function(m) {
  x <- toss[toss$matches >= m, ]
  c(min_matches = m,
    teams       = nrow(x),
    observed_sd = sd(x$pct_toss_won),
    predicted   = sqrt(mean((50 / sqrt(x$matches))^2)))
}))

round(lln_toss, 2)
```

```
#>      min_matches teams observed_sd predicted
#> [1,]           0    94        8.46      7.91
#> [2,]          50    49        4.92      4.77
#> [3,]         100    27        3.03      3.16
#> [4,]         200    16        2.10      2.05
#> [5,]         500    10        0.85      1.32
```

The spread of toss-win percentages falls from 8.5 percentage points across all teams to 0.9 among the ten most experienced — and at every step it tracks the value the theory predicts. No survey was designed and no experiment was run. The convergence is simply there in the record of the game.

Yet Figure 1.6 reveals another feature that the Law of Large Numbers does not explain. As the sample size increases, the sampling distributions not only become narrower — they also become increasingly bell-shaped. This is surprising because the underlying distribution of earnings is strongly skewed.

Why should averages of skewed data look approximately normal?

The answer is provided by an even more remarkable result: the **Central Limit Theorem**.

1.6 The Central Limit Theorem

> *Why do sample means look bell-shaped?*

The Law of Large Numbers explains why the sample mean moves closer to the population mean as the sample size increases. It tells us where the sample mean is centred, but not how it is distributed for any given sample size. Without knowing that distribution, we cannot quantify the uncertainty associated with a single survey.

The Central Limit Theorem answers precisely this question. Its remarkable feature is its generality.

> **REMEMBER THIS**
>
> **Central Limit Theorem.** For sufficiently large samples, the sampling distribution of the sample mean is approximately normal, regardless of the shape of the population from which the sample is drawn.

The theorem will be stated more precisely once we have developed the necessary notation. For now, its essential idea is already visible in Figure 1.6. Even when the population itself is far from bell-shaped, the distribution of the sample mean becomes increasingly bell-shaped as the sample size grows.

The strength of this result is easy to underestimate. The population may be binary, highly skewed, or concentrated in only one part of its range. It makes remarkably little difference. Average enough observations together, and the distribution of those averages approaches the familiar bell-shaped curve.

A simulation

The easiest way to appreciate the theorem is to watch it happen. Figure 1.7 compares the sampling distributions of the sample mean for four populations with very different shapes, using sample sizes of one, five and thirty.

```r
set.seed(3)

draw_mean <- function(parent, n) {
  switch(parent,
    "Bernoulli" = mean(rbinom(n, 1, 0.3)),
    "Poisson"   = mean(rpois(n, 2)),
    "Uniform"   = mean(runif(n)),
    "Earnings"  = mean(sample(earn, n, replace = TRUE)))
}

parents <- c("Bernoulli", "Poisson", "Uniform", "Earnings")
```

```
sizes   <- c(1, 5, 30)

clt <- do.call(rbind, lapply(parents, function(p)
  do.call(rbind, lapply(sizes, function(n)
    data.frame(parent = p,
               n      = factor(paste("n =", n), levels = paste("n =",
               ↪  sizes)),
               m      = replicate(4000, draw_mean(p, n)))))))

# Standardise within each panel, so all twelve are on one scale and the
# comparison is purely about shape.
clt$z <- ave(clt$m, clt$parent, clt$n,
             FUN = function(x) (x - mean(x)) / sd(x))
```

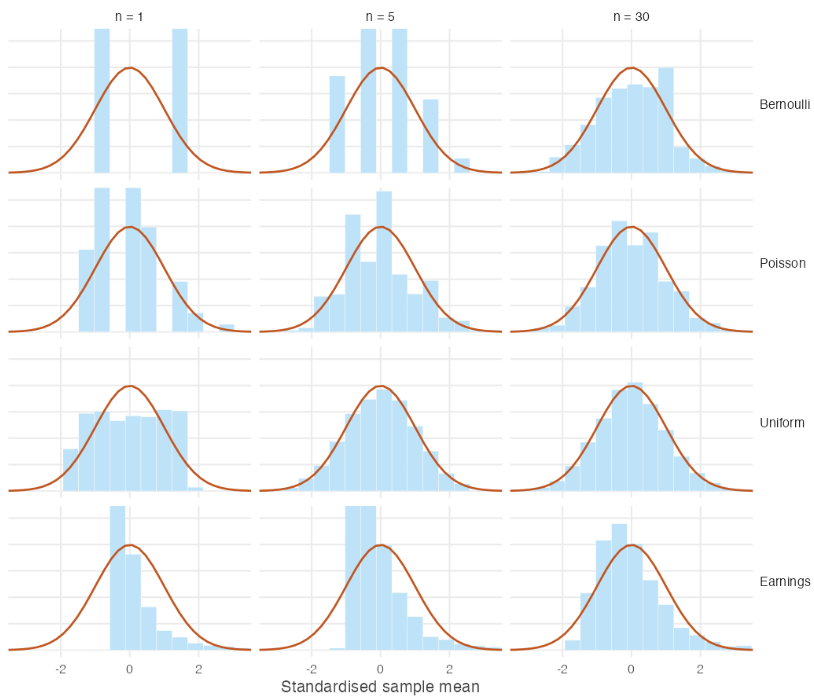

Figure 1.7: Standardised sampling distributions of the sample mean for four populations, against the standard normal curve.

The first column shows the four underlying populations. None resembles a normal distribution. By the third column, however, the distributions of the sample means are all much closer to the familiar bell-shaped curve.

The theorem guarantees that this convergence eventually occurs, but it says nothing about how quickly it happens. Populations that are already fairly symmetric require only modest sample sizes, whereas heavily skewed populations may require substantially larger samples before the normal approximation becomes accurate. In Figure 1.7, the uniform population is already close to normal when $n = 5$, while the earnings distribution remains visibly asymmetric even when $n = 30$.

An intuition

Why should averaging observations produce a bell-shaped distribution?

A simple example provides the intuition. Roll a fair die once and every outcome from one to six is equally likely. Now roll two dice and add the results.

```
sums <- outer(1:6, 1:6, "+")

table(sums)
```

```
#> sums
#>  2  3  4  5  6  7  8  9 10 11 12
#>  1  2  3  4  5  6  5  4  3  2  1
```

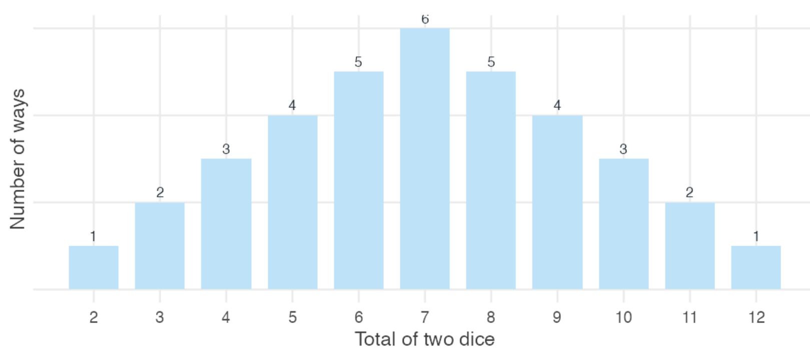

Figure 1.8: The number of ways each total can be obtained when two dice are rolled.

Seven can be obtained in six different ways: $1 + 6,\ 2 + 5,\ 3 + 4,\ 4 + 3,\ 5 + 2,\ 6+1$. By contrast, a total of two can be obtained in only one way. The middle values have many more possible combinations than the extremes, so they occur more frequently.

The same intuition extends far beyond dice. To obtain an unusually large average, most observations must themselves be unusually large. Likewise, an unusually small average requires most observations to be unusually small. Both situations are relatively rare. A value close to the middle, however, can arise from countless combinations of larger and smaller observations offsetting one another.

As more observations are averaged together, the number of ways to obtain values near the centre grows much faster than the number of ways to obtain values at the extremes. The result is the familiar bell-shaped distribution.

How large is "large enough"?

Many textbooks suggest that a sample size of at least thirty is sufficient for the Central Limit Theorem to provide a good approximation. This is a useful rule of thumb, but it is not a mathematical guarantee.

How quickly the approximation becomes accurate depends largely on the shape of the population. Populations that are approximately symmetric require relatively few observations, whereas heavily skewed populations often require much larger samples.

```r
set.seed(9)

skewness <- function(x) mean((x - mean(x))^3) / sd(x)^3

sizes <- c(5, 30, 100, 500)

sk <- c(population = skewness(earn),
        sapply(sizes, function(n)
          skewness(replicate(2000, mean(sample(earn, n, replace =
           ↪  TRUE))))))
names(sk)[-1] <- paste0("n=", sizes)

round(sk, 2)
```

```
#> population        n=5       n=30      n=100      n=500
#>       7.32       2.99       1.20       0.76       0.27
```

The earnings distribution used throughout this chapter has a skewness of approximately 7.3. When the sample size is five, the sampling distribution remains strongly right-skewed. Even at thirty observations, some asymmetry is still visible. By one hundred observations, however, the distribution is much closer to symmetric.

This matters because many economic variables — including incomes, firm sizes, land holdings and city populations — are themselves highly skewed. For such variables, the normal approximation may still be poor when the sample size is only thirty. In these situations, larger samples or simulation-based checks are often preferable to relying mechanically on a rule of thumb.

Why this matters

The Central Limit Theorem, together with the Law of Large Numbers, explains why statistical inference is possible.

The Law of Large Numbers tells us that repeated samples become increasingly centred on the population mean as the sample size grows. The Central Limit Theorem tells us that the distribution of those sample means approaches a normal distribution. Together, these results reveal that although different samples produce different answers, those answers are not arbitrary. They vary in systematic and predictable ways.

> **REMEMBER THIS**
>
> This insight lies at the heart of inferential statistics. We observe only one sample, but by understanding how the same statistic would behave across many hypothetical samples, we can quantify the uncertainty associated with the sample we actually observe.

So far we have developed these ideas largely through intuition, simulation and visualisation. The next section introduces the notation and concepts needed to express them more precisely and to build the methods of statistical inference developed in the remainder of this book.

1.7 A language for randomness

> *Why do we need a mathematical language?*

Throughout this chapter we have described repeated sampling almost entirely in plain English. We have spoken about rolling dice, drawing surveys, computing averages and repeating experiments thousands of times. That language has helped us build intuition, but it is no longer sufficient. To develop statistical inference, we need a concise and precise way of expressing these ideas.

The first concept we need is that of a random variable.

Random variables

Some quantities are known before we observe them. If you measure your height today, it is a fixed number. It may be unknown to you, but it is not changing from one repetition to the next.

Other quantities are different. Roll a die and you do not know what number will appear. Randomly select a worker from a population and you do not know that person's earnings. Draw a sample of one hundred households and you do not know what the sample mean will be until the sample has actually been collected.

Statisticians describe such quantities as **stochastic**. A stochastic quantity is one whose value is determined by the outcome of a random process.

The mathematical object used to describe a stochastic quantity is called a **random variable**. We usually denote random variables by capital letters such as X, Y and Z. For example, X may denote the earnings of a randomly selected worker, Y the annual rainfall in a particular district, and D the outcome of rolling a fair die.

The sample mean deserves special attention. Every time we draw a different sample, we obtain a different average. The sample mean is therefore itself a random variable. Before observing the data, we denote it by $\bar{X}$.

> **REMEMBER THIS**
>
> This observation is fundamental to inferential statistics. Since $\bar{X}$ is a random variable, it has a probability distribution. That distribution tells us how the sample mean behaves across repeated samples. Once we understand its centre and spread, we can quantify the uncertainty associated with the single sample we actually observe.

Realisations

Eventually every random variable takes a particular observed value.

Suppose X represents the outcome of rolling a fair die. Before the die is rolled, X could take any value from one to six. After the die lands showing four, we have observed one **realisation** of the random variable, written as $x = 4$.

The same distinction applies to the sample mean. Before collecting a sample, the sample mean is the random variable $\bar{X}$. After collecting the data and computing the average, we observe one realisation of that random variable, written as $\bar{x}$.

Throughout the remainder of this book, capital letters denote random variables, while lower-case letters denote their observed values.

The long-run average

Every random variable has a centre. The centre is called its **expected value** and is written as

$$E(X) = \sum_i x_i \, P(X = x_i).$$

The term *expected value* can be misleading. It is not the value we expect to observe on any particular repetition. Rather, it is the average value the random variable would take over a very large number of repetitions.

For a fair die,

$$E(X) = \frac{1+2+3+4+5+6}{6} = 3.5.$$

Of course, no single roll can ever produce 3.5. It is a theoretical average, not an individual outcome.

This distinction becomes clearer when we compare theory with data.

```
mean(rolls)                # average of 10,000 simulated rolls
```

```
#> [1] 3.514
```

```
sum(1:6 * rep(1/6, 6))     # theoretical expected value
```

```
#> [1] 3.5
```

The close agreement between these two numbers is exactly what the Law of Large Numbers predicts. The expected value is calculated from the probability model, while the sample average is calculated from observed data. As the number of observations grows, the latter approaches the former.

When the random variable represents a randomly selected member of a population, its expected value is simply the population mean. For this reason we denote both by the same symbol,

$$E(X) = \mu.$$

Measuring variation

Knowing the centre of a distribution is not enough. Two random variables can have the same expected value but very different amounts of variation.

The spread of a random variable around its expected value is measured by its **variance**,

$$\mathrm{Var}(X) = E[(X - \mu)^2],$$

and the square root of the variance is the **standard deviation**,

$$\sigma = \sqrt{\mathrm{Var}(X)}.$$

The expected value tells us where a random variable is centred. The variance and standard deviation tell us how much it fluctuates around that centre.

Parameters and statistics

Statistics is concerned with learning about populations from samples. It is therefore essential to distinguish between quantities that describe a population and quantities computed from a sample.

A **parameter** is a numerical quantity describing the population. The population mean is written μ, the population variance σ^2, and the population standard deviation σ. These quantities are fixed — they do not change — but they are usually unknown, because we rarely observe the entire population.

A **statistic**, by contrast, is computed from a sample. The sample mean, sample variance and sample standard deviation are all statistics. Unlike parameters, statistics change from one sample to the next, because they depend on which observations happen to be selected.

The entire objective of inferential statistics can now be expressed in a single sentence:

> **REMEMBER THIS**
>
> Use statistics computed from a sample to learn about unknown parameters describing the population.

Everything that follows in this book — estimation, confidence intervals, hypothesis testing and regression analysis — is an extension of this single idea.

A summary of notation

The notation introduced in this section will be used throughout the book.

Symbol	Meaning
X, Y, Z	random variables — quantities not yet observed

Symbol	Meaning
$x,\ y,\ z$	realisations — values actually observed
$\bar{X}$	the sample mean, before the sample is drawn
$\bar{x}$	the sample mean, once the sample is observed
n	the sample size
$E(X)$	expected value: the long-run average of X
$\mathrm{Var}(X)$	variance: the spread of X about its expected value

And, separating what describes the population from what is computed from a sample:

Quantity	Population (parameter)	Sample (statistic)
Mean	μ	$\bar{x}$
Variance	σ^2	s^2
Standard deviation	σ	s

The left-hand column is fixed and unknown. The right-hand column is observed and changes with every sample. Almost all of statistics is the effort to say something about the first using only the second.

1.8 Before any mathematics can help

> *What must be true of the data before any of this applies?*

Everything in this unit assumes that the sample carries information about the population we care about. That assumption is not guaranteed by any formula, and when it fails, no amount of statistical sophistication will announce the failure. The arithmetic runs perfectly on data that cannot answer the question.

Four ways it fails, none of which any calculation will warn you about.

The sample is not from the population you mean. The Annual Survey of Industries covers *registered* factories. Most Indian manufacturing employment is unregistered. Conclusions about "Indian manufacturing" drawn from that source are conclusions about a formal sliver of it, however large the sample.

The sample is not random. The class survey in Section 1.2 has thirty-two heights. They are not a random sample of students at the university, still less of young Indians: they are the people who took the course and attended that day. A confidence interval computed from them is arithmetically correct and tells you nothing about anyone else.

The data are measured badly. Incomes reported from memory are systematically understated. Rainfall attributed to a district may have been recorded at a station a hundred kilometres away. National accounts are revised for years after first publication. A sampling distribution describes how an estimate varies across samples; it says nothing about whether the quantity was measured correctly in the first place. Careful statistics performed on poor measurements remain poor statistics.

The question is causal but the data are not. Suppose states with more industry have higher literacy. Does industry raise literacy? Does literacy attract industry? Does something else drive both? The correlation is identical in all three cases, and no statistical procedure separates them without an argument about how the data came to exist.

> **REMEMBER THIS**
>
> These are not edge cases. They are the ordinary condition of applied economics, and recognising them is a skill quite separate from computation. Good inference begins with good data and thoughtful study design, not with clever mathematics.

1.9 Summary

> **REMEMBER THIS**
>
> 1. **Randomness creates variation, and variation is the raw material of statistics.** Where nothing varies there is no distribution, no uncertainty, and nothing to infer.
> 2. **Descriptive statistics** summarises the data in hand. **Inferential statistics** uses that data to make claims about a population it did not observe, and reports how uncertain those claims are.
> 3. Different samples give different answers. That is **sampling variation —** unavoidable, and not a defect in the data.
> 4. A quantity whose value changes from one repetition to the next is a **random variable**, written X. Its observed value is a **realisation**, written x. The sample mean is $\bar{X}$ before the sample is drawn and $\bar{x}$ after.

5. $E(X) = \sum_i x_i P(X = x_i)$ is the **expected value** — a long-run average, not a value you expect to see. For a population draw, $E(X) = \mu$.
6. $\mathrm{Var}(X) = E[(X-\mu)^2]$ measures spread, and $\sigma = \sqrt{\mathrm{Var}(X)}$ is the standard deviation.
7. **Parameters** (μ, σ^2, σ) describe the population and are fixed but unknown. **Statistics** ($\bar{x}$, s^2, s) are computed from a sample and change with every sample. Inference uses the second to learn about the first.
8. The distribution of a statistic across repeated samples is its **sampling distribution**. Every statistic has one, not only the mean.
9. **Law of Large Numbers:** as n grows, the sample mean becomes increasingly likely to lie close to μ. It does not say any particular large sample is correct, and it offers no comfort to a gambler.
10. **Central Limit Theorem:** for sufficiently large n, the sampling distribution of $\bar{X}$ is approximately normal whatever the shape of the population. The rule $n \geq 30$ is a rule of thumb, not a guarantee — a heavily skewed population needs far more.
11. No formula warns you when the sample is drawn from the wrong population, drawn non-randomly, measured badly, or asked a causal question it cannot answer.

Taken together, these amount to a single change of perspective. We stopped asking what a particular sample says, and began asking how a statistic behaves across the many samples that might have been drawn. Everything that follows — estimation, confidence intervals, hypothesis testing, regression — rests on that shift.

1.10 Practice problems

YOUR TURN

These are for learning. Work each one before opening the solution; the answer is worth very little if you have not first tried to produce it yourself. The question bank that follows is for testing yourself and has no answers at all.

Concept checks

Short questions. A sentence or two is enough.

1.1 Why does repeating the same survey produce a different average each time?

SOLUTION

Because a different set of people is surveyed each time. Every average is computed from whoever happened to be selected, and selection involves chance. The population has not changed and no arithmetic error has occurred — only the membership of the sample.

1.2 A die with a three painted on all six faces is rolled ten thousand times. What is the distribution of the outcome? Why is this example in the chapter at all?

SOLUTION

There is no distribution to speak of: every outcome is three, and the standard deviation is zero. The example is there to show that statistics requires variation. Where outcomes never differ, there is nothing to describe, nothing to be uncertain about, and no inference to perform.

1.3 Distinguish between $\bar{X}$ and $\bar{x}$.

SOLUTION

$\bar{X}$ is the sample mean regarded as a random variable, before the sample is drawn: it has a distribution, a centre and a spread. $\bar{x}$ is the single number obtained once a particular sample has been collected — one realisation of $\bar{X}$.

1.4 A survey of 400 households gives a mean monthly income of ₹18,000. Is ₹18,000 a parameter or a statistic? What is the parameter here?

SOLUTION

₹18,000 is a statistic: it was computed from a sample and would change with a different sample. The parameter is the mean monthly income of the whole population, μ, which is fixed but unknown.

Explain

One paragraph each. These reward clarity rather than calculation.

1.5 Explain to a friend who has never studied statistics why the Law of Large Numbers does not mean that a large sample is guaranteed to give the right answer.

SOLUTION

The law says that as the sample grows, the sample mean becomes increasingly likely to lie close to the population mean — not that it will equal it. Any particular large sample may still be unlucky. What changes with sample size is the *probability* of being far off, which shrinks; the possibility never disappears. A large sample makes a badly wrong answer improbable, not impossible.

1.6 A friend argues that because a coin has landed heads six times running, tails is now more likely. Using the Law of Large Numbers, explain why this is wrong.

SOLUTION

The coin has no memory, and the law promises nothing of the kind. What converges is the *proportion* of heads, and it converges by dilution rather than by correction. The six extra heads are never cancelled out; they simply become a smaller and smaller share of a growing total. The absolute surplus of heads typically grows with the number of tosses even as the proportion approaches one half.

1.7 Two researchers study earnings in the same district. One surveys 100 households, the other 2,500. Both report a mean. Which should be trusted more, and what would you need to know before answering?

SOLUTION

Other things equal, the larger survey has the smaller standard deviation of the sample mean — by a factor of five, since $\sqrt{2500/100} = 5$. But this holds only if both samples were drawn from the population of interest and drawn at random. A carefully designed survey of 100 households is worth more than a badly drawn survey of 2,500. Sample size governs precision; sampling design governs whether precision is about the right quantity at all.

Numerical problems

Work each of these with a pen first. Each ends by asking you to check your answer by simulation — the point being that theory and simulation should agree, and that when they do not, one of them is wrong.

1.8 The earnings data used throughout this chapter has a population mean of about ₹54,200 and a population standard deviation of about ₹100,100.

Suppose a survey team draws random samples of (a) 50, (b) 100, (c) 200 and (d) 1,000 workers.

i. Find the expected value of the sample mean in each case.
ii. Find the standard deviation of the sample mean in each case.
iii. What happens to that standard deviation as the sample size increases, and why?
iv. Verify your answers by simulation.

SOLUTION

The expected value of the sample mean equals the population mean whatever the sample size, so $E(\bar{X}) =$ ₹54,200 in all four cases. Sample size affects precision, not location.
The standard deviation of the sample mean is $\sigma/\sqrt{n} = 100{,}100/\sqrt{n}$:

```r
sigma <- 100100
n     <- c(50, 100, 200, 1000)

round(data.frame(n, expected = 54200, se = sigma / sqrt(n)))
```

```
#>      n expected    se
#> 1   50    54200 14156
#> 2  100    54200 10010
#> 3  200    54200  7078
#> 4 1000    54200  3165
```

It falls in proportion to $1/\sqrt{n}$. Each additional worker contributes information, so the average over many of them is pulled more tightly around the population value. Note that raising the sample from 50 to 1,000 — twenty times the data — reduces it only by a factor of $\sqrt{20} \approx 4.5$.
Because we hold the whole population, the answer can be checked directly:

```r
set.seed(4)
earn <- read.csv("data/wages-india-synthetic.csv")$annual_earnings

sim <- sapply(n, function(k)
  sd(replicate(2000, mean(sample(earn, k, replace = TRUE)))))

round(data.frame(n, theoretical = sd(earn) / sqrt(n), simulated =
↪  sim))
```

```
#>      n theoretical simulated
#> 1   50       14162     14219
#> 2  100       10014      9740
#> 3  200        7081      7204
#> 4 1000        3167      3120
```

The two columns agree closely. The small discrepancies are themselves sampling variation: each simulated standard deviation is estimated from 2,000 samples rather than infinitely many.

1.9 An auto-rickshaw driver's daily earnings vary from day to day, with a mean of ₹800 and a standard deviation of ₹200. In a particular month the driver works 30 days.

i. What are the mean and standard deviation of total earnings for the month?
ii. Approximate the probability that the month's total exceeds ₹25,000.
iii. Approximate the probability that it falls below ₹22,000.
iv. Approximate the probability that it lies between ₹22,000 and ₹25,000.
v. What assumption does this calculation require, and is it plausible here?
vi. Check your answers by simulating 10,000 months.

SOLUTION

The monthly total is a sum of 30 daily earnings. Its mean is $30 \times 800 =$ ₹$24{,}000$, and because variances add for independent quantities, its standard deviation is $200\sqrt{30} =$ ₹$1{,}095$.
By the Central Limit Theorem the total is approximately normal, so:

```
mu_day <- 800; sd_day <- 200; days <- 30

mu_tot <- mu_day * days
sd_tot <- sd_day * sqrt(days)

c(mean = mu_tot, sd = sd_tot,
  above_25000 = pnorm(25000, mu_tot, sd_tot, lower.tail = FALSE),
  below_22000 = pnorm(22000, mu_tot, sd_tot),
  between     = pnorm(25000, mu_tot, sd_tot) - pnorm(22000, mu_tot,
  ↪  sd_tot))
```

```
#>        mean          sd above_25000 below_22000     between
#> 24000.00000  1095.44512     0.18066     0.03394     0.78540
```

So roughly an 18% chance of clearing ₹25,000, a 3% chance of falling below ₹22,000, and a 79% chance of landing between the two.
The assumption is independence. The calculation treats each day's earnings as unrelated to every other day's. That is questionable. A week of heavy monsoon suppresses earnings on consecutive days; a festival week raises them together. When days are positively correlated, the true standard deviation of the monthly total is *larger* than $200\sqrt{30}$, and the probability of an unusually bad month is correspondingly higher than this calculation suggests.

This is worth dwelling on. The arithmetic is not wrong; the assumption is. No amount of care in the computation would reveal the problem. Simulating under the stated assumption:

```
set.seed(5)
months <- replicate(10000, sum(rnorm(days, mu_day, sd_day)))

c(mean = mean(months), sd = sd(months),
  above_25000 = mean(months > 25000),
  below_22000 = mean(months < 22000),
  between     = mean(months >= 22000 & months <= 25000))
```

```
#>        mean          sd above_25000 below_22000     between
#>  23987.5096   1090.6065      0.1788      0.0321      0.7891
```

The simulated proportions match the normal approximation closely — as they must, since the simulation was itself built on the assumption of independent daily earnings. Simulation confirms the arithmetic; it cannot confirm the assumption.

R exercises

1.10 Simulate the Central Limit Theorem starting from an exponential distribution, which is strongly right-skewed. Draw 5,000 sample means for sample sizes of 1, 5, 30 and 200 from `rexp(n, rate = 1)`, and plot the four distributions. At which sample size does the result begin to look normal?

SOLUTION

```
set.seed(1)

sizes <- c(1, 5, 30, 200)
sim <- do.call(rbind, lapply(sizes, function(n)
  data.frame(n = factor(paste("n =", n), levels = paste("n =",
  ↪  sizes)),
             m = replicate(5000, mean(rexp(n, rate = 1))))))

ggplot(sim, aes(m)) +
  geom_histogram(bins = 50, fill = book_palette$fill, colour =
  ↪  "white",
                 linewidth = 0.1) +
  facet_wrap(~ n, scales = "free") +
  labs(x = "Sample mean", y = NULL) +
  theme_book() +
  theme(axis.text.y = element_blank())
```

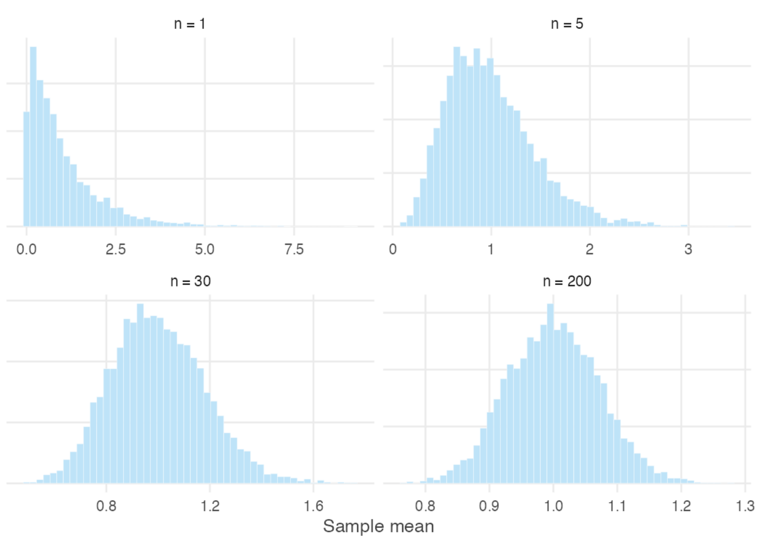

Figure 1.9: Sampling distributions of the mean from an exponential population.

At $n = 1$ the picture is the exponential distribution itself — a sharp peak at zero with a long right tail. By $n = 5$ it has a recognisable mound but remains clearly skewed. By $n = 30$ it is close to symmetric, and by $n = 200$ it is difficult to distinguish from a normal distribution by eye.

The exponential distribution has skewness 2, which is milder than the earnings data used in this chapter. That is why $n = 30$ works reasonably well here and less well there.

1.11 Using `data/cricket-toss.csv`, plot each team's toss-win percentage against its number of matches. What shape does the scatter take, and why?

SOLUTION

```
toss <- read.csv("data/cricket-toss.csv")

band <- data.frame(n = 5:2200)
band$hi <- 50 + 2 * 50 / sqrt(band$n)
band$lo <- 50 - 2 * 50 / sqrt(band$n)

ggplot(toss, aes(matches, pct_toss_won)) +
  geom_ribbon(data = band, aes(x = n, ymin = lo, ymax = hi),
              inherit.aes = FALSE, fill = book_palette$fill, alpha =
               ↪  0.45) +
  geom_hline(yintercept = 50, colour = book_palette$reject,
   ↪  linewidth = 0.6) +
  geom_point(alpha = 0.7, size = 1.6, colour = book_palette$ink) +
  scale_x_log10(labels = scales::comma) +
  labs(x = "Matches played (log scale)", y = "Toss-win percentage") +
  theme_book()
```

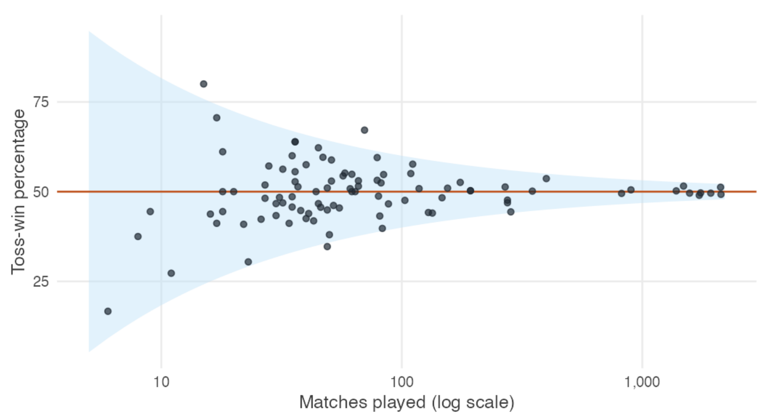

Figure 1.10: Toss-win percentage against number of matches played, with the theoretical limits of two standard deviations.

The scatter forms a funnel: wide on the left, where teams have played few matches, and narrowing towards 50% as the number of matches grows.

The shaded band shows two standard deviations either side of 50, computed as $50/\sqrt{n}$ — the same formula that governs the sample mean. Almost every team falls inside it.
This is the Law of Large Numbers made visible. The teams with extreme records are not lucky; they are simply the teams with few observations.

1.11 Question bank

WATCH OUT

No answers are given for this section, by design. These questions are for testing yourself once the practice problems are complete. There are no hints and no worked solutions — exactly as in an examination.

Multiple choice

QB 1.1 Which of the following is a random variable? *(select all that apply)*

a. The number showing after a die is rolled
b. The number showing on a die with a three painted on every face
c. The mean earnings of a sample of 500 workers, before the sample is drawn
d. The mean earnings of a sample of 500 workers, after the sample is drawn

QB 1.2 A population has mean $\mu = 40$ and standard deviation $\sigma = 12$. For samples of size 36, the standard deviation of the sample mean is

a. 12
b. 2
c. 0.33
d. 6

QB 1.3 Which statements about the Law of Large Numbers are correct? *(select all that apply)*

a. As the sample grows, the sample mean becomes more likely to be close to the population mean
b. As the sample grows, the sample mean becomes certain to equal the population mean
c. It applies only when the population is normally distributed

d. It explains why a survey need not interview everyone

QB 1.4 A researcher doubles the size of a survey from 500 to 1,000 households. The standard deviation of the sample mean falls by a factor of approximately

a. 2
b. 4
c. 1.41
d. It does not change

QB 1.5 The Central Limit Theorem states that

a. Large samples are normally distributed
b. The population becomes normal as the sample grows
c. The sampling distribution of the sample mean is approximately normal for sufficiently large samples
d. Any variable is approximately normal if enough observations are collected

QB 1.6 Which of the following would make the normal approximation to the sampling distribution *less* accurate at a given sample size? *(select all that apply)*

a. A heavily right-skewed population
b. A symmetric population
c. A population with a small number of extreme outliers
d. A larger population standard deviation

QB 1.7 A statistic differs from a parameter because

a. A statistic is always correct
b. A statistic is computed from a sample and changes from sample to sample
c. A parameter is computed from a sample
d. A parameter changes when a different sample is drawn

Short reasoning

QB 1.8 A newspaper reports that average household income in a district is ₹32,000, based on a survey of 800 households, and describes this as "the income of a typical household". Give two separate objections to that description.

QB 1.9 An analyst has a sample of 50,000 observations and concludes that because the sample is very large, the results must be reliable. Under what circumstances would this reasoning be wrong?

QB 1.10 Explain why the sampling distribution of the sample mean is narrower than the distribution of the individual observations, using the idea of offsetting variation rather than any formula.

Numerical

QB 1.11 The weights of rice bags filled by a machine have mean 25 kg and standard deviation 0.6 kg. A quality inspector weighs 36 bags.

(a) What is the expected value of the sample mean weight?
(b) What is its standard deviation?
(c) How many bags would need to be weighed to halve that standard deviation?

QB 1.12 Daily wages of agricultural labourers in a district have mean ₹350 and standard deviation ₹90. For a random sample of 100 labourers, approximate the probability that the sample mean wage

(a) exceeds ₹360, (b) falls below ₹340, (c) lies between ₹340 and ₹360.

QB 1.13 A population is strongly right-skewed with mean 12 and standard deviation 15. A student proposes to use the normal approximation for the sampling distribution of the mean with a sample of 20 observations. Comment on whether this is advisable, and describe how you would check.

1.12 Looking ahead

We now understand why different samples produce different answers, and why those answers follow predictable patterns.

Suppose you collect a sample of fifty households and find an average monthly income of ₹24,000.

Is that evidence that the population average is ₹24,000?

Could the true average instead be ₹25,000? Or ₹23,500? How different would your sample have to be before you concluded that the population itself is different?

These questions motivate the next unit.

Chapter 2

Estimation: Why the Sample Mean?

2.1 The problem of estimation

> *We want to know something about a population we cannot observe. What exactly are we going to do?*

The previous unit was about uncertainty. It established that repeated samples produce different averages, that those averages have a sampling distribution of their own, and that this distribution becomes increasingly concentrated as the sample grows. All of that described how a statistic *behaves*.

It did not tell us what to *do*.

Consider the position of anyone conducting a real survey. There is a quantity they want to know — the average annual earnings of Indian wage workers, the proportion of households with piped water, the average yield per hectare of paddy in Punjab. That quantity is a fixed number. It exists. It is simply not observed, because nobody has surveyed all 1.4 billion people, and nobody ever will.

What they have instead is a sample, and a decision to make: which number computed from that sample should be put forward as the answer?

> **REMEMBER THIS**
>
> A **parameter** is a fixed, unknown feature of the population, such as μ or σ^2.
> An **estimator** is a rule for computing a guess at a parameter from a sample.

> Because it is computed from a sample, it is a random variable, and we write it with a capital letter: $\bar{X}$.
> An **estimate** is the number that rule produces from the sample actually drawn: $\bar{x} = 54{,}300$.

The distinction between the last two is worth pausing on, because the words are often used interchangeably in ordinary speech. An estimator is the rule: take the observations, add them together and divide by n. An estimate is the number produced after the rule has been applied to a particular sample.

The difference matters. An estimate can never be judged directly, because we almost never know the true population mean. If a survey estimates average monthly income to be ₹50,000, we do not know whether the truth is ₹49,800, ₹50,100 or ₹55,000. The truth is precisely what we are trying to discover.

The estimator, however, can be judged. Although we never learn whether one particular estimate is close to the truth, we can ask how the estimator behaves across many hypothetical samples. Does it systematically overestimate the population mean? Does it systematically underestimate it? Does it produce very different answers from one sample to the next? These are questions about the procedure, not about any single outcome.

> **REMEMBER THIS**
>
> Since the parameter is never known, an estimator cannot be judged by any single estimate. It is judged by how it behaves across the many samples that might have been drawn.
> Evaluating procedures rather than outcomes is the idea that drives the whole of this unit.

There are many possible estimators of the population mean. We could use the sample median. We could take the midpoint of the largest and smallest observations. We could even ignore the data entirely and always report ₹50,000. Each is an estimator of μ, because each provides a rule for producing an estimate from a sample.

Why, then, has statistics settled almost universally on the sample mean?

That is the question for the rest of this unit.

2.2 When can a sample represent a population?

> *Before comparing estimators — can a sample tell us anything about a population at all?*

There is an objection to deal with first. Everything above assumed that a sample carries information about the population it came from. That is not automatic. It depends on how the sample was collected, and it can fail badly.

Rather than assert this, we can watch it happen. Below is a population of 20,000 households, spread across 200 localities. Sixty of those localities are urban, and incomes there are higher. Because we build the population ourselves, we know its mean exactly.

```
set.seed(2)

n_loc <- 200                 # localities
per   <- 100                 # households in each

urban    <- rep(c(TRUE, FALSE), times = c(60, 140))
loc_mean <- 18000 + 9000 * urban + rnorm(n_loc, sd = 4000)

pop <- data.frame(
  locality = rep(1:n_loc, each = per),
  urban    = rep(urban,    each = per),
  income   = round(rnorm(n_loc * per, rep(loc_mean, each = per), sd =
  ↪  5000))
)

mu    <- mean(pop$income)
sigma <- sd(pop$income)

c(mu = mu, sigma = sigma, se_if_random = sigma / sqrt(100))
```

```
#>          mu       sigma se_if_random
#>     20724.8      7974.4        797.4
```

The population mean is 20,725. Each survey below collects 100 households, and is repeated a thousand times. Only the *method of collection* differs.

```
# Survey A: 100 households drawn at random from the whole population
# Survey B: 100 households drawn at random, but only from urban localities
# Survey C: five localities chosen at random, 20 households in each
# Survey D: five *urban* localities chosen at random, 20 households in each

cluster_sample <- function(pool) {
  chosen <- sample(pool, 5)
```

```r
  unlist(lapply(chosen, function(l) sample(pop$income[pop$locality == l],
   ↪  20)))
}

urban_pop <- pop$income[pop$urban]
urb_loc   <- which(urban)

set.seed(11)
designs <- list(
  A = replicate(1000, mean(sample(pop$income, 100))),
  B = replicate(1000, mean(sample(urban_pop,  100))),
  C = replicate(1000, mean(cluster_sample(1:n_loc))),
  D = replicate(1000, mean(cluster_sample(urb_loc)))
)

round(t(sapply(designs, function(x) c(centre = mean(x), spread = sd(x)))))
```

```
#>   centre spread
#> A  20702    788
#> B  27503    667
#> C  20774   2843
#> D  27538   2140
```

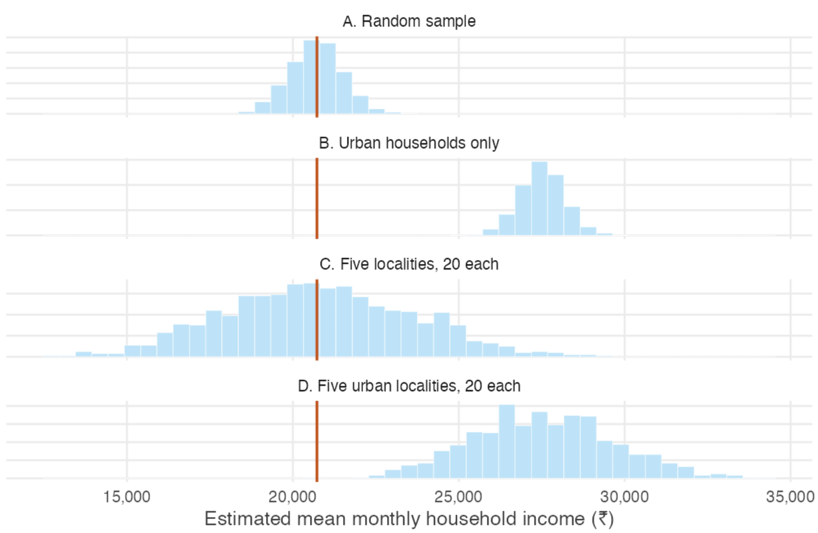

Figure 2.1: One thousand surveys under each design. The vertical line is the true population mean. Only the first design is both centred on it and as precise as theory promises.

Read the four rows carefully, because they fail in different ways.

Survey A works. The thousand estimates are centred on the population mean, and their spread is almost exactly the $\sigma/\sqrt{n}$ predicted in the previous unit.

Survey B is wrong, and looks excellent. Every estimate is far too high — the surveyor never visits a village, so village incomes cannot appear in the answer. Notice the second column: the spread is *smaller* than in Survey A. Urban households resemble one another, so the estimates agree closely with each other while agreeing on the wrong number. Consistency is not accuracy. This is the more dangerous failure, because a narrow spread is easily mistaken for a reliable result.

Survey C is centred correctly but far too variable. Choosing five localities and surveying twenty households in each is cheap and common — an interviewer travels to a place and works through it. But households in a locality resemble one another, so the second household adds much less new information than the first. The sample has 100 rows and nothing like 100 households' worth of information. The spread is over three times what $\sigma/\sqrt{n}$ predicts, so a researcher using that formula would report confidence they have not earned.

Survey D fails both ways at once, which is what most convenience samples do in practice.

> *What did the first survey have that the others did not?*

Two things, and each failure above corresponds to losing one of them.

> **REMEMBER THIS**
>
> **Identically distributed** — every observation is drawn from the same distribution, and it is the distribution we want to learn about. Survey B violated this: urban households are drawn from a different distribution than the population of Indian households.
> **Independent** — knowing the value of one observation tells us nothing about another. Survey C violated this: neighbours share a locality, a labour market and often an occupation.
> Together the two are abbreviated **i.i.d.**

These assumptions are not mathematical decoration. Sections 2.4 and 2.5 derive the two properties that justify using the sample mean, and each derivation uses exactly one of them — in one identifiable line. Break the assumption and the property it supports disappears, which is precisely what the four surveys show.

> **WATCH OUT**
>
> No amount of mathematical sophistication repairs a badly collected sample. If the households you surveyed are not drawn from the population you are asking about, there is no formula that recovers the ones you

> missed.
> Good inference begins with study design, not with statistics.

2.3 How can we judge an estimator?

> *If μ is never known, how can we ever tell whether an estimator is any good?*

We now know what a sample must satisfy. The original question remains: of the many rules available, why the sample mean?

The obstacle is the one identified at the start. To see whether an estimator lands near the truth we would have to know the truth, and in any real application we do not. If we survey Indian households we do not know true average household income; if we sample firms we do not know true average productivity. There is nothing to compare against.

The way around this is to build a laboratory. We construct one artificial population whose properties are known exactly, and study repeated sampling from it. Because the true mean is known in advance, every sample mean can be measured against it. Nothing is being discovered about the world here — the point is to watch a procedure work under conditions where its performance is visible.

The population

Take a deck of playing cards, scored A = 1, J = 11, Q = 12, K = 13, and 2 through 10 at face value.

```
deck <- rep(1:13, times = 4)

c(N = length(deck), mu = mean(deck), sigma = sd(deck))
```

```
#>      N     mu  sigma
#> 52.000  7.000  3.778
```

This population contains 52 values. Its mean is exactly $\mu = 7$ and its standard deviation is $\sigma = 3.78$. Both are known with certainty, which is exactly what real data never offers.

One sample from it

Draw ten cards and compute their average.

```
set.seed(7)
mean(sample(deck, size = 10, replace = TRUE))
```

```
#> [1] 5.7
```

The result is not 7. Draw again and it would differ again — sampling variation, as before. What is new is that we can see how far this sample strayed, because we know where it should have been.

> **WATCH OUT**
>
> Note `replace = TRUE`: each card is returned to the deck before the next draw.
> That is what makes the draws **independent**, in the sense of the previous section. Drawing without replacement changes what remains after each card is removed, so the draws are dependent and the sample mean varies slightly less than theory predicts.
> It is also the better model of what we usually mean. Surveying 2,000 people does not meaningfully deplete a population of a billion.

A thousand samples

One sample tells us little about a procedure. Repeating the exercise a thousand times shows the whole sampling distribution — and, for the first time, shows it alongside the truth it is supposed to be tracking.

```
set.seed(7)
deck_means <- replicate(1000, mean(sample(deck, size = 10, replace =
  TRUE)))

c(mean_of_sample_means = mean(deck_means),
  sd_of_sample_means   = sd(deck_means))
```

```
#> mean_of_sample_means   sd_of_sample_means
#>                6.932                1.201
```

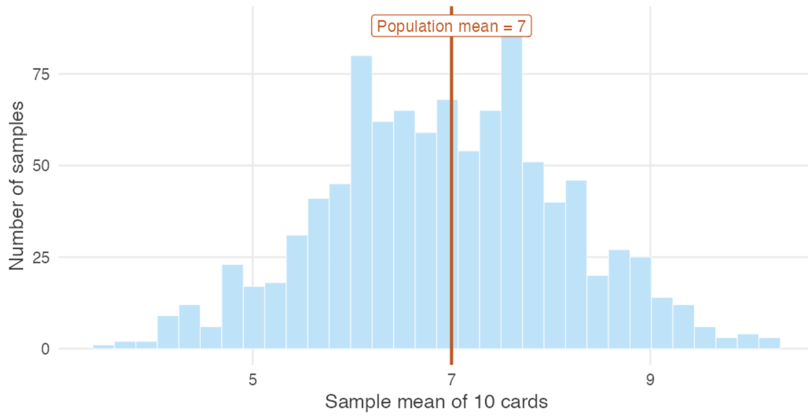

Figure 2.2: One thousand sample means, each computed from 10 cards drawn from a population whose mean is exactly 7.

Two features of this picture are worth separating, and the next two sections take them one at a time: where the distribution sits, and how wide it is.

2.4 Is the sample mean centred on the truth?

> *The sample means cluster around 7. Is that a coincidence of this simulation?*

No individual sample produced 7. Yet the average of the thousand sample means is 6.932 — essentially the population value.

It is not a coincidence, and it does not depend on the deck. It follows from the definition of the sample mean and one line of algebra.

WHERE THIS COMES FROM

$$\begin{aligned} E(\bar{X}) &= E\left(\frac{X_1 + X_2 + \cdots + X_n}{n}\right) \\ &= \frac{1}{n}\Big(E(X_1) + E(X_2) + \cdots + E(X_n)\Big) \\ &= \frac{1}{n}\underbrace{(\mu + \mu + \cdots + \mu)}_{n \text{ times}} \\ &= \frac{n\mu}{n} \; = \mu \end{aligned}$$

> The third line is the one that matters. It replaces every $E(X_i)$ with the same μ, and it is entitled to do so only because the observations are **identically distributed** — each drawn from the population we are asking about. Averaging n of them then leaves μ unchanged.

Look back at Survey B. Its observations were identically distributed among themselves, but not drawn from the population of interest, so the μ substituted in that third line was the urban mean rather than the national one. The algebra was never wrong; it answered a question nobody had asked.

This is the first of the two properties promised earlier, and it now has a name.

> **REMEMBER THIS**
>
> An estimator is **unbiased** if its expected value equals the parameter it estimates. The sample mean is unbiased for the population mean:
>
> $$E(\bar{X}) = \mu$$
>
> Unbiasedness is a statement about the procedure, not about any particular estimate. It says the rule is not systematically too high or too low. It does not say your $\bar{x}$ is close to μ — every one of the thousand samples above missed.

2.5 Why do larger samples fluctuate less?

> *Unbiasedness cannot be the whole story. What does a larger sample actually buy?*

Unbiasedness alone is worth remarkably little. Consider the rule "take the first observation and ignore the rest." It is perfectly unbiased — $E(X_1) = \mu$ — and nobody would use it, because it never improves. Collect a million households and it still reports one household's income.

So the second property has to concern spread. Run the deck experiment across a range of sample sizes.

```r
set.seed(7)

sizes <- c(5, 10, 20, 40, 80)
results <- t(sapply(sizes, function(n) {
  m <- replicate(2000, mean(sample(deck, size = n, replace = TRUE)))
  c(n = n, mean = mean(m), sd = sd(m), theoretical_sd = sd(deck) / sqrt(n))
}))
```

```r
round(results, 3)
```

```
#>        n  mean    sd theoretical_sd
#> [1,]   5 6.932 1.688          1.690
#> [2,]  10 7.022 1.165          1.195
#> [3,]  20 7.009 0.845          0.845
#> [4,]  40 7.019 0.597          0.597
#> [5,]  80 6.996 0.415          0.422
```

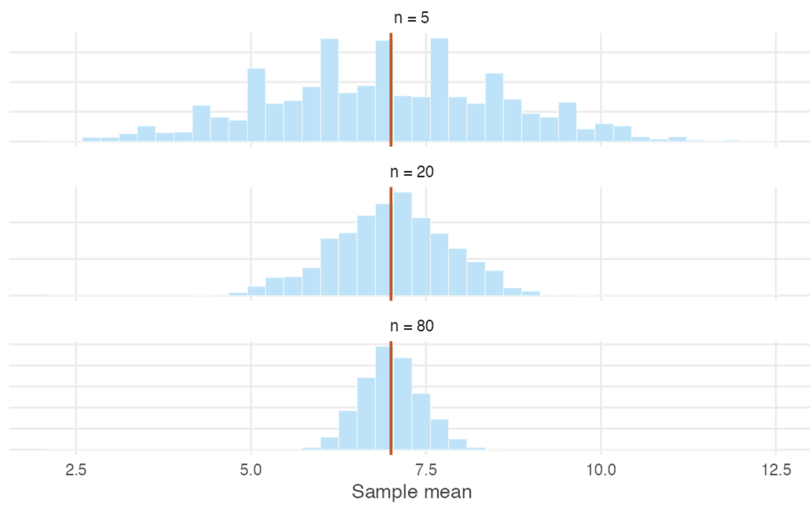

Figure 2.3: Sampling distributions of the mean for growing sample sizes. The centre never moves. The spread collapses.

The second column confirms the previous section: every sample size is centred on 7, and even samples of five are centred correctly. Centring does not improve with n, because it was never wrong.

The third column is what changes. At $n = 5$ the sample means have a standard deviation of about 1.69; at $n = 80$ it is about 0.42. Compare it against the fourth column: the observed spread matches $\sigma/\sqrt{n}$ almost exactly at every sample size.

WHERE THIS COMES FROM

$$\begin{aligned}
\mathrm{Var}(\bar{X}) &= \mathrm{Var}\Big(\frac{1}{n}(X_1 + \cdots + X_n)\Big) \\
&= \frac{1}{n^2}\,\mathrm{Var}(X_1 + \cdots + X_n) \\
&= \frac{1}{n^2}\,\underbrace{(\sigma^2 + \sigma^2 + \cdots + \sigma^2)}_{n \text{ times}} \\
&= \frac{n\sigma^2}{n^2} \;=\; \frac{\sigma^2}{n}
\end{aligned}$$

The second line uses $\mathrm{Var}(aX) = a^2\mathrm{Var}(X)$. The third line is the one that matters: the variance of a sum equals the sum of the variances only when the observations are **independent**. This is the single point in the entire argument at which independence is required.

Survey C is now explained. Its households were drawn from the right population, so it stayed centred — but they were not independent, the third line above did not hold, and the true spread was more than three times what $\sigma/\sqrt{n}$ claimed. A researcher who applied the formula anyway would have reported a precision the data did not contain.

The square root of this variance is used so constantly that it has its own name.

REMEMBER THIS

The **standard error** of the sample mean is the standard deviation of its sampling distribution:

$$\mathrm{se}(\bar{X}) = \sqrt{\mathrm{Var}(\bar{X})} = \frac{\sigma}{\sqrt{n}}$$

It measures how much the sample mean moves from sample to sample. A small standard error means that whatever answer we happened to get, a different sample would have given something similar.

The word *error* is unfortunate, and worth guarding against. It does not measure a mistake, and it is not the distance between your estimate and the truth — that distance is unknowable. Like unbiasedness, it is a property of the procedure.

2.6 Why does the sample mean become more reliable?

> *Two properties, established separately. What do they amount to together?*

We can now answer the question this unit opened with.

The sample mean is used almost universally because its sampling distribution sits in the right place and tightens as the sample grows. It is centred on μ at every sample size, and the width of that distribution, $\sigma/\sqrt{n}$, falls towards zero as n increases. A distribution centred on the correct value and becoming ever narrower must concentrate on that value. Large samples are therefore increasingly likely to produce averages close to μ.

That last sentence is the Law of Large Numbers, which Section 1.5 introduced as a description of observed behaviour. It is no longer a description. Averages settle down because the procedure that produces them is unbiased and increasingly precise, and there is nothing else to the result.

Both halves are load-bearing, and it is worth seeing why neither would do alone.

An estimator that was centred but whose spread never shrank would remain permanently unreliable — the "use the first observation" rule from the previous section, which stays honest and stays useless however much data you give it.

An estimator whose spread shrank around a value that was not μ would be worse still. It would become *reliably wrong*, converging with increasing confidence on the incorrect answer, and every additional observation would harden the mistake rather than expose it. Survey B is exactly this: had the urban-only surveyor collected ten thousand households instead of a hundred, their estimates would have clustered even more tightly around a number that was still ₹6,800 too high.

REMEMBER THIS

The sample mean earns its place through two properties together:

$$E(\bar{X}) = \mu \qquad\qquad \mathrm{Var}(\bar{X}) = \frac{\sigma^2}{n}$$

Centred on the truth, and increasingly precise. Neither is sufficient on its own, and both depend on assumptions about how the sample was collected.

2.7 Why is precision expensive?

> *If larger samples are better, why does anyone stop collecting data?*

One practical consequence deserves its own section, because it shapes how survey work is actually done.

Return to the square root. Doubling the sample size does not halve the standard error; it divides it by $\sqrt{2} \approx 1.41$. To halve the standard error, the sample must grow *fourfold*.

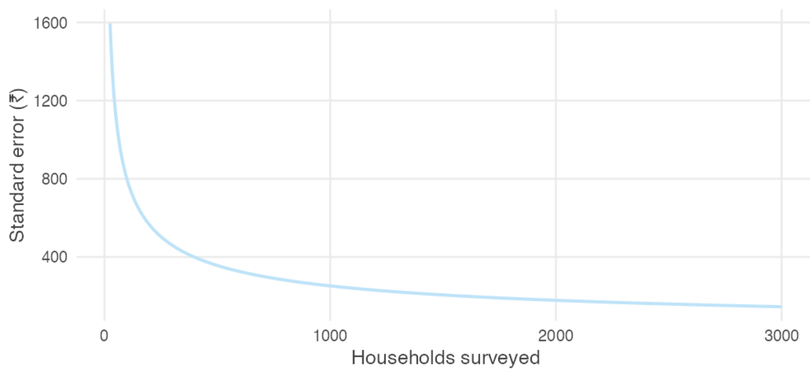

Figure 2.4: The standard error of the sample mean against sample size. Almost all of the available precision is bought in the first few hundred observations.

Precision is therefore expensive, and grows more expensive the more of it you buy. Each additional household costs the same to interview as the last, but contributes less than the last. Going from 100 households to 400 halves the standard error; going from 400 to 1,600 halves it again, at four times the cost.

This is why national surveys run to a few thousand households rather than a few hundred thousand. Past a certain point the additional precision is not worth the additional money, and the money is better spent on the things no sample size can fix — better questions, better interviewers, better coverage of the households that are hardest to reach.

That last point is worth holding onto. The failures in Section 2.2 were all failures of design, and none of them would have been helped by a larger sample. Survey B with ten thousand urban households is still wrong. When a result is disappointing, the instinct is to ask for more data; often the honest answer is that more of the same data will not help.

2.8 Summary

REMEMBER THIS

- A population has fixed but unknown **parameters**.
- An **estimator** is a rule; an **estimate** is the result of applying it to one sample.
- Since the truth is never observed, estimators are judged by their behaviour across repeated samples, not by any single estimate they produce.
- A sample speaks for a population only if it is drawn **independently** and from **the population of interest**. Neither assumption is decoration: break the first and the spread is far wider than any formula predicts, break the second and the estimates are centred on the wrong number while often looking more precise than the honest ones.
- Under random sampling the sample mean is **unbiased**: $E(\bar{X}) = \mu$.
- Its **standard error**, $\sigma/\sqrt{n}$, falls as $1/\sqrt{n}$ — so larger samples give more precise estimates, at a rising price per unit of precision.
- Centred and increasingly precise together are what make the sample mean trustworthy, and together they are the Law of Large Numbers.

Everything in this unit has been about choosing a good estimator. The next unit asks how to use that estimator to make statements about an unknown population from a single observed sample.

2.9 Practice problems

YOUR TURN

These are for learning. Work each one before opening the solution; the answer is worth very little if you have not first tried to produce it yourself. The question bank that follows is for testing yourself and has no answers at all.

Several problems treat `data/wages-india-synthetic.csv` as a *population* of 20,000 wage earners. This is artificial, and deliberately so — it is the same device as the deck of cards, at realistic Indian magnitudes. The dataset is simulated but calibrated to IHDS-II estimates, so the returns to schooling and the urban premium in it are close to the real ones.

Concept checks

Short questions. A sentence or two is enough.

2.1 A survey of 800 households reports mean monthly income of ₹18,400. Which part of that sentence is the estimator, and which is the estimate?

SOLUTION

The estimator is the sample mean — the rule "add the 800 incomes and divide by 800." The estimate is ₹18,400, the number that rule returned from this particular sample.
The estimator exists before any data are collected. The estimate exists only afterwards.

2.2 Why can a statistician never check whether a particular estimate is close to the truth?

SOLUTION

Because checking would require knowing the population mean, and if the population mean were known there would have been no reason to conduct the survey. The truth is precisely what the estimate is trying to discover.
This is why the subject evaluates procedures rather than results.

2.3 Consider the estimator "ignore the data entirely and always report ₹50,000." Is it unbiased? What is its standard error?

SOLUTION

Its standard error is zero — it returns the same number from every possible sample, so it does not vary at all.
It is unbiased if and only if the population mean happens to be exactly ₹50,000, which we have no way of knowing. In general it is biased by whatever amount μ differs from ₹50,000.
The problem is worth doing because it separates the two properties completely. Perfect precision is compatible with being arbitrarily wrong, and a small standard error is therefore no evidence that an estimate is any good.

2.4 In Section 2.2, Survey B produced estimates with a *smaller* spread than Survey A. Why is that alarming rather than reassuring?

SOLUTION

Because the small spread came from surveying households that resemble one another, not from surveying more of India. The estimates agreed closely with each other while agreeing on a number ₹6,800 above the truth.
A researcher who saw only their own survey would observe a tight, well-behaved sample and conclude that the estimate was reliable. Spread measures agreement among samples drawn the same way; it says nothing about whether that way was the right one.

2.5 A report states: "mean earnings were ₹54,000 with a standard error of ₹10,000, so the true figure is within ₹10,000 of our estimate." What is wrong with the second half of that sentence?

SOLUTION

The standard error is not the distance between the estimate and the truth. That distance is unknowable, since the truth is unobserved.
The standard error describes the *procedure*: it says that repeating this survey would produce sample means with a standard deviation of about ₹10,000. Any particular estimate may lie much closer to μ than that, or much further from it.

Explain

A paragraph each.

2.6 Explain why unbiasedness alone is not a good enough reason to use an estimator. Give an example of an unbiased estimator nobody would use.

SOLUTION

Unbiasedness says the estimator is right *on average across repeated samples*. It says nothing about how far any single sample strays, and a rule can be centred correctly while being wildly variable.
The clearest example is "use the first observation and ignore the rest." Since $E(X_1) = \mu$, it is exactly unbiased. It is also useless, because its standard error is σ regardless of sample size. Collecting a million households does not improve it by any amount at all.
Usefulness requires a second property: the spread must fall as the sample grows. Unbiasedness places the distribution correctly; the variance is what makes the placement worth anything.

2.7 An interviewer travels to six villages and surveys thirty households in each.

Which i.i.d. assumption is threatened, which of the two derivations in this unit stops working, and what happens to the results?

SOLUTION

Independence is threatened. Households in a village share a labour market, a cropping pattern, and often an occupation, so once the first few are surveyed the rest carry much less new information.
The derivation that fails is the variance one. Its third line replaces $\mathrm{Var}(X_1 + \cdots + X_n)$ with $\sigma^2 + \cdots + \sigma^2$, and that step requires independence. Nothing goes wrong with $E(\bar{X}) = \mu$, whose third line uses only identical distribution.
So the estimates remain centred on the population mean — the design is not biased — but they vary far more from survey to survey than $\sigma/\sqrt{n}$ predicts. A researcher who applied that formula would report a precision the data does not contain. In Survey C the true spread was more than three times the formula's claim.

2.8 Explain why the Law of Large Numbers needs *both* properties of the sample mean. Describe what would go wrong if each were missing in turn.

SOLUTION

The Law of Large Numbers says large samples are increasingly likely to produce averages close to μ. That requires the sampling distribution both to sit in the right place and to narrow.
Without shrinking variance, the estimator stays centred but never improves. "Use the first observation" is centred on μ forever and never gets closer to it, so no sample size makes it likely to land near the truth.
Without correct centring, shrinking variance actively makes matters worse. The estimator becomes *reliably wrong* — it converges with increasing confidence on the incorrect value, and every extra observation hardens the error rather than exposing it. The urban-only survey is exactly this case: ten thousand urban households would have produced a tighter cluster around a number still ₹6,800 too high.

Numerical problems

Pen and paper. Calculators are fine; R is not needed until the next section.

2.9 In the population of wage earners used below, annual earnings have standard deviation $\sigma = ₹100{,}141$.

i. Find the standard error of the sample mean for $n = 100$, $n = 400$ and $n = 2{,}500$.

ii. How large a sample is needed for the standard error to fall below ₹5,000?
iii. A colleague proposes to improve an $n = 400$ survey by adding 100 households. By how much does the standard error fall?

SOLUTION

(i) se $= \sigma/\sqrt{n}$:

$$n = 100 : \quad \frac{100{,}141}{10} = ₹10{,}014$$
$$n = 400 : \quad \frac{100{,}141}{20} = ₹5{,}007$$
$$n = 2{,}500 : \quad \frac{100{,}141}{50} = ₹2{,}003$$

(ii) Require $100{,}141/\sqrt{n} < 5{,}000$, so $\sqrt{n} > 20.03$ and $n > 401.1$. A sample of **402** suffices.
(iii) At $n = 500$ the standard error is $100{,}141/\sqrt{500} =$ ₹4,479, down from ₹5,007 — a fall of about ₹528, or roughly 11%, for 25% more fieldwork. The returns are already thin at this sample size.

2.10 A researcher with n observations proposes the estimator $T = \frac{1}{2}(X_1 + X_n)$: average the first and last household only.

i. Show that T is unbiased.
ii. Find Var(T).
iii. Using $\sigma =$ ₹100,141, compare the standard error of T with that of the sample mean of all 100 observations.
iv. What does this show about unbiasedness?

SOLUTION

(i) $E(T) = \frac{1}{2}(E(X_1) + E(X_n)) = \frac{1}{2}(\mu + \mu) = \mu$. Both observations are drawn from the same population, so each has expectation μ.
(ii) Using Var$(aX) = a^2$Var(X) and independence,

$$\text{Var}(T) = \tfrac{1}{4}(\sigma^2 + \sigma^2) = \frac{\sigma^2}{2}$$

which is σ^2/n with $n = 2$ — unsurprising, since T *is* the sample mean of two observations.
(iii) se$(T) = 100{,}141/\sqrt{2} =$ ₹70,810, against $100{,}141/10 =$ ₹10,014 for the full sample mean. T is about seven times more variable.
(iv) That unbiasedness is a low bar. T is exactly as unbiased as $\bar{X}$ and vastly worse, because it discards 98 of the 100 observations collected.

Choosing between unbiased estimators is decided on variance.

2.11 A survey costs ₹900 per household in fieldwork. The current design surveys 400 households.

i. What does the current survey cost?
ii. What would it cost to halve the standard error?
iii. The funder instead offers to double the budget. By what factor does the standard error fall?

SOLUTION

(i) $400 \times$ ₹$900 =$ ₹$360{,}000$.
(ii) Halving the standard error requires four times the sample, so 1,600 households at a cost of ₹1,440,000 — an extra ₹1,080,000 to buy one factor of two.
(iii) Doubling the budget buys 800 households. The standard error falls by a factor of $\sqrt{2} \approx 1.41$, a reduction of about 29%.
Twice the money does not buy twice the precision, and this gets worse the larger the survey already is. It is often a better use of the second ₹360,000 to improve coverage of hard-to-reach households than to double the sample — because the errors that arise from missing those households are not reduced by sample size at all.

R exercises

2.12 Treat `data/wages-india-synthetic.csv` as a population of 20,000 wage earners. Compute its true mean and standard deviation. Then draw 2,000 samples of 100 workers each, and check both properties of the sample mean against theory.

SOLUTION

```r
wages <- read.csv("data/wages-india-synthetic.csv")
earn  <- wages$annual_earnings

mu_pop    <- mean(earn)
sigma_pop <- sd(earn)

set.seed(3)
sim <- replicate(2000, mean(sample(earn, 100, replace = TRUE)))

round(c(population_mean = mu_pop,
        centre_of_estimates = mean(sim),
        theoretical_se = sigma_pop / sqrt(100),
        observed_se = sd(sim)))
```

```
#>     population_mean centre_of_estimates      theoretical_se
 ↪  observed_se
#>               54181               54072               10014
 ↪  9939
```

The estimates are centred on the population mean, and their spread is close to $\sigma/\sqrt{n}$. Both properties hold.
This population is severely right-skewed — its mean is 54,181 against a median of 24,220 — yet neither property needed the population to be well behaved. Unbiasedness followed from each observation having expectation μ, and the variance result from independence. Shape never entered either derivation.

2.13 Repeat the previous exercise sampling only from urban workers, as an interviewer working exclusively in cities would. Report the centre and spread, and say which assumption has been broken.

SOLUTION

```r
urban_earn <- earn[wages$residence == "Urban"]

set.seed(3)
sim_urban <- replicate(2000, mean(sample(urban_earn, 100, replace =
 ↪  TRUE)))

round(c(population_mean = mu_pop,
        centre_of_estimates = mean(sim_urban),
        observed_se = sd(sim_urban)))
```

```
#>     population_mean centre_of_estimates         observed_se
#>               54181              101270               15099
```

The estimates are centred near ₹102,000 against a true population mean of about ₹54,000 — almost double. The assumption broken is *identically distributed*: urban workers are drawn from a different distribution than Indian wage earners as a whole, so the μ appearing in the third line of the $E(\bar{X})$ derivation is the urban mean, not the national one.
Note also that no amount of extra data would help. The estimator is converging efficiently on the wrong number.

2.14 Draw 2,000 sample means for $n = 25, 100, 400$ and $1,600$ from the same population, and plot the observed standard error against the theoretical $\sigma/\sqrt{n}$. Does the $1/\sqrt{n}$ rule hold?

SOLUTION

```
set.seed(3)
ns <- c(25, 100, 400, 1600)

se_tab <- data.frame(
  n = ns,
  observed = sapply(ns, function(n) sd(replicate(4000,
  ↪  mean(sample(earn, n, replace = TRUE))))),
  theory   = sigma_pop / sqrt(ns)
)

round(se_tab)
```

```
#>      n observed theory
#> 1   25    20143  20028
#> 2  100    10024  10014
#> 3  400     4961   5007
#> 4 1600     2510   2504
```

```
ggplot(se_tab, aes(n)) +
  geom_line(aes(y = theory), colour = book_palette$reject, linewidth
  ↪  = 0.9) +
  geom_point(aes(y = observed), size = 2.6, colour =
  ↪  book_palette$fill) +
  labs(x = "Sample size", y = "Standard error (₹)",
       subtitle = "Line: theoretical. Points: simulated.") +
  theme_book()
```

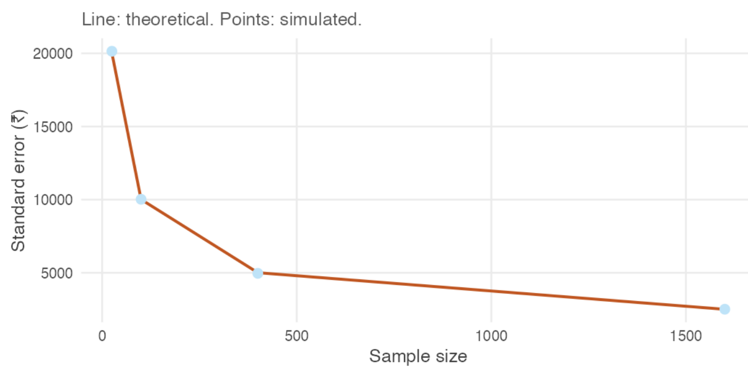

Figure 2.5: Observed and theoretical standard errors for growing sample sizes.

The points sit on the curve at every sample size, agreeing with theory to within about one per cent. Quadrupling n halves the standard error each time, exactly as $1/\sqrt{n}$ requires.
Note the `replace = TRUE`, which matters more here than it did with the deck. Drop it and the observed spread at $n = 1{,}600$ comes out roughly five per cent *below* theory, because 1,600 is eight per cent of this population of 20,000 — sample a substantial fraction of a finite population without replacement and the draws are no longer independent, so $\sigma/\sqrt{n}$ stops being right. Real surveys of India are nowhere near this fraction, which is why the assumption is harmless in practice and not in a simulation.

2.15 Do the two properties depend on the population being well behaved? Compare the sampling distribution of the mean at $n = 10$ and $n = 400$ for this heavily skewed earnings population, and comment on what changes and what does not.

SOLUTION

```r
set.seed(3)
sk <- do.call(rbind, lapply(c(10, 400), function(n)
  data.frame(n = factor(paste("n =", n), levels = c("n = 10", "n =
  ↪  400")),
             m = replicate(4000, mean(sample(earn, n, replace =
              ↪  TRUE))))))

ggplot(sk, aes(m)) +
  geom_histogram(bins = 60, fill = book_palette$fill, colour =
  ↪  "white",
                 linewidth = 0.1) +
  geom_vline(xintercept = mu_pop, colour = book_palette$reject,
             linewidth = 0.8) +
  facet_wrap(~ n, scales = "free") +
  scale_x_continuous(labels = function(x) format(x, big.mark = ","))
  ↪  +
  labs(x = "Sample mean of annual earnings (₹)", y = NULL) +
  theme_book() +
  theme(axis.text.y = element_blank())
```

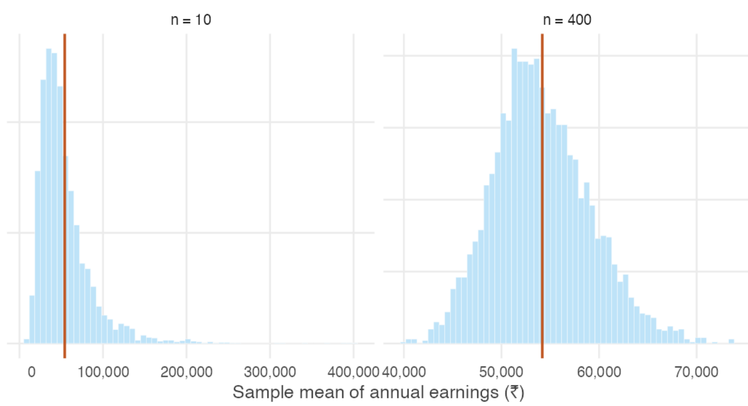

Figure 2.6: Sampling distribution of mean earnings at two sample sizes. The red line is the population mean.

Both panels are centred on the population mean. Unbiasedness holds at $n = 10$ just as at $n = 400$, and it required nothing of the population's shape — the derivation used only that each observation has expectation μ.
What changes is the *shape*. At $n = 10$ the sampling distribution inherits the population's long right tail and is clearly skewed. At $n = 400$ it is

close to symmetric. That is the Central Limit Theorem from Section 1.6 at work, and it is a reminder that "large enough" depends on the population: for earnings data, $n = 30$ would not have been remotely sufficient.

2.16 Using the same population, compare three estimators of μ across 2,000 samples of 50: the sample mean, the sample median, and the midpoint of the smallest and largest observations. Which would you use, and why?

SOLUTION

```r
set.seed(3)
midrange <- function(x) (min(x) + max(x)) / 2

comp <- replicate(2000, {
  s <- sample(earn, 50, replace = TRUE)
  c(mean = mean(s), median = median(s), midrange = midrange(s))
})

round(cbind(centre = rowMeans(comp),
            bias   = rowMeans(comp) - mu_pop,
            spread = apply(comp, 1, sd)))
```

```
#>          centre   bias spread
#> mean      53896   -285  14550
#> median    24653 -29528   5399
#> midrange 250648 196467 172624
```

Only the sample mean is centred on μ. The median sits far below it, because this population is strongly right-skewed and the median estimates the population *median*, not the mean — it is an excellent estimator of a different parameter. The midrange is wildly variable, since it depends entirely on the two most extreme observations drawn and discards the other 48.

The sample mean is the only one of the three that is unbiased for μ and tightens dependably as the sample grows.

Worth noting, though: if the question of interest were "what does a typical wage earner make?", the median would be the better summary. Choosing an estimator requires first being clear about which parameter you actually want.

2.10 Question bank

WATCH OUT

No answers are given for this section, by design. These questions are for testing yourself once the practice problems are complete. There are no hints and no worked solutions — exactly as in an examination.

Multiple choice

QB 2.1 Which of the following is an estimate rather than an estimator?

a. $\bar{X}$
b. The rule "take the median of the sample"
c. ₹18,400
d. $\sigma/\sqrt{n}$

QB 2.2 An estimator is unbiased if

a. every estimate it produces equals the parameter
b. its expected value equals the parameter
c. its variance falls as the sample size grows
d. the sample is drawn at random

QB 2.3 A population has $\sigma = 60$. For the standard error of the sample mean to equal 2, the sample size must be

a. 30
b. 120
c. 900
d. 3,600

QB 2.4 A survey samples only households with telephones. Which is the most accurate description of the consequence?

a. The estimates will be centred correctly but too variable
b. The estimates will be centred on the wrong value, and a larger sample will not fix it
c. The estimates will be centred on the wrong value, but a larger sample will fix it
d. Nothing goes wrong provided the sample is large

QB 2.5 The variance derivation $\text{Var}(\bar{X}) = \sigma^2/n$ uses independence at exactly one step. Which one?

a. Taking the constant $1/n$ outside the variance
b. Writing $\text{Var}(aX) = a^2\text{Var}(X)$
c. Replacing the variance of the sum with the sum of the variances
d. Cancelling n against n^2

QB 2.6 Two surveys estimate the same quantity. Survey X has a standard error of ₹4,000; Survey Y has a standard error of ₹1,000. Which statement follows? *(select all that apply)*

a. Survey Y's estimate is closer to the truth
b. Survey Y's estimates vary less across repeated sampling
c. Survey Y used a larger sample, if both were drawn the same way
d. Survey Y is unbiased

QB 2.7 Doubling a survey's sample size reduces its standard error by a factor of

a. 4
b. 2
c. $\sqrt{2}$
d. It depends on σ

Short reasoning

QB 2.8 A student writes: "the standard error tells us how much our estimate differs from the population mean." Correct the statement, and explain in one sentence why the original cannot be right.

QB 2.9 Explain why an estimator with zero variance is not necessarily a good estimator. Give an example.

QB 2.10 A researcher collects 500 households by standing outside a government hospital on a weekday morning and interviewing whoever passes. Identify at least two distinct problems with this sample, and state for each whether it affects the centre of the estimates, their spread, or both.

QB 2.11 Two rules for estimating μ from n observations are proposed: the sample mean, and the average of the first $n/2$ observations. Both are unbiased. Explain how you would decide between them, and what the answer is.

QB 2.12 An economist reports that increasing a survey from 1,000 to 1,200 households "substantially improved the precision of our estimates." Evaluate that claim quantitatively.

QB 2.13 Section 2.2 showed a badly designed survey producing estimates with a *smaller* spread than a well-designed one. Explain how this is possible, and what it implies about using a reported standard error to judge whether a study is trustworthy.

Numerical

QB 2.14 A population of farm households has mean annual income $\mu = ₹84{,}000$ and standard deviation $\sigma = ₹36{,}000$.

i. Find the standard error of the sample mean for a survey of 144 households.
ii. What sample size would be needed to reduce the standard error to ₹1,500?
iii. Fieldwork costs ₹1,200 per household. What is the additional cost of the sample in (ii) relative to the one in (i)?

QB 2.15 For a population with variance σ^2, consider the estimator $W = \frac{1}{3}X_1 + \frac{2}{3}X_2$ computed from two independent observations.

i. Show that W is unbiased.
ii. Find $\text{Var}(W)$.
iii. Compare it with the variance of the ordinary sample mean of the same two observations. Which is preferable, and what general principle does this illustrate?

QB 2.16 A state has 40 districts. A survey selects 4 districts at random and interviews 250 households in each, reporting a standard error of $\sigma/\sqrt{1000}$.

i. State the assumption this calculation requires.
ii. Explain why it is unlikely to hold here.
iii. Say whether the reported standard error is likely to be too large or too small, and why that direction is the dangerous one.

QB 2.17 Using `data/wages-india-synthetic.csv` as a population, write R code to estimate how large a sample is needed before the standard error of mean annual earnings falls below ₹4,000. Verify your answer by simulation as well as by formula, and explain any discrepancy between the two.

2.11 Looking ahead

We opened by asking why statistics has settled on the sample mean, and the answer is now in hand. It is centred on the truth, its variability falls as the sample grows, and larger samples therefore tend to produce estimates closer to the population mean.

But notice what every argument in this unit relied on. We drew a thousand samples and looked at how they scattered. We compared four survey designs by running each a thousand times. The sampling distribution has been visible on every page.

In practice it never is. Nobody draws a thousand samples. A survey is conducted once, at considerable expense, and yields exactly one sample mean.

The sample mean has earned our trust as an estimator. But we see only one of them. The next question is therefore unavoidable: what can a single sample tell us about an unknown population?

Chapter 3

Point and Interval Estimation

3.1 What can a single sample tell us?

> *We have one sample and one number. What are we entitled to say?*

The previous unit answered an important question: why do statisticians use the sample mean to estimate the population mean? We found that it has two desirable properties. First, it is unbiased — over repeated samples it is centred on the true population mean. Second, its variability decreases as the sample size grows. Larger samples therefore tend to produce estimates closer to the truth.

There was, however, one feature of that argument which is impossible in practice.

To judge the behaviour of the sample mean we repeatedly drew thousands of samples from a population whose true mean we already knew. By watching the sampling distribution emerge, we could see that the sample mean was centred correctly and became more precise as the sample grew.

Real surveys do not work that way.

A labour force survey, a household expenditure survey, a school attendance audit — each is conducted once. We observe one sample, calculate one estimate, and the data collection ends. We never see the thousands of alternative samples that might have been drawn.

Suppose a survey of students reports an average attendance of 26.1 classes.

Is the population mean exactly 26.1? Almost certainly not. How close is it? The data alone do not tell us.

Simply reporting the estimate is therefore incomplete. We already know that every sample mean varies from sample to sample; some will be larger than the population mean and others smaller. Without a measure of that variability, the estimate has no context.

The natural response is to report the estimate together with its standard error. For the sample mean we found that

$$\mathrm{se}(\bar{X}) = \frac{\sigma}{\sqrt{n}}$$

This is a useful result, and it immediately raises another difficulty. The formula contains the population standard deviation σ, which is unknown. If we are estimating the population mean because it is unknown, we can hardly assume we know the population standard deviation.

One missing quantity has become two.

Nor are all questions about population means. Many surveys seek to estimate proportions instead.

- What proportion of households live below the poverty line?
- What proportion of workers are employed in the informal sector?
- What proportion of children are fully vaccinated?

These require estimators of a different kind.

Finally, even once we can calculate both an estimate and its standard error, we still face the central question of statistical inference. Suppose two surveys both estimate average attendance at 26.1 classes. One has a standard error of 0.4; the other has a standard error of 3.2.

How should those numbers change our confidence in the estimate? How uncertain is "uncertain"?

Answering that requires moving beyond a single number. Rather than reporting only a point estimate, we will construct **interval estimates —** ranges of plausible values for the unknown parameter, together with a statement of how reliable such ranges are.

> **REMEMBER THIS**
>
> This unit develops the three ingredients needed for statistical inference.
>
> 1. How to estimate unknown population variability.
> 2. How to extend estimation from means to proportions.
> 3. How to combine point estimates and standard errors into confidence intervals.

By the end of the unit we will be able to replace a statement such as

> The average attendance is 26.1 classes.

with one that is considerably more informative:

> We estimate the average attendance to lie between 24.4 and 27.8 classes, with 95% confidence.

That is the central goal of this unit.

3.2 Estimating the standard error

> *The standard error is the number we want. Where is it supposed to come from?*

At the end of the previous section we identified the first obstacle to statistical inference. The standard error of the sample mean is

$$\mathrm{se}(\bar{X}) = \frac{\sigma}{\sqrt{n}}$$

but the population standard deviation σ is unknown.

This should not surprise us. Every quantity in that formula except n belongs to the population, while all we possess is a sample. If the population mean must be estimated, the population standard deviation must be estimated too.

The obvious strategy is the one that already worked. Replace the unknown population quantity by its sample analogue. For the mean this gave

$$\mu \quad \longrightarrow \quad \bar{X}$$

For the variance we might therefore try

$$\sigma^2 \quad \longrightarrow \quad \frac{1}{n}\sum_{i=1}^{n}(X_i - \bar{X})^2$$

It is an appealing estimator. It measures the average squared distance of the observations from their centre, exactly as the population variance does.

Unfortunately it is wrong. Not disastrously wrong, but systematically wrong: it underestimates the population variance.

Before trying to understand why, let us see the problem.

A statistical laboratory

As in the previous unit, we need a population where the truth is known. A fair die is ideal, because its variance is known exactly:

$$\mu = 3.5 \qquad \sigma^2 = \frac{35}{12} = 2.9167$$

Draw samples of ten rolls, compute the sum of squared deviations from the sample mean, and divide it by four different numbers. If any of these is a good estimator, its average across thousands of samples should land on 2.9167.

```
set.seed(11)
n <- 10

ss <- replicate(20000, {
  d <- sample(1:6, n, replace = TRUE)
  sum((d - mean(d))^2)                     # the sum of squared deviations
})

divisors <- c(n, n - 1, n - 2, n - 3)

tab <- data.frame(
  divisor      = divisors,
  avg_variance = sapply(divisors, function(k) mean(ss / k)),
  avg_sd       = sapply(divisors, function(k) mean(sqrt(ss / k)))
)

round(tab, 4)
```

```
#>   divisor avg_variance avg_sd
#> 1      10        2.630  1.601
#> 2       9        2.922  1.688
#> 3       8        3.287  1.790
#> 4       7        3.757  1.914
```

```
c(true_variance = 35 / 12, true_sd = sqrt(35 / 12))
```

```
#> true_variance       true_sd
#>         2.917         1.708
```

Notice what the table shows. Dividing by n is always too small. Dividing by $n-2$ is too large, and $n-3$ is worse. Only one divisor consistently lands on the truth, and it is not n.

It is $n-1$, which gives 2.9219 against a true 2.9167.

That observation raises two questions, and the rest of this section answers them in turn.

1. Why does dividing by n underestimate the variance?
2. Why is the correction exactly $n-1$, rather than $n-2$ or some other number?

The second question is the one that matters. Any downward adjustment would move the estimate in the right direction; only one gets the size right.

Why dividing by n fails

The reason is that $\bar{x}$ is computed from the very observations whose spread we are measuring.

The sample mean sits, by construction, in the middle of the sample. It is the value that makes $\sum(x_i - \bar{x})^2$ as small as it can possibly be — no other number, including the true μ, gives a smaller total. So when we measure the spread of the sample around its own mean, we measure it around the most flattering point available, and the answer comes out too small.

> **THE IDEA IN PLAIN WORDS**
>
> Once $\bar{x}$ is known, the deviations are not free to take any values. They must sum to zero:
>
> $$\sum_{i=1}^{n}(x_i - \bar{x}) = 0$$
>
> Given the first $n-1$ of them, the last is determined. Only $n-1$ of the n deviations carry independent information about the spread.
> The quantity $n-1$ is called the **degrees of freedom**: the number of observations, less the number of quantities estimated from them along the way.

Why the correction is exactly n − 1

The argument above explains the direction of the error. It does not explain its size, and the size is what distinguishes $n-1$ from $n-2$. For that we need to take an expectation.

> **WHERE THIS COMES FROM**
>
> Write $\widetilde{S}^2$ for the divide-by-n version, and simplify it first. Expanding the square,

$$\begin{aligned}
\widetilde{S}^2 &= \frac{1}{n}\sum_{i=1}^{n}(X_i - \bar{X})^2 \\
&= \frac{1}{n}\left(\sum_{i=1}^{n} X_i^2 - 2\bar{X}\sum_{i=1}^{n} X_i + n\bar{X}^2\right) \\
&= \frac{1}{n}\left(\sum_{i=1}^{n} X_i^2 - 2n\bar{X}^2 + n\bar{X}^2\right) \qquad \text{since } \textstyle\sum X_i = n\bar{X} \\
&= \frac{1}{n}\sum_{i=1}^{n} X_i^2 \; - \; \bar{X}^2
\end{aligned}$$

Now take expectations:

$$E(\widetilde{S}^2) = \frac{1}{n}E\left(\sum_{i=1}^{n} X_i^2\right) - E(\bar{X}^2)$$

Both terms follow from rearranging the definition of variance, $\text{Var}(X) = E(X^2) - [E(X)]^2$, into $E(X^2) = \text{Var}(X) + [E(X)]^2$.
For a single observation, $E(X_i) = \mu$ and $\text{Var}(X_i) = \sigma^2$, so

$$E(X_i^2) = \sigma^2 + \mu^2 \qquad \Longrightarrow \qquad E\left(\sum_{i=1}^{n} X_i^2\right) = n(\sigma^2 + \mu^2)$$

For the sample mean, the previous unit established $E(\bar{X}) = \mu$ and $\text{Var}(\bar{X}) = \sigma^2/n$, so

$$E(\bar{X}^2) = \frac{\sigma^2}{n} + \mu^2$$

Substituting both,

$$\begin{aligned}
E(\widetilde{S}^2) &= \frac{1}{n}\,n(\sigma^2 + \mu^2) - \left(\frac{\sigma^2}{n} + \mu^2\right) \\
&= \sigma^2 + \mu^2 - \frac{\sigma^2}{n} - \mu^2 \\
&= \sigma^2\left(1 - \frac{1}{n}\right) \;=\; \sigma^2\,\frac{n-1}{n}
\end{aligned}$$

So the divide-by-n estimator is too small by exactly the factor $(n-1)/n$ — not approximately, and not only for dice. Multiplying it by $n/(n-1)$ removes the factor, and multiplying by $n/(n-1)$ is the same as dividing by $n-1$.

> **REMEMBER THIS**
>
> The **sample variance** and **sample standard deviation** are
>
> $$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2 \qquad\qquad s = \sqrt{s^2}$$
>
> s^2 is an unbiased estimator of σ^2: $E(s^2) = \sigma^2$. The adjustment from n to $n-1$ is called **Bessel's correction**.
> In R these are `var()` and `sd()`, both of which already use $n-1$.

Check the theory against the simulation. With $n = 10$ the predicted shortfall is $(n-1)/n = 0.9$, so dividing by n should return $0.9 \times 2.9167 = 2.625$. The simulation returned 2.6297.

When the population mean is known

Occasionally μ is known even though σ^2 is not — a machine calibrated to a specification, or a process with a target value. Then no degrees of freedom are spent estimating the centre, nothing is flattered, and the divisor is n:

$$\hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n}(x_i - \mu)^2$$

Note that this uses μ, not $\bar{x}$. Since $\bar{x}$ minimises the sum of squared deviations, the total measured from μ is *larger* than the total measured from $\bar{x}$ — which is precisely why it can be divided by the larger n and still come out right.

> **THE IDEA IN PLAIN WORDS**
>
> The rule is easier to remember as a question than as two formulas. How many quantities did I have to estimate from this sample before I could measure its spread?
> Estimated the centre from the data: divide by $n-1$. Knew the centre already: divide by n.

Back to the standard error

We can now return to where the section began. The standard error of the sample mean is $\sigma/\sqrt{n}$, and with σ unknown we substitute s:

$$\widehat{\text{se}}(\bar{X}) = \frac{s}{\sqrt{n}}$$

This is the quantity reported by every statistical package. Note the hat: it is itself an estimate, depending on the random sample through s, and therefore carrying sampling variability of its own.

That substitution looks harmless. It is not, and Section 3.5 measures the damage.

> **GOING A LITTLE FURTHER**
>
> s^2 is unbiased for σ^2, but s is **not** unbiased for σ. Taking a square root does not preserve unbiasedness.
>
> The evidence is already in the table above. Look at the `avg_sd` column: the $n-1$ row gives 1.6876 against a true σ of 1.7078 — close, but low, and it stays low however many samples are run. On that column no divisor in the table is right; the best would fall between $n-1$ and $n-2$, and would depend on the distribution.
>
> The bias is small, shrinks quickly with n, and is not worth correcting in practice. What is worth taking away is that unbiasedness attaches to a specific estimator of a specific parameter, and does not survive being passed through a nonlinear function.

3.3 How do we estimate a proportion?

> *What proportion of Indian wage earners work in urban areas? That is not an average. Do we need new machinery?*

Many important quantities are proportions. The share of households below the poverty line, the share of children enrolled in school, the share of a workforce that is female, the unemployment rate. None of these is obviously an average, and it would be reasonable to expect a separate theory.

There is none, and the reason is a small trick.

Write down, for each person in the sample, a variable that equals 1 if they have the characteristic and 0 if they do not.

```
wages <- read.csv("data/wages-india-synthetic.csv")

is_urban <- as.numeric(wages$residence == "Urban")
head(is_urban, 20)
```

```
#>  [1] 0 0 1 1 0 1 0 0 0 1 0 1 0 1 1 0 0 0 0 1
```

The *average* of that column is the *proportion* of people with the characteristic. Adding up a column of ones and zeros counts the ones, and dividing by n turns the count into a share.

```r
c(mean_of_indicator = mean(is_urban),
  proportion_urban  = sum(wages$residence == "Urban") / nrow(wages))
```

```
#> mean_of_indicator  proportion_urban
#>            0.2684            0.2685
```

REMEMBER THIS

A proportion is a mean. The **sample proportion**

$$\hat{p} = \frac{x}{n}$$

where x is the number of observations with the characteristic, is exactly the sample mean of a variable coded 1 and 0.
Everything established about $\bar{X}$ therefore applies to $\hat{p}$ unchanged. It is unbiased, its variability falls with $\sqrt{n}$, and the Central Limit Theorem makes it approximately normal in large samples.

This is worth pausing on, because it is the first time in the book that a result has been obtained for free. We did not prove anything new. We noticed that a question we had not met was a question we had already answered, wearing different clothes.

How much does a sample proportion vary?

One thing does simplify. For a general population we need σ from somewhere. For a 0/1 variable, σ is determined by p itself.

WHERE THIS COMES FROM

Let X_i equal 1 with probability p and 0 with probability $1 - p$. Then

$$E(X_i) = 1 \cdot p + 0 \cdot (1 - p) = p$$

so the population mean of the indicator *is* the population proportion. For the variance, note that $X_i^2 = X_i$, since $0^2 = 0$ and $1^2 = 1$. Hence $E(X_i^2) = p$ and

$$\mathrm{Var}(X_i) = E(X_i^2) - \left[E(X_i)\right]^2 = p - p^2 = p(1-p)$$

Applying $\mathrm{Var}(\bar{X}) = \sigma^2/n$ from the previous unit,

$$\mathrm{Var}(\hat{p}) = \frac{p(1-p)}{n} \qquad \mathrm{se}(\hat{p}) = \sqrt{\frac{p(1-p)}{n}}$$

In practice p is unknown — it is what we are estimating — so we substitute $\hat{p}$:

$$\widehat{\mathrm{se}}(\hat{p}) = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

```r
set.seed(8)
sample_urban <- sample(is_urban, 500)

p_hat <- mean(sample_urban)
se_p  <- sqrt(p_hat * (1 - p_hat) / 500)

round(c(p_hat = p_hat, se = se_p, true_p = mean(is_urban)), 4)
```

```
#>  p_hat     se true_p
#> 0.2820 0.0201 0.2684
```

THE IDEA IN PLAIN WORDS

The quantity $p(1-p)$ is largest at $p = 0.5$ and falls away towards zero at either end.

That has a sensible meaning. A population evenly split is the hardest to pin down, because samples can land either way. A population where 99% share a characteristic is easy — almost every sample says 99%. Rare and near-universal things are estimated precisely; genuinely contested ones are not.

The consequence for survey design is that $p = 0.5$ is the worst case, and a sample large enough for a 50-50 split is large enough for anything.

3.4 From a point estimate to an interval

> *Our best guess is a single number. How often is a single number right?*

Everything we have estimated so far has been a **point estimate**: $\bar{x}$ for the population mean μ, s^2 for the population variance σ^2, and $\widehat{p}$ for the population proportion p. Each is a single number extracted from the sample and presented as our best estimate of an unknown population quantity.

A point estimate, however, is only a guess. It is the best guess the sample can provide, but it is almost certainly not exactly equal to the population parameter.

This is worth stating explicitly because it runs against our intuition. Suppose a survey reports that students attend an average of 26.1 classes. It is tempting to conclude that the population mean attendance is 26.1. In reality that conclusion is almost certainly wrong. The estimate is not poorly calculated or misleading — it is simply unlikely to equal the true population mean exactly. The real question is not whether the estimate is correct, but **how far from the truth it is likely to be**.

To explore that question, let us return to a setting where we know the truth. Treat the full dataset of 680 students as the population, whose mean attendance is 26.147 classes. We repeatedly draw random samples of 40 students and compute the sample mean for each one. Comparing these estimates with the known population mean allows us to see how much a point estimate typically differs from the truth.

```
students <- read.csv("data/attendance-grades.csv")
att      <- students$attendance
mu_pop   <- mean(att)

set.seed(2)
guesses <- replicate(10000, mean(sample(att, 40, replace = TRUE)))

error <- abs(guesses - mu_pop)

c(exactly_right    = sum(guesses == mu_pop),
  typical_miss     = median(error),
  average_miss     = mean(error),
  missed_by_over_1 = mean(error > 1))
```

```
#>    exactly_right     typical_miss     average_miss missed_by_over_1
#>           0.0000           0.5779           0.6895           0.2464
```

Not one of the ten thousand estimates was exactly right. The typical estimate missed the population mean by about 0.58 classes, and roughly a quarter of them were off by more than a full class.

None of this indicates that anything has gone wrong. Each of those ten thousand estimates was computed correctly, from an honestly drawn sample, using an estimator we spent the whole of the previous unit justifying. Missing the truth is simply what point estimates do.

> *So how do we say something that stands a chance of being true?*

By widening the guess.

Instead of naming a single value, we name a *range* and claim the truth lies inside it. A range can be right, and how often it is right is something we can control: the wider we make it, the more often it succeeds.

Notice that this is not a retreat into vagueness. We are not saying "somewhere around 26" because we have given up on precision. We are making a **calculated guess** — calculated in the literal sense, because the previous units told us exactly how much $\bar{X}$ varies from sample to sample. It is $\sigma/\sqrt{n}$. Knowing that, we can widen the guess by a specific, defensible amount and state in advance how often such a range will contain the truth.

> **REMEMBER THIS**
>
> A **point estimate** names one value. It is the best single guess available, and it is essentially never exactly correct.
> An **interval estimate** names a range, together with the proportion of the time such ranges contain the parameter. It trades precision for the ability to be right.
> The standard error is what makes the second possible. Without a measure of how much the estimate moves from sample to sample, there would be no principled way to choose the width.

The rest of this section works out how wide the range must be.

Building the interval

Suppose a survey of 40 students reports an average attendance of 26.4 classes.

This is our point estimate of the unknown population mean. It is the single best guess the sample can provide.

But how much faith should we place in that guess?

From the previous unit we know that if the survey were repeated many times, each sample would produce a slightly different mean. Some would fall above the population mean and others below it. The point estimate we happen to hold is only one of many values that could have been observed.

Fortunately we also know something about how those sample means behave.

The Central Limit Theorem tells us that, for reasonably large samples, the sampling distribution of the sample mean is approximately normal. Its centre is the population mean μ, and its spread is the standard error $\sigma/\sqrt{n}$.

Suppose previous studies tell us the population standard deviation is $\sigma = 5.45$ classes. With $n = 40$,

$$\mathrm{se}(\bar{X}) = \frac{5.45}{\sqrt{40}} = 0.86$$

So although we do not know the value of the population mean, we do know how much sample means typically fluctuate around it.

The normal distribution turns that fluctuation into a probability. Its probabilities are determined entirely by distance from the centre: about 68% of values lie within one standard deviation, about 95% within 1.96, and about 99.7% within three.

The middle one is what we need.

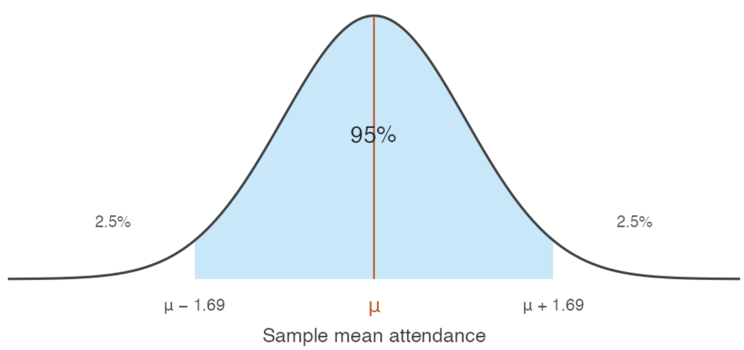

Figure 3.1: The sampling distribution of the sample mean. The shaded region holds 95% of all sample means that could be drawn.

The shaded region represents the 95% of all possible sample means lying within 1.96 standard errors of the population mean. If we repeatedly drew samples of 40 students, roughly 95 out of every 100 sample means would fall inside this region and about 5 would fall outside.

The relationship between distance and probability is worth exploring rather than memorising.

Notice what happens as the distance grows. Widening the region always captures more of the distribution, but never all of it, and the gain slows sharply. Moving from 1.96 to 2.58 standard errors buys the last four percentage points, and no finite distance reaches 100%.

Returning to our survey: with a standard error of 0.86, the distance corresponding to 95% is

Table 3.1: How much of the distribution lies within a given number of standard errors of the centre. The web edition of this book presents the same relationship as a slider.

from centre	inside	outside
0.500	38.29%	61.71%
1.000	68.27%	31.73%
1.500	86.64%	13.36%
1.960	95.00%	5.00%
2.000	95.45%	4.55%
2.500	98.76%	1.24%
2.576	99.00%	1.00%
3.000	99.73%	0.27%
3.500	99.95%	0.05%

$$1.96 \times 0.86 \approx 1.69 \text{ classes}$$

Reversing the question

Now comes the key idea.

If almost all sample means lie within 1.69 classes of the population mean, then the sample mean we actually observed is usually no more than 1.69 classes away from the truth. So instead of asking

> How far is the sample mean from the population mean?

we ask

> Given the sample mean we observed, which values of the population mean are consistent with it?

The answer is obtained by taking the observed sample mean and moving 1.69 classes in each direction:

$$26.4 - 1.69 \qquad \text{to} \qquad 26.4 + 1.69$$

which gives

$$(24.71,\ 28.09)$$

Rather than offering a single best guess, we now report a range of plausible values for the unknown population mean.

WHERE THIS COMES FROM

The reversal above was stated in words. Here it is as algebra, which is where both the $\pm$ formula and the number 1.96 come from.

For large enough samples the sample mean is approximately normal with mean μ and standard deviation $\sigma/\sqrt{n}$. Standardising it — subtracting its mean and dividing by its standard deviation — gives a quantity with a distribution containing no unknowns at all:

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \;\sim\; N(0,1)$$

This is the step that makes everything else possible. Whatever the population, whatever its units, whatever n, the standardised sample mean has the *same* distribution — so a single number read off that distribution serves every problem of this kind.

That number is the solution to $P(|Z| \le c) = 0.95$, which is $c = 1.96$. In R it is `qnorm(0.975)`, the value with 2.5% of the distribution above it.

Now unwind the standardisation. Each line is the previous one with the same operation applied to all three parts of the inequality:

$$\begin{aligned}
P(|Z| \le 1.96) &= 0.95 \\
P\left(-1.96 \;\le\; \frac{\bar{X}-\mu}{\sigma/\sqrt{n}} \;\le\; 1.96\right) &= 0.95 \quad \text{def. of } Z \\
P\left(-1.96\frac{\sigma}{\sqrt{n}} \;\le\; \bar{X}-\mu \;\le\; 1.96\frac{\sigma}{\sqrt{n}}\right) &= 0.95 \quad \times\,\sigma/\sqrt{n} \\
P\left(-\bar{X}-1.96\frac{\sigma}{\sqrt{n}} \;\le\; -\mu \;\le\; -\bar{X}+1.96\frac{\sigma}{\sqrt{n}}\right) &= 0.95 \quad -\bar{X} \\
P\left(\bar{X}-1.96\frac{\sigma}{\sqrt{n}} \;\le\; \mu \;\le\; \bar{X}+1.96\frac{\sigma}{\sqrt{n}}\right) &= 0.95 \quad \times\,(-1)
\end{aligned}$$

The last step multiplies through by -1, which reverses both inequality signs and therefore swaps the two endpoints.

Nothing has been added and nothing has been assumed beyond the Central Limit Theorem. The same statement now has μ in the middle instead of $\bar{X}$, which is what turns a fact about how sample means scatter into an interval around the one we happened to observe.

WATCH OUT

The second line above must be read as a **single event** — Z lying between two bounds — and not as two separate probabilities added together. It is tempting to split $P(|Z| \leq 1.96)$ into $P(Z \leq 1.96) + P(-Z \leq 1.96)$. Those two probabilities are each 0.975, and they sum to 1.95, which is not a probability at all. The two events overlap almost entirely, and adding them counts the overlap twice.

Only the constant changes for other confidence levels. Replacing 1.96 by $z_{\alpha/2}$, the value with $\alpha/2$ of the distribution above it, gives the general result at once — 1.645 for 90%, 2.576 for 99%.

This interval is called a **95% confidence interval**, and the remaining question is what that phrase actually means — a question the next section answers.

REMEMBER THIS

A **confidence interval** for μ, when σ is known, is

$$\bar{x} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

The quantity $z_{\alpha/2}\,\sigma/\sqrt{n}$ is the **margin of error**. The confidence level is $1 - \alpha$.

Confidence	α	$z_{\alpha/2}$
90%	0.10	1.645
95%	0.05	1.960
99%	0.01	2.576

In R these come from `qnorm(1 - alpha/2)` — there is no need for a printed table.

```
x_bar <- 26.4
sigma <- 5.45
n     <- 40

for (level in c(0.90, 0.95, 0.99)) {
  z  <- qnorm(1 - (1 - level) / 2)
  ci <- x_bar + c(-1, 1) * z * sigma / sqrt(n)
  cat(sprintf("%2.0f%%  z = %.3f   [%.2f, %.2f]   width %.2f\n",
              100 * level, z, ci[1], ci[2], diff(ci)))
}
```

```
#> 90%  z = 1.645   [24.98, 27.82]   width 2.83
```

```
#> 95%  z = 1.960   [24.71, 28.09]   width 3.38
#> 99%  z = 2.576   [24.18, 28.62]   width 4.44
```

Higher confidence buys a wider interval, and this is the central trade-off of the whole method. A 99% interval is more likely to contain μ precisely because it commits to less. An interval certain to contain the truth would run from minus infinity to infinity and tell us nothing at all.

What does 95% actually refer to?

The interval either contains μ or it does not. Nothing is random about it once the sample is drawn — μ is a fixed number, and so are the two endpoints sitting in front of us.

The 95% describes the **procedure**. If we drew sample after sample and constructed an interval each time, about 95% of those intervals would contain μ. Since we cannot know whether ours is among them, we report the interval and its coverage rate together.

We can watch this happen. Treat the 680 students as a population, so μ is known, and construct 100 intervals from 100 samples.

```r
students <- read.csv("data/attendance-grades.csv")
att      <- students$attendance

mu_pop    <- mean(att)
sigma_pop <- sqrt(mean((att - mean(att))^2))

set.seed(1)
ci_sim <- t(replicate(100, {
  x <- sample(att, 40, replace = TRUE)
  mean(x) + c(-1, 1) * 1.96 * sigma_pop / sqrt(40)
}))

covers <- ci_sim[, 1] <= mu_pop & ci_sim[, 2] >= mu_pop

d <- data.frame(id = 1:100, lo = ci_sim[, 1], hi = ci_sim[, 2],
                covers = ifelse(covers, "Contains μ", "Misses μ"))

ggplot(d, aes(y = id, xmin = lo, xmax = hi, colour = covers)) +
  geom_errorbarh(height = 0, linewidth = 0.45) +
  geom_vline(xintercept = mu_pop, linewidth = 0.7, colour = "grey20") +
  scale_colour_manual(values = c(book_palette$fill, book_palette$reject),
                      name = NULL) +
  labs(x = "Mean attendance", y = NULL) +
  theme_book() +
  theme(axis.text.y = element_blank(), legend.position = "top")
```

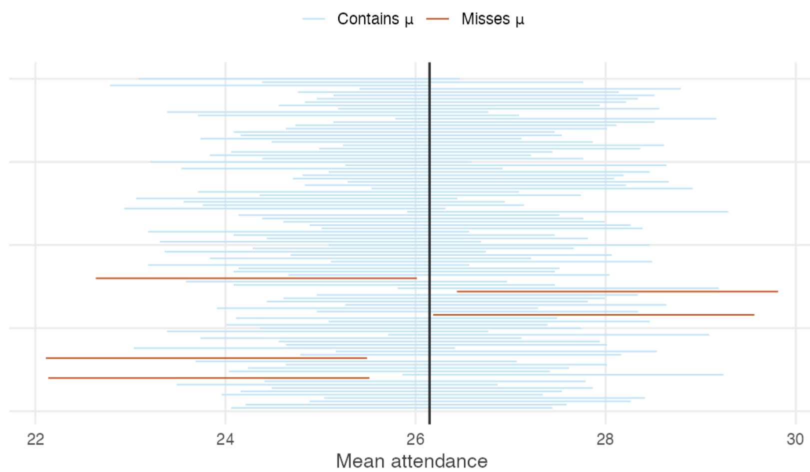

Figure 3.2: One hundred 95% confidence intervals, each from a fresh sample of 40 students. The vertical line is the true population mean.

Of these 100 intervals, 95 contain the population mean and 5 do not. The misses are not mistakes. They are the 5% the method openly budgets for, and a procedure that never missed would be one that had stopped saying anything.

Note also that the misses are not near-misses in any special sense — they are simply the samples whose means happened to fall furthest from μ. Nothing about such a sample looks wrong from the inside. A researcher holding one of those five would see a perfectly ordinary set of forty students.

3.5 What changes when σ is unknown?

> *Every interval so far used σ. We never know it. Can we just use s instead?*

In the previous section we constructed confidence intervals assuming that the population standard deviation σ was known. This allowed us to calculate the standard error exactly,

$$\mathrm{se}(\bar{X}) = \frac{\sigma}{\sqrt{n}}$$

and to use the normal distribution to determine how far the interval should extend on either side of the sample mean.

In practice the population standard deviation is almost never known. This is not a new problem — Section 3.2 showed how to estimate it with the sample standard deviation s. So a natural question arises.

Can we simply replace σ with s and continue as before?

At first sight the answer appears to be yes. We replace $\sigma/\sqrt{n}$ with $s/\sqrt{n}$ and construct the interval exactly as before.

Unfortunately, that interval is no longer a 95% confidence interval.

Why does the interval change?

The interval developed in the previous section relied on one important fact: the only random quantity in it was the sample mean. The width was fixed, because σ was assumed known.

Once σ is replaced by s, the sample determines both the centre of the interval **and** its width.

Most samples produce reasonable estimates of the population standard deviation, but some do not. Occasionally a sample happens to contain observations that are unusually similar, giving a value of s that is too small, and the resulting interval is then narrower than it should be. At the same time, the sample mean may lie further from the population mean than usual.

These two errors can occur together, and when they do the interval is both badly placed and too narrow to reach the truth. The result is that the interval misses more often than we intended.

How serious is the problem?

The easiest way to find out is by simulation.

Suppose the population is normal with mean 26 and standard deviation 5.45. We repeatedly draw samples of size $n = 10$, estimate the standard deviation with s, and construct intervals in two ways: the first using the familiar normal critical value of 1.96, the second using the appropriate critical value from the t distribution.

```r
set.seed(6)
n <- 10

cover <- replicate(50000, {
  x  <- rnorm(n, mean = 26, sd = 5.45)
  se <- sd(x) / sqrt(n)
  c(using_1.96       = abs(mean(x) - 26) < 1.96 * se,
    using_t_critical = abs(mean(x) - 26) < qt(0.975, df = n - 1) * se)
```

```
})

round(rowMeans(cover), 4)
```

```
#>      using_1.96 using_t_critical
#>          0.9181           0.9495
```

The results are striking. Using the normal critical value, the interval captures the population mean only about 91.8% of the time rather than the advertised 95%. One statement in twenty was supposed to be wrong; nearly one in twelve is.

Simply replacing σ by s has made the interval too optimistic.

The t distribution

To restore the intended coverage we must make the interval slightly wider. This is achieved by replacing the normal critical value with one from **Student's t distribution**, which has heavier tails than the normal.

> **REMEMBER THIS**
>
> When σ is unknown, the confidence interval for μ is
>
> $$\bar{x} \pm t_{n-1,\,\alpha/2}\,\frac{s}{\sqrt{n}}$$
>
> where $t_{n-1,\,\alpha/2}$ is the critical value from the t distribution with $n-1$ degrees of freedom.
> In R: `qt(1 - alpha/2, df = n - 1)`.

Notice that the same quantity $n - 1$ appears here as in the definition of the sample variance. Both arise for the same reason: the sample standard deviation is itself estimated from the data, and estimating the centre used up one observation's worth of information.

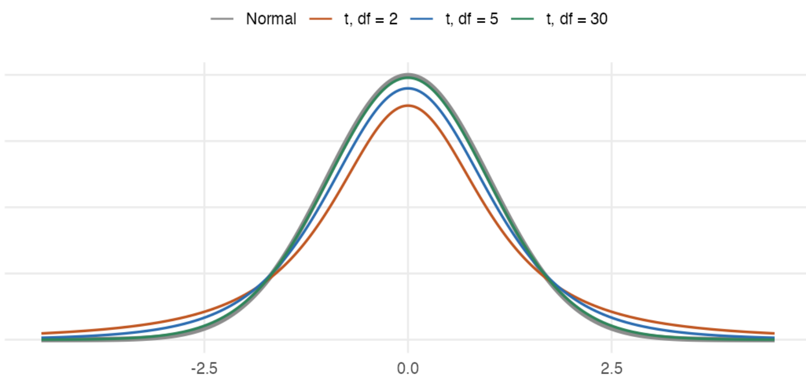

Figure 3.3: The t distribution against the normal. Lower degrees of freedom put more probability in the tails, which is what widens the interval.

The heavier tails are the whole mechanism. Because more of the t distribution's probability sits far from the centre, reaching 95% of it requires travelling further — and travelling further is exactly what widens the interval.

How much difference does it make?

The t distribution depends on the sample size. Small samples carry greater uncertainty about the population standard deviation and therefore need wider intervals. As the sample grows, s becomes a more reliable estimate of σ, the t distribution approaches the normal, and the difference between the two intervals gradually disappears.

```
ns <- c(5, 10, 20, 30, 60, 120)
round(data.frame(n = ns, df = ns - 1,
                 t_critical = qt(0.975, ns - 1),
                 z_critical = 1.96,
                 pct_wider  = 100 * (qt(0.975, ns - 1) / 1.96 - 1)), 2)
```

```
#>     n  df t_critical z_critical pct_wider
#> 1   5   4       2.78       1.96     41.66
#> 2  10   9       2.26       1.96     15.42
#> 3  20  19       2.09       1.96      6.79
#> 4  30  29       2.05       1.96      4.35
#> 5  60  59       2.00       1.96      2.09
#> 6 120 119       1.98       1.96      1.03
```

For a sample of five the correction is substantial — the interval is over 40% wider than the normal one. By the time the sample reaches around a hundred observations the two critical values are almost identical, which is why the distinction is rarely important in large samples and matters a great deal in a study with thirty.

One final point

The t distribution corrects only for the fact that the population standard deviation is unknown.

It does not correct for departures from normality.

If the underlying population is strongly non-normal and the sample is very small, even the t interval may fall short of its advertised coverage. Repeat the coverage experiment above on the real attendance data rather than a normal population, and at $n = 10$ the t interval achieves about 90% rather than 95%. That remaining shortfall has nothing to do with estimating σ. It is the sampling distribution of the sample mean not yet being well approximated by a normal distribution.

> **WATCH OUT**
>
> Two separate issues are in play, and only one of them has been fixed here.
>
> - **Estimating an unknown standard deviation** — addressed by the t distribution.
> - **Approximating the sampling distribution by a normal** — addressed by the Central Limit Theorem, and only when n is large enough for the population in question.
>
> Keeping these distinct is essential for knowing when a confidence interval can be trusted. A t interval computed from twelve observations of a heavily skewed variable is not made trustworthy by the t correction; the correction was never aimed at that problem.

3.6 How large a sample do we need?

> *A survey has not yet been run. How many households should it cover?*

Everything so far took the sample as given. In practice the question usually arrives earlier, from someone who has to budget for the fieldwork.

The margin of error for a mean is

$$E = z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

which can be solved for n:

$$n = \left(\frac{z_{\alpha/2}\, \sigma}{E} \right)^2$$

The squaring is the $\sqrt{n}$ rule from the previous unit, arriving from the other direction: halving the margin of error quadruples the sample.

For a proportion the same rearrangement gives

$$n = \frac{z_{\alpha/2}^2\, p(1-p)}{E^2}$$

with one difficulty. It requires p, which is what the survey exists to find out. The standard resolution is to use the worst case. Since $p(1-p)$ is maximised at $p = 0.5$, setting $p = 0.5$ gives a sample size large enough whatever the truth turns out to be.

```
sample_size_prop <- function(margin, level = 0.95, p = 0.5) {
  z <- qnorm(1 - (1 - level) / 2)
  ceiling(z^2 * p * (1 - p) / margin^2)
}

data.frame(margin_of_error = c("5%", "3%", "2%", "1%"),
           n_at_95 = sapply(c(0.05, 0.03, 0.02, 0.01), sample_size_prop),
           n_at_90 = sapply(c(0.05, 0.03, 0.02, 0.01), sample_size_prop,
                            level = 0.90))
```

```
#>   margin_of_error n_at_95 n_at_90
#> 1              5%     385     271
#> 2              3%    1068     752
#> 3              2%    2401    1691
#> 4              1%    9604    6764
```

These numbers explain something otherwise puzzling about published surveys. A national opinion poll covering 1.4 billion people typically samples one or two thousand — and the table shows why that is sufficient rather than negligent. The population size does not appear anywhere in the formula. What determines precision is the number of people asked, not the fraction of the country they represent.

WATCH OUT

This calculation answers a narrow question: how large a sample is needed for the standard error to reach a target, *assuming the sample is drawn properly*.

It says nothing about the failures of Section 2.2, and the sample size it returns will not repair any of them. A survey of two thousand urban households is not a survey of India, however carefully the two thousand was computed.

Intervals for a proportion

Collecting the pieces, an interval for a population proportion is

$$\hat{p} \;\pm\; z_{\alpha/2}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

Note that z appears here rather than t. The t correction exists because σ was estimated separately from μ; for a proportion, $\hat{p}$ determines the standard error on its own, so there is no second estimate to account for.

Suppose that in a random sample of 1,000 secondary school teachers in Karnataka, 518 are women.

```
x <- 518
n <- 1000

p_hat <- x / n
se    <- sqrt(p_hat * (1 - p_hat) / n)

round(c(p_hat = p_hat, se = se,
        lower = p_hat - 1.96 * se,
        upper = p_hat + 1.96 * se), 4)
```

```
#>  p_hat     se  lower  upper
#> 0.5180 0.0158 0.4870 0.5490
```

So we report that between 48.7% and 54.9% of secondary school teachers in Karnataka are women, with 95% confidence.

That interval contains 0.5, which is worth noticing. The sample proportion is above half, but the data are also consistent with women being a minority of the profession. A headline reading "most secondary school teachers are women" would be going beyond what a sample of a thousand can support.

3.7 When only one direction matters

> *Sometimes we care about one end only. Why pay for the other?*

A confidence interval spends half its error budget at each end. Often only one end is of interest.

A regulator checking whether a pollutant exceeds a limit does not care how low it might be. A funder asking whether attendance is at least 24 classes does not care how high it might be. In these cases the whole of α can be spent on the side that matters, which buys a tighter bound.

REMEMBER THIS

A **one-sided confidence bound** uses z_α rather than $z_{\alpha/2}$:

$$\text{upper bound:}\quad \bar{x} + z_\alpha \frac{\sigma}{\sqrt{n}} \qquad\qquad \text{lower bound:}\quad \bar{x} - z_\alpha \frac{\sigma}{\sqrt{n}}$$

At 95% confidence, $z_\alpha = 1.645$ rather than 1.960.

```r
x_bar <- 26.4; sigma <- 5.45; n <- 40

round(c(two_sided_lower = x_bar - qnorm(0.975) * sigma / sqrt(n),
        two_sided_upper = x_bar + qnorm(0.975) * sigma / sqrt(n),
        one_sided_upper = x_bar + qnorm(0.950) * sigma / sqrt(n),
        one_sided_lower = x_bar - qnorm(0.950) * sigma / sqrt(n)), 3)
```

```
#> two_sided_lower two_sided_upper one_sided_upper one_sided_lower
#>           24.71           28.09           27.82           24.98
```

The one-sided bound is closer to the estimate than the two-sided one at the same confidence level. Nothing has been gained for free: we have given up any claim about the other direction entirely.

WATCH OUT

The choice between one-sided and two-sided must be made **before** looking at the data, on the basis of what the question is.
Computing a two-sided interval, noticing it just fails to exclude some value of interest, and switching to a one-sided bound to obtain a tighter result is not a calculation. It is a way of arranging to be right, and the stated confidence level no longer describes what was done.

3.8 What a confidence interval does not mean

> *Four statements about the interval we just built. Which of them are correct?*

By now you know how to construct a confidence interval and how to interpret it through repeated sampling. Yet confidence intervals are among the most misunderstood ideas in statistics, and experienced researchers interpret them incorrectly often enough that the errors have names.

Consider the following four statements. Decide what you think of each before reading on.

Statement 1: the probability the mean is inside

> There is a 95% probability that the population mean lies between 24.4 and 27.8.

Incorrect. This is the most common misconception.

Once the sample has been collected, the confidence interval is fixed. The population mean is also fixed. Either the interval contains it or it does not — there is no longer any randomness to attach a probability to.

The 95% refers to the *method*, not to this particular interval. If we repeatedly collected samples and constructed intervals in exactly the same way, about 95% of them would contain the true population mean.

The uncertainty lies in the sampling process, not in the value of the population mean.

Statement 2: the spread of the population

> Ninety-five percent of the population lies within this confidence interval.

Incorrect.

A confidence interval is about the location of the population **mean**, not about the spread of individual observations.

Our interval for average attendance runs from 24.4 to 27.8 classes — a few classes wide. Individual students in the same data attended anywhere from almost none of their classes to nearly all of them.

The clearest way to see that these are different quantities is to watch them respond to sample size. As n increases the interval narrows, because the mean is estimated more precisely. The variability among individual students does not change at all.

Statement 3: two intervals that overlap

> The confidence intervals for two groups overlap, so the groups are not significantly different.

Not necessarily.

Two confidence intervals can overlap even when the difference between the two population means is statistically significant.

The reason is straightforward. Each interval describes uncertainty about a single mean. Whether two groups differ depends on the uncertainty surrounding **their difference**, which is a different quantity with a different standard error.

Later in the book we construct confidence intervals for differences between means. Those intervals, and not visual overlap, are the correct basis for comparison.

Statement 4: a wider interval

> A wider confidence interval means the estimate is worse.

Not necessarily.

A wider interval means the estimate is *less precise*, and precision is not the same thing as quality.

A carefully designed survey with a small sample may produce a wide interval simply because the available information is limited. That interval is an honest report of genuine uncertainty.

A poorly designed survey may produce a very narrow interval by repeatedly sampling the wrong population. Section 2.2 showed exactly this: Survey B, which interviewed only urban households, produced estimates with a *smaller* spread than the correct design while centring on a figure ₹6,800 too high. The interval was precise, and precisely wrong.

Confidence intervals measure uncertainty due to **sampling variation**. They do not detect bias arising from poor sampling, non-response, or measurement error.

> **REMEMBER THIS**
>
> A confidence interval tells us what the data can reasonably support about an unknown population parameter.
> It does **not** give the probability that a particular interval contains the truth, it does **not** describe the spread of individual observations, it does **not** settle whether two groups differ by whether they overlap, and it does

not guarantee that the data were collected well.
Understanding these limitations is as much a part of using the interval as knowing how to calculate it.

3.9 Summary

REMEMBER THIS

- The **sample variance** $s^2 = \frac{1}{n-1}\sum(x_i - \bar{x})^2$ is unbiased for σ^2. The divisor is $n-1$ because $\bar{x}$ was estimated from the same data, leaving $n-1$ free deviations. If μ is known, the divisor is n and the deviations are taken from μ.
- A **proportion is a mean** — of a variable coded 1 and 0 — so every result about $\bar{X}$ transfers to $\hat{p} = x/n$ without new proof.
- $\text{se}(\hat{p}) = \sqrt{p(1-p)/n}$, largest at $p = 0.5$. Evenly split populations are the hardest to estimate; near-universal ones are easy.
- A **confidence interval** is $\bar{x} \pm z_{\alpha/2}\,\sigma/\sqrt{n}$ when σ is known, and $\bar{x} \pm t_{n-1,\,\alpha/2}\,s/\sqrt{n}$ when it is not. For a proportion, $\hat{p} \pm z_{\alpha/2}\sqrt{\hat{p}(1-\hat{p})/n}$.
- The t distribution compensates for estimating σ from the same sample. It does not compensate for a non-normal population — two different assumptions, one correction.
- The confidence level is a property of the **procedure**: across repeated samples, that proportion of intervals contains the parameter. A particular interval either contains it or does not.
- Higher confidence means a wider interval. Halving the margin of error quadruples the required sample, and $n = (z_{\alpha/2}\sigma/E)^2$ makes that explicit before any fieldwork is commissioned.
- What determines precision is how many people were asked, not what fraction of the population they are. Population size appears in none of these formulas.

Everything in this unit produced a *range* of values consistent with the data. The next unit asks a sharper question: given a specific claim about the population, is our sample consistent with it?

3.10 Practice problems

YOUR TURN

These are for learning. Work each one before opening the solution. The question bank that follows has no answers at all.

Concept checks

3.1 Why does the sample variance divide by $n-1$ rather than n?

SOLUTION

Because the deviations are measured from $\bar{x}$, which was itself computed from the same observations. The sample mean is the value that makes $\sum(x_i-\bar{x})^2$ as small as possible, so measuring spread around it gives an answer that is too small.
Formally, the deviations must sum to zero, so only $n-1$ of them are free to vary. Dividing by $n-1$ corrects the shortfall exactly, making s^2 unbiased for σ^2.

3.2 A dataset records whether each of 900 households owns a bicycle. Why is no new theory needed to estimate the proportion that do?

SOLUTION

Because coding "owns" as 1 and "does not own" as 0 makes the proportion the mean of that column. Summing ones counts the owners, and dividing by 900 turns the count into a share.
So $\hat{p}$ *is* a sample mean, and everything already established — unbiasedness, standard error falling with $\sqrt{n}$, approximate normality by the Central Limit Theorem — applies without modification.

3.3 Explain why a 99% confidence interval is wider than a 95% one.

SOLUTION

Higher confidence requires the interval to succeed more often across repeated samples, and the only way to succeed more often is to claim less. Widening the interval makes it easier to capture μ.
The trade-off is unavoidable. An interval guaranteed to contain the truth would have to span every possible value, and would carry no information.

3.4 For which value of p is a proportion hardest to estimate precisely, and why does that make sense?

SOLUTION

$p = 0.5$, since $p(1-p)$ is maximised there.
It makes sense because an evenly split population is the one where samples can most easily land either way. If 99% of households have electricity, almost any sample says so. If 50% do, the sample proportion has real room to move.
This is why survey planners use $p = 0.5$ when computing sample sizes: it is the worst case, so it is safe whatever the truth turns out to be.

3.5 A student reports a 95% interval of $[24.4, 27.8]$ and writes: "so 95% of students attend between 24.4 and 27.8 classes." What has gone wrong?

SOLUTION

The interval is about the *population mean*, not about individual students. It says the average attendance is plausibly between 24.4 and 27.8. Individual students vary far more widely — the standard deviation is about 5.5 classes, so many students lie well outside that range.
Note also that the interval narrows as n grows, whereas the spread of students does not change at all. The two quantities behave differently, which is a sign they are measuring different things.

Explain

3.6 Explain what the "95%" in a 95% confidence interval refers to, and why it cannot refer to the particular interval you have computed.

SOLUTION

It refers to the procedure. If samples were drawn repeatedly and an interval constructed from each by this method, about 95% of those intervals would contain the population parameter.
It cannot refer to the interval in front of you because nothing about it is random any more. The population mean is a fixed number, and once the sample is drawn, the two endpoints are fixed numbers too. Either μ is between them or it is not; there is no probability left in the situation.
The randomness lived in which sample was drawn, and that was spent. The 95% is a statement about the method's track record, which is why we report the interval and its confidence level together — the level describes how much such statements can generally be trusted.

3.7 Why is the t distribution needed at all, and why does it stop mattering in large samples?

SOLUTION

When σ is known, the only quantity estimated from the sample is $\bar{x}$. When σ is unknown, s is estimated from the same data, so both the centre and the width of the interval are subject to sampling error. A sample with an unusually small s produces an interval that is narrow *and* poorly positioned, and those errors compound. Using 1.96 anyway gives about 92% coverage at $n = 10$ instead of 95%.
The t distribution has heavier tails, widening the interval by exactly enough to restore the advertised coverage.
It stops mattering in large samples because s becomes an increasingly reliable estimate of σ. The critical value falls from 2.776 at $n = 5$ to 1.984 at $n = 100$, approaching 1.96 as the degrees of freedom grow. Once s is essentially known, there is nothing left to correct for.

3.8 A colleague computes a two-sided 95% interval, notices that it barely includes zero, and recomputes it as a one-sided bound so that zero falls outside. Explain the problem.

SOLUTION

The confidence level describes a procedure fixed in advance. Once the choice between one-sided and two-sided is made *after* seeing where the data fell, the procedure being followed is no longer the one whose 95% property was established.
The real procedure here is "compute a two-sided interval, and if it fails to give the answer I want, switch." That procedure covers the truth far less than 95% of the time, because it has two chances to produce the conclusion sought.
One-sided bounds are legitimate when the question is genuinely one-directional — a regulator who cares only about exceeding a limit. What makes them illegitimate is choosing the direction from the data.

Numerical problems

3.9 A sample of 10 professors reported the following hours worked in a week: 48, 22, 19, 65, 72, 37, 55, 60, 49, 28.

i. Estimate the population mean.
ii. Estimate the population variance and standard deviation.
iii. Estimate them again given that the population mean is known to be 46 hours.

iv. Why does (iii) use a different divisor?

SOLUTION

(i) $\bar{x} = 455/10 = 45.5$ hours.
(ii) With μ unknown, deviations are taken from $\bar{x} = 45.5$ and $\sum(x_i - \bar{x})^2 = 3034.5$:

$$s^2 = \frac{3034.5}{9} = 337.17 \qquad s = 18.36$$

(iii) With $\mu = 46$ known, deviations are taken from 46 and $\sum(x_i - \mu)^2 = 3037$:

$$\hat{\sigma}^2 = \frac{3037}{10} = 303.7 \qquad \hat{\sigma} = 17.43$$

(iv) In (ii) the centre was estimated from the sample, costing one degree of freedom. In (iii) it was known, so none was spent.
Notice that the sum of squares is *larger* in (iii) — 3037 against 3034.5 — because $\bar{x}$ minimises that sum and μ is not $\bar{x}$. The larger total divided by the larger n is what keeps both estimates unbiased.

3.10 The average life of a sample of 10 tyres was 28,400 km. Lifetimes are normally distributed with $\sigma = 3{,}300$ km.

i. Construct 90%, 95% and 99% confidence intervals for the mean life.
ii. Find a value that, with 95% confidence, is larger than the population mean.
iii. Find a value that, with 99% confidence, is smaller than the population mean.

SOLUTION

$\sigma/\sqrt{n} = 3300/\sqrt{10} = 1043.6$.
(i)

$$\begin{aligned}
90\% :&\quad 28400 \pm 1.645(1043.6) = [26683,\ 30117] \\
95\% :&\quad 28400 \pm 1.960(1043.6) = [26355,\ 30445] \\
99\% :&\quad 28400 \pm 2.576(1043.6) = [25712,\ 31088]
\end{aligned}$$

(ii) A one-sided upper bound at 95% uses $z_{0.05} = 1.645$: $28400 + 1.645(1043.6) = 30{,}117$ km.
(iii) A one-sided lower bound at 99% uses $z_{0.01} = 2.326$: $28400 - 2.326(1043.6) = 25{,}973$ km.
Note that (ii) uses 1.645 rather than 1.960: the whole 5% sits in one tail,

because we are making a claim in one direction only.

3.11 In a random sample of 400 death certificates of college students, 98 recorded a motorcycle accident.

i. Estimate the proportion of such deaths due to motorcycle accidents.
ii. Estimate the standard error of that estimate.
iii. Construct a 95% confidence interval.

SOLUTION

(i) $\hat{p} = 98/400 = 0.245.$
(ii)

$$\widehat{\mathrm{se}}(\hat{p}) = \sqrt{\frac{0.245 \times 0.755}{400}} = \sqrt{0.000462} = 0.0215$$

(iii) $0.245 \pm 1.96(0.0215) = [0.203,\ 0.287].$
So between about 20% and 29%, with 95% confidence.

3.12 In a random sample of 1,000 secondary school teachers in Karnataka, 518 identify as women.

i. Construct a 95% confidence interval for the population proportion.
ii. Construct a 90% upper confidence bound.
iii. Can the state conclude that women are a majority of the profession?

SOLUTION

$\hat{p} = 0.518$ and $\widehat{\mathrm{se}} = \sqrt{0.518 \times 0.482/1000} = 0.0158.$
(i) $0.518 \pm 1.96(0.0158) = [0.487,\ 0.549].$
(ii) $0.518 + 1.282(0.0158) = 0.538.$
(iii) No. The 95% interval contains 0.5, so the data are also consistent with women being slightly under half of secondary school teachers. The point estimate is above half, but a sample of a thousand cannot separate 51.8% from 50% with confidence.
This is a good illustration of why the interval is reported rather than the point estimate alone. “51.8% of teachers are women” and “between 48.7% and 54.9% are” support very different headlines.

3.13 A state wants to estimate the proportion of households with a functioning water connection to within 2 percentage points at 95% confidence.

i. How many households must be surveyed?

ii. A pilot suggests the proportion is near 0.85. How many are needed now?
iii. Why is (i) larger, and when should it still be preferred?

SOLUTION

(i) With no prior information, use the worst case $p = 0.5$:

$$n = \frac{1.96^2(0.5)(0.5)}{0.02^2} = \frac{0.9604}{0.0004} = 2401$$

(ii) With $p = 0.85$, $p(1-p) = 0.1275$:

$$n = \frac{1.96^2(0.1275)}{0.0004} = 1225$$

Roughly half as many.
(iii) Because $p(1-p)$ is largest at 0.5, so that choice always gives the biggest requirement and therefore a safe one.
It should still be preferred when the pilot is thin, or drawn from a non-representative area, or when the true proportion might differ across the state. If the real figure turns out to be 0.5 and the survey was sized for 0.85, the margin of error will exceed the 2 points promised.

R exercises

3.14 Using `data/attendance-grades.csv`, compute point estimates of the mean, variance and standard deviation of attendance, semester GPA and final score. Then construct 95% confidence intervals for each mean.

SOLUTION

```
students <- read.csv("data/attendance-grades.csv")
vars     <- c("attendance", "sem_gpa", "final_score")

est <- t(sapply(vars, function(v) {
  x  <- students[[v]]
  n  <- length(x)
  se <- sd(x) / sqrt(n)
  tc <- qt(0.975, df = n - 1)
  c(mean = mean(x), var = var(x), sd = sd(x), se = se,
    lower = mean(x) - tc * se, upper = mean(x) + tc * se)
}))

round(est, 3)
```

```
#>                 mean    var    sd    se  lower  upper
```

```
#> attendance  26.147 29.757 5.455 0.209 25.736 26.558
#> sem_gpa      6.503  3.391 1.841 0.071  6.364  6.641
#> final_score 90.856 21.455 4.632 0.178 90.507 91.205
```

The intervals are narrow because $n = 680$ is large. Note that `var()` and `sd()` already use the $n-1$ divisor, so no manual correction is needed.

3.15 Verify the $n-1$ correction yourself. Treat attendance as a population, draw 20,000 samples of 8, and compare the average of $\frac{1}{n}\sum(x_i-\bar{x})^2$ with the average of $\frac{1}{n-1}\sum(x_i-\bar{x})^2$.

SOLUTION

```r
att <- students$attendance
sigma2 <- mean((att - mean(att))^2)

set.seed(21)
n <- 8
out <- replicate(20000, {
  x <- sample(att, n, replace = TRUE)
  c(over_n = mean((x - mean(x))^2), over_n_1 = var(x))
})

round(c(population_variance = sigma2,
        avg_over_n          = mean(out[1, ]),
        avg_over_n_1        = mean(out[2, ]),
        predicted_over_n    = sigma2 * (n - 1) / n), 3)
```

```
#> population_variance          avg_over_n        avg_over_n_1
 ↪  predicted_over_n
#>               29.71               26.18               29.91
 ↪  26.00
```

Dividing by n understates the population variance, and it does so by exactly the predicted factor $(n-1)/n = 7/8$. Dividing by $n-1$ recovers the right answer on average.
The bias is not small at this sample size — about 12% — which is why the correction is built into `var()` rather than left to the user.

3.16 Treat `wages-india-synthetic.csv` as a population and let p be the proportion of workers in urban areas. Draw 1,000 samples of 400, construct a 95% interval from each, and count how many contain the true p.

SOLUTION

```
wages    <- read.csv("data/wages-india-synthetic.csv")
is_urban <- as.numeric(wages$residence == "Urban")
p_true   <- mean(is_urban)

set.seed(4)
ci <- t(replicate(1000, {
  s  <- sample(is_urban, 400, replace = TRUE)
  ph <- mean(s)
  ph + c(-1, 1) * 1.96 * sqrt(ph * (1 - ph) / 400)
}))

c(true_p = p_true,
  coverage = mean(ci[, 1] <= p_true & ci[, 2] >= p_true))
```

```
#>   true_p coverage
#>   0.2684   0.9580
```

Close to the advertised 95%. The method delivers what it promises for a proportion of this size at this sample size.
It would perform less well for a very rare characteristic — with $p = 0.01$ and $n = 400$ the expected number of successes is 4, the normal approximation is poor, and coverage falls short. A common rule of thumb requires both np and $n(1-p)$ to be at least 10.

3.17 Show that the t correction matters. On a normal population with $\mu = 50$ and $\sigma = 12$, compare the coverage of intervals built with 1.96 against those built with the t critical value, for $n = 5, 10, 30$ and 100.

SOLUTION

```
set.seed(30)
cover <- t(sapply(c(5, 10, 30, 100), function(n) {
  r <- replicate(20000, {
    x  <- rnorm(n, 50, 12)
    se <- sd(x) / sqrt(n)
    c(abs(mean(x) - 50) < 1.96 * se,
      abs(mean(x) - 50) < qt(0.975, n - 1) * se)
  })
  c(n = n, using_z = mean(r[1, ]), using_t = mean(r[2, ]))
}))

round(cover, 4)
```

```
#>        n using_z using_t
#> [1,]   5  0.8792  0.9510
```

```
#> [2,]  10  0.9193  0.9486
#> [3,]  30  0.9404  0.9498
#> [4,] 100  0.9484  0.9508
```

The t intervals hold their 95% at every sample size. The z intervals fall well short when n is small, and the gap closes as n grows — by $n = 100$ the two are nearly indistinguishable.
This is the whole case for the t distribution in one table, and also the reason nobody worries about it in large samples.

3.18 How does interval width respond to sample size and confidence level? Using the attendance population, plot the width of the confidence interval for the mean against n, for the 90%, 95% and 99% levels.

SOLUTION

```r
sigma_att <- sqrt(mean((att - mean(att))^2))

grid <- expand.grid(n = seq(10, 500, by = 5),
                    level = c(0.90, 0.95, 0.99))
grid$width <- 2 * qnorm(1 - (1 - grid$level) / 2) * sigma_att /
 ↪  sqrt(grid$n)
grid$level <- factor(paste0(100 * grid$level, "%"))

ggplot(grid, aes(n, width, colour = level)) +
  geom_line(linewidth = 0.8) +
  labs(x = "Sample size", y = "Interval width (classes)", colour =
   ↪  NULL) +
  theme_book() +
  theme(legend.position = "top")
```

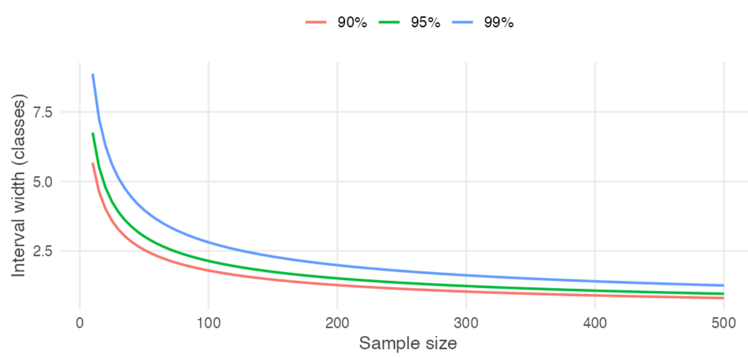

Figure 3.4: Confidence interval width against sample size, at three confidence levels.

Width falls as $1/\sqrt{n}$, steeply at first and then hardly at all. Most of the available precision is bought in the first hundred observations.
The three curves never cross, and the gap between them narrows in absolute terms as n grows. Choosing 99% over 95% is expensive in a small survey and nearly free in a large one.

3.11 Question bank

WATCH OUT

No answers are given for this section, by design. These are for testing yourself once the practice problems are complete.

Multiple choice

QB 3.1 The divisor $n - 1$ in the sample variance is used because

a. it makes the variance larger, which is safer
b. $\bar{x}$ was estimated from the same data, leaving $n - 1$ free deviations
c. the population is finite
d. the sample standard deviation would otherwise be negative

QB 3.2 A 95% confidence interval for μ is $[112, 128]$. Which statement is correct?

a. 95% of the population lies between 112 and 128
b. There is a 95% probability that μ lies between 112 and 128
c. 95% of intervals built this way would contain μ
d. The sample mean has a 95% chance of lying between 112 and 128

QB 3.3 Holding everything else fixed, moving from 95% to 99% confidence

a. narrows the interval
b. widens the interval
c. leaves the width unchanged but shifts the centre
d. requires a larger sample

QB 3.4 For a sample proportion, the standard error is largest when $\hat{p}$ is closest to

a. 0
b. 0.25
c. 0.5
d. 1

QB 3.5 A survey of 600 households finds 30% own a two-wheeler. The approximate 95% margin of error is

a. 0.4 percentage points
b. 1.9 percentage points
c. 3.7 percentage points
d. 7.5 percentage points

QB 3.6 The t distribution is used instead of the normal when

a. the sample is small
b. σ is estimated from the sample
c. the population is skewed
d. the confidence level exceeds 95%

QB 3.7 To halve the margin of error of a survey, the sample size must be multiplied by

a. $\sqrt{2}$
b. 2
c. 4
d. It depends on the confidence level

QB 3.8 Which of the following would *not* narrow a confidence interval for μ? *(select all that apply)*

a. Increasing the sample size
b. Lowering the confidence level
c. Sampling from a population with smaller σ
d. Sampling only from a subgroup that happens to be homogeneous

Short reasoning

QB 3.9 A report gives a 95% confidence interval for mean household income of [₹18,200, ₹19,600] and concludes: "almost all households in this district earn between ₹18,200 and ₹19,600." Identify the error and state what the interval does say.

QB 3.10 Explain why the population size does not appear in the formula for the margin of error, and why this surprises people.

QB 3.11 Two districts are surveyed. District A's 95% interval for mean income is [₹19,000, ₹23,000]; District B's is [₹22,000, ₹26,000]. A journalist writes that the districts do not differ significantly because the intervals overlap. Evaluate.

QB 3.12 A researcher has a sample of 12 observations from a strongly skewed population and constructs a t interval. State two distinct assumptions being relied on, and say which one the t distribution addresses.

QB 3.13 Explain why s is not an unbiased estimator of σ even though s^2 is unbiased for σ^2, and why this is rarely mentioned in practice.

QB 3.14 A polling organisation reports a margin of error of ±3 points but sampled only households with landline telephones. Explain what the ±3 does and does not cover.

Numerical

QB 3.15 A sample of 16 bags of rice has mean weight 24.6 kg and sample standard deviation 1.2 kg.

i. Construct a 95% confidence interval for the mean weight.
ii. Construct a 99% confidence interval.
iii. The label claims 25 kg. Comment.
iv. How would your answer to (i) change if the standard deviation of 1.2 kg were known to be the population value?

QB 3.16 In a random sample of 250 farm households in a district, 145 reported using irrigation.

i. Estimate the proportion and its standard error.
ii. Construct a 90% confidence interval.
iii. Construct a 95% lower confidence bound.
iv. A scheme guarantees that at least half of households have irrigation. Is the sample consistent with that?

QB 3.17 A state government wishes to estimate mean annual household expenditure to within ₹500 at 95% confidence. Previous rounds suggest $\sigma \approx$ ₹9,000.

i. What sample size is required?
ii. Fieldwork costs ₹1,100 per household. What is the budget?
iii. The budget is capped at ₹15 lakh. What margin of error is achievable?

QB 3.18 Using `data/wages-india-synthetic.csv`, write R code to construct 95% confidence intervals for mean annual earnings separately for urban and rural workers, using samples of 300 from each. Report both intervals, state whether they overlap, and explain carefully what that does and does not tell you about the difference between the two groups.

3.12 Looking ahead

In this unit we learned how to use a sample to estimate an unknown population parameter. Rather than reporting a single number, we reported a **confidence interval** — a range of values consistent with the observed data.

For many problems that is exactly what is needed. Opinion polls report margins of error, medical studies estimate treatment effects, and government surveys publish confidence intervals around unemployment rates, inflation and average income. In each case the question begins with the data:

> Given this sample, what values of the population parameter are plausible?

Many real questions, however, begin somewhere else.

A school claims that its students attend an average of at least 28 classes. A manufacturer advertises that its batteries last 500 hours. A government announces that unemployment has fallen below 5%.

These claims are made *before* any data are collected.

The task is no longer to estimate an unknown quantity. It is to decide whether a sample provides enough evidence to support — or to challenge — a specific claim that someone else has put forward.

This requires turning the question around. Instead of asking

> What values are consistent with the data?

we ask

> Are the data consistent with a particular value?

At first sight these look like different problems. One starts with the data and searches for plausible parameter values; the other starts with a proposed value and asks whether the data agree with it.

As the next unit shows, the two are closely connected. Hypothesis tests and confidence intervals are built from the same ingredients — the sampling distribution, the standard error, and the probability model laid on top of them — but they are pointed in opposite directions.

Confidence intervals ask **what values are plausible**.

Hypothesis tests ask **whether a particular claim is plausible**.

Chapter 4

Hypothesis Testing: One Sample

In the previous unit we learned that every estimate comes with uncertainty. A confidence interval tells us which population values are plausible given our sample.

But many real statistical questions begin with a specific claim. Has average monsoon rainfall changed? Is the unemployment rate below 5%? Does a new drug lower blood pressure?

Rather than asking *which values are plausible*, these questions ask *is this particular claim consistent with the data?* Answering that is the purpose of hypothesis testing.

Consider a claim made about a population. You collect a sample. Your estimate is not exactly what the claim predicted.

Should you conclude that the claim is wrong?

Or is this simply the kind of difference that appears because every random sample is slightly different?

That question is the starting point for hypothesis testing, and it is the engine behind much of modern statistics.

4.1 The problem, before any formulas

The APU administration claims the average height of all students is 165 cm.

You are sceptical. You could settle it by measuring all 3,500 students, but that costs weeks of work. So instead you measure 36 students chosen at random, and you find their average height is **168 cm**.

The claim said 165. Your sample says 168. Is the administration wrong?

> **WATCH OUT**
>
> The tempting answer is "yes, obviously — 168 is not 165." That answer is wrong, and understanding *why* it is wrong is the whole chapter.
> You did not measure everyone. You measured 36 people. Even if the true average really were exactly 165, your particular 36 students would almost never average exactly 165. Some samples come out high, some low. **A gap between the sample and the claim is guaranteed to exist even when the claim is perfectly true.**

Let us actually watch that happen.

We obviously do not know whether the administration is correct. But imagine, just for a moment, that they are. If the true average really were 165 cm, with a standard deviation of 8 cm, what kinds of sample means would we expect to see? Draw a sample of 36 students, ten separate times, and look.

```r
set.seed(42)

ten_samples <- replicate(10, mean(rnorm(n = 36, mean = 165, sd = 8)))
round(ten_samples, 2)
```

```
#>  [1] 165.5 165.1 165.6 163.0 164.8 164.4 165.3 165.4 164.1 164.5
```

Every one of those numbers came from a population whose mean is *exactly* 165. Not one of them is 165. Several are further from 165 than a full centimetre.

So the real question is not "is 168 different from 165?" It obviously is. The real question is:

> **THE IDEA IN PLAIN WORDS**
>
> **If the administration's claim were true, how often would a sample of 36 students come out as far from 165 as ours did?**
> If the answer is "quite often" — then our sample is unremarkable, and we have no grounds to call anyone a liar.
> If the answer is "almost never" — then either something very unusual happened to us, or the administration's claim is not correct. In statistics we usually regard the second explanation as the more convincing one.

We can answer that question by brute force before we do any algebra at all. Let's draw not ten samples but fifty thousand, still assuming the claim is true, and count how many stray at least 3 cm away from 165 (as ours did).

```r
set.seed(42)

many_means <- replicate(50000, mean(rnorm(n = 36, mean = 165, sd = 8)))

# How far did our real sample fall from the claim?
observed_gap <- abs(168 - 165)

mean(abs(many_means - 165) >= observed_gap)
```

```
#> [1] 0.02424
```

About 2.4% of the time. So if the administration is telling the truth, a sample like ours turns up roughly once in forty tries.

We have now answered the question using simulation alone, and without naming a single technical term. If the claim were true, a sample mean at least 3 cm away from 165 would appear about once in forty surveys.

That makes our sample unusual. But how unusual is unusual *enough*? Once in forty is rare; would once in ten have been rare enough? Once in a hundred? The rest of this chapter builds a systematic way of answering exactly that — and by the end of it you will be able to obtain the number above exactly, in one line, with no simulation at all.

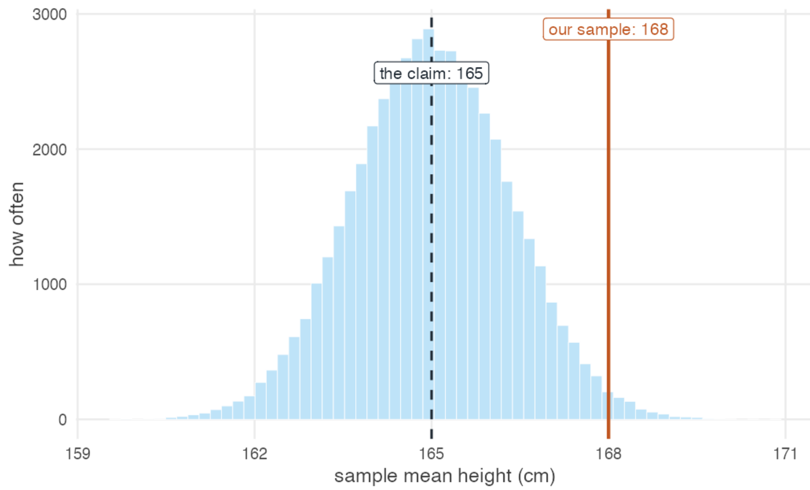

Figure 4.1: Fifty thousand sample means, each from 36 students, drawn from a population whose true mean is 165 cm (dashed line). The solid line marks our observed sample mean of 168 cm. Only about 2.4% of the simulated samples fall at least this far from the claimed value in either direction, which is what makes our sample surprising if the claim is true.

4.2 Naming the two hypotheses

> *We have the reasoning. What are the two positions we are reasoning between?*

The previous section did the whole of a hypothesis test without naming anything. We assumed the administration's claim, asked how often a sample would stray as far from it as ours did, and found the answer was about once in forty.

Let us give the two positions in that argument their names.

We start from a sample $x_1, x_2, \dots, x_n$ drawn from a population with unknown mean μ and standard deviation σ. The claim under scrutiny is the **null hypothesis**:

$$H_0 : \mu = 165$$

The position we would move to if the claim collapses is the **alternative hypothesis**:

$$H_A : \mu \neq 165$$

Why begin by assuming the claim is true?

At first this looks backwards. If we suspect the claim is wrong, why start by supposing it is right?

Because the assumption is what gives us something to measure against. Notice what we actually did in the previous section: we could only simulate those fifty thousand samples *because* we had a specific value of μ to simulate from. Had we begun with "the mean is not 165," there would have been nothing to draw samples from — not 165 covers 164, and 170, and 3,000, and each would give a different answer.

So we grant the claim, temporarily, and see where it leads. If our sample would be quite ordinary under it, we have no reason to doubt it. If our sample would be extraordinarily unlikely under it, then the claim itself becomes hard to believe.

> **REMEMBER THIS**
>
> The null hypothesis is always the *boring* option — nothing is going on, the claim stands, the difference is zero. We never *prove* it. We only ever fail to knock it down.
> This is why we say "**fail to reject** H_0" and never "accept H_0." A court that acquits has not proven innocence; it has failed to prove guilt.

4.2.1 Which hypothesis does the claim go in?

> *A claim arrives. Does it become H_0 or H_A?*

This is one of the most common places for a beginner to go wrong, and the way out is not to memorise a rule but to ask a question first.

Before writing anything down, ask yourself:

> What result would convince me?

Suppose a report claims that the mean CAT score of applicants is *more than* 499. What would persuade you? Almost everyone answers the same way: a sample mean comfortably **above** 499. A sample averaging 470 would not support the report, and neither would one averaging exactly 499.

That answer has already determined the test. The thing you would find convincing — a mean above 499 — is the thing you are gathering evidence *for*, and a test only ever produces evidence *against* H_0. So the claim has to sit in the alternative:

$$H_0 : \mu \leq 499 \qquad H_A : \mu > 499$$

The direction follows too. You would be convinced by large values, so it is large values that must count against the null, and the test looks in the upper tail.

> **REMEMBER THIS**
>
> Two questions settle every case.
> **What am I trying to gather evidence for?** That goes in H_A. A test can only accumulate evidence against the null, so whatever you hope to establish must be what survives when the null falls.
> **Which direction would count as evidence?** That fixes the inequality in H_A, and with it the tail the test looks in.

Examination questions usually signal this in their phrasing, and the signal is worth recognising.

Table 4.1: How the usual phrasings map onto the two questions above.

The question asks	The claim becomes	Because
"Is there evidence to **support** the claim?"	H_A	The claim is what you are gathering evidence for.
"Can you **reject** the claim?"	H_0	The claim is what you are challenging, and you are asking whether the data can knock it down.

Where the equality goes

One structural rule holds in every case, and it will save you from a good deal of trouble:

> **REMEMBER THIS**
>
> **The equality always belongs in the null hypothesis.**
> H_0 takes $=$, $\leq$ or $\geq$. H_A takes $\neq$, $<$ or $>$, and never equality.

The reason is the one from earlier in this section. Every test begins by assuming one specific value of μ and computing how likely our sample would be under it. Only a hypothesis containing an equality supplies such a value — $\mu \leq 499$ can be tested at its boundary, 499, whereas $\mu > 499$ names no particular number to compute with.

> **WATCH OUT**
>
> A claim phrased with a *strict* inequality — "the mean is more than 499" — therefore cannot go into H_0 as stated. If you are asked for evidence supporting it, the pair is
>
> $$H_0 : \mu \leq 499 \qquad H_A : \mu > 499$$
>
> Writing $H_0 : \mu \geq 499$ is the common error. It puts the claim in the null *and* points the test in the wrong direction, so a sample mean **above** 499 — evidence *for* the claim — ends up producing a p-value near 1 and a "fail to reject." The arithmetic is fine; the conclusion is backwards.

The CAT report, worked through

A report claims the mean CAT score of applicants is more than 499. A sample of 100 applicants has mean 502, with $\sigma = 10.6$. Is there enough evidence at $\alpha = 0.01$ to support the report?

We settled the hypotheses above by asking what would convince us:

$$H_0 : \mu \leq 499 \qquad H_A : \mu > 499$$

Right-tailed, so the critical value is $Z_{0.01} = 2.33$ — positive, not negative.

Before computing anything, check the direction. The sample mean is 502 and the claim is "more than 499." The sample points the *same way* as the claim, so whatever the arithmetic returns, it must come out favourable to the report. If it does not, the tail has been pointed the wrong way.

```r
xbar_c <- 502; mu_c <- 499; sigma_c <- 10.6; n_c <- 100

z_c <- (xbar_c - mu_c) / (sigma_c / sqrt(n_c))
c(z_calc = z_c,
  z_crit = qnorm(0.99),
  p_value = pnorm(z_c, lower.tail = FALSE))   # upper tail: H_A is ">"
```

```
#>   z_calc   z_crit  p_value
#> 2.830189 2.326348 0.002326
```

$Z_{calc} = 2.83 > 2.33$, and $p = 0.0023 < 0.01$. We **reject** H_0, which means the data **support** the report's claim — and the answer points the way the sanity check said it would.

THE IDEA IN PLAIN WORDS

Make that check a habit, and make it before you trust any p-value rather than after.
Ask which way the sample points relative to the claim, and decide what answer you should therefore expect. If the arithmetic hands back something else — a p-value of 0.998 and "no support" when the sample sits above the claimed value — the mistake is almost always in H_A, not in the calculation.

4.3 How far is too far?

Write μ_0 for the value the null claims (here, 165). We will reject H_0 when the sample mean lands too far from μ_0 — that is, when

$$|\bar{x} - \mu_0| \geq c$$

for some cutoff c. Everything now hinges on choosing c.

WHERE THIS COMES FROM

We choose c so that, **if the null were true**, a gap that large would occur with probability only α — a small risk we decide in advance, typically 0.05:

$$P[|\bar{x} - \mu_0| \geq c] = \alpha$$

From the Central Limit Theorem (Chapter 4) we know $\bar{x}$ is normally distributed with mean μ and standard deviation $\sigma/\sqrt{n}$. Dividing through by that standard error turns the left-hand side into a standard normal:

$$P\left[\frac{|\bar{x} - \mu_0|}{\sigma/\sqrt{n}} \geq \frac{c}{\sigma/\sqrt{n}}\right] = \alpha$$
$$P\left[|Z| \geq \frac{c}{\sigma/\sqrt{n}}\right] = \alpha$$

The event $|Z| \geq k$ splits into two symmetric tails, each carrying half the probability:

$$2 \cdot P\left[Z \geq \frac{c}{\sigma/\sqrt{n}}\right] = \alpha \qquad \Longrightarrow \qquad P\left[Z \geq \frac{c}{\sigma/\sqrt{n}}\right] = \frac{\alpha}{2}$$

By definition $P[Z \geq Z_{\alpha/2}] = \alpha/2$, so the bracketed quantity *is* $Z_{\alpha/2}$, and therefore

$$\boxed{c = Z_{\alpha/2}\,\frac{\sigma}{\sqrt{n}}}$$

Substituting that c back into the rejection rule and dividing both sides by the standard error gives the form you will actually use:

$$\frac{|\bar{x} - \mu_0|}{\sigma/\sqrt{n}} \geq Z_{\alpha/2}$$

The quantity on the left is the **test statistic**, written Z_{calc}. The value on the right is the **critical value**. The rule is simply: *if the statistic exceeds the critical value, reject the null.*

DOING IT IN R

Those critical values are the ones you have been reading off a printed table. There is no need — `qnorm()` gives you any of them to as many decimals as you like.

```
alpha <- c(0.10, 0.05, 0.01)

# Two-tailed: we want the value leaving alpha/2 in the upper tail
qnorm(1 - alpha / 2)
```

```
#> [1] 1.645 1.960 2.576
```

Those three numbers are the 1.645, 1.96 and 2.58 you have memorised. Now you know where they come from, and you will never need the table again.

4.4 The worked example, both ways

Back to the students. We have $\bar{x} = 168$, $\mu_0 = 165$, $\sigma = 8$, $n = 36$, and we will work at $\alpha = 0.05$.

On paper

Step 1 — Hypotheses.

$$H_0 : \mu = 165 \qquad H_A : \mu \neq 165$$

Step 2 — Critical value. Two-tailed at $\alpha = 0.05$, so $Z_{0.025} = 1.96$.

Step 3 — Test statistic.

$$Z_{calc} = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} = \frac{168 - 165}{8/\sqrt{36}} = \frac{3}{8/6} = \frac{3}{1.333} = 2.25$$

Step 4 — Decision. $|2.25| > 1.96$, so we **reject** H_0 at the 5% level.

Step 5 — The p-value.

$$P(|Z| > 2.25) = 2 \times P(Z > 2.25) = 2 \times 0.01222 = 0.0244$$

If the administration's claim were true, only 2.4% of samples would sit as far from 165 as ours. That is unlikely enough to reject the claim.

The same thing in R

Now watch every one of those numbers reappear.

```r
xbar  <- 168
mu0   <- 165
sigma <- 8
n     <- 36

se     <- sigma / sqrt(n)          # standard error
z_calc <- (xbar - mu0) / se        # the test statistic
z_crit <- qnorm(1 - 0.05 / 2)      # the critical value
p_val  <- 2 * pnorm(-abs(z_calc))  # the p-value

c(se = se, z_calc = z_calc, z_crit = z_crit, p_value = p_val)
```

```
#>      se  z_calc  z_crit p_value
#> 1.33333 2.25000 1.95996 0.02445
```

The standard error is 1.3333, the test statistic is 2.25, and the p-value is 0.0244 — exactly the hand calculation, and near enough exactly the 0.0242 we got by brute-force simulation at the start of the chapter.

> **THE IDEA IN PLAIN WORDS**
>
> Notice what just happened. The simulation, the algebra, and the R code are three descriptions of one idea. The formula is not a rule handed down from above — it is a shortcut for the counting exercise we did by hand in Section 4.1.

And here is the picture the numbers are describing:

```
plot_rejection(stat = 2.25, crit = 1.96, tails = "two",
               stat_label = "Z = 2.25")
```

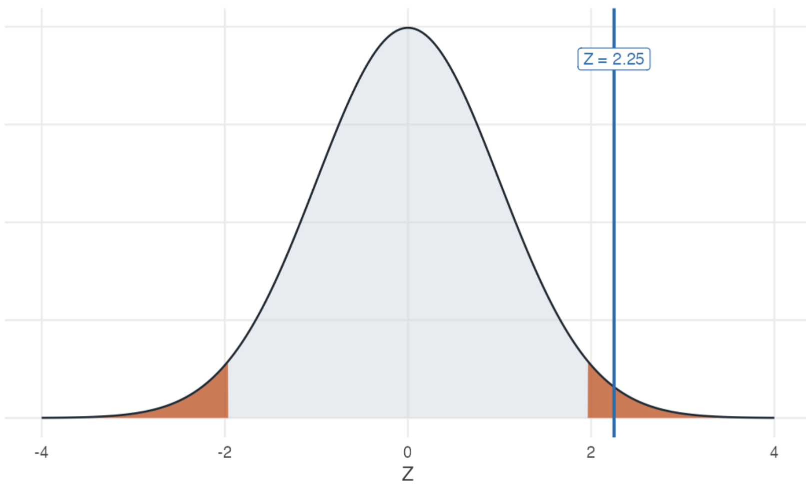

Figure 4.2: The rejection region at $\alpha = 0.05$ (shaded), with our test statistic of 2.25 falling inside it.

Letting a package do it

For a real dataset rather than summary statistics, `z.test()` from the **BSDA** package does the whole thing at once. Since our problem quoted only summary statistics, we first build a sample consistent with them:

```
library(BSDA)

heights <- make_sample(n = 36, mean = 168, sd = 8)

z.test(heights, mu = 165, sigma.x = 8, conf.level = 0.95)
```

```
#>
#>  One-sample z-Test
#>
#> data:  heights
#> z = 2.3, p-value = 0.02
#> alternative hypothesis: true mean is not equal to 165
#> 95 percent confidence interval:
#>  165.4 170.6
#> sample estimates:
```

```
#> mean of x
#>       168
```

Read that output against the hand calculation: `z = 2.25`, `p-value = 0.0244`, and a confidence interval that we are about to explain.

> **WATCH OUT**
>
> `z.test()` needs `sigma.x` — the **population** standard deviation. If you do not know it (and in real work you almost never do), you must not use a Z-test. Use `t.test()` instead; see Section 4.7.

4.5 What a p-value actually says

The p-value is the single most misquoted number in empirical work, so let us be exact.

> **REMEMBER THIS**
>
> The p-value is the probability of seeing a sample **at least as extreme as ours**, *given that the null hypothesis is true*.
> It is **not** the probability that the null is true. It is **not** the probability that we made a mistake. It is a statement about the data, computed under an assumption about the world — not a statement about the world.

Small p-value means the data would be surprising if the null held, which is evidence against the null. Large p-value means the data are unremarkable under the null, so we have no case.

Table 4.2: Reading a p-value.

p-value	Read it as	At $\alpha = 0.05$
0.60	Completely ordinary under the null.	Fail to reject
0.18	A bit high, but this happens roughly 1 time in 6.	Fail to reject
0.049	Would happen about 1 time in 20.	Reject
0.0001	Would happen 1 time in 10,000. Strong evidence.	Reject

4.6 The same test, viewed as an interval

> *Unit 3 built confidence intervals; this unit builds tests. Are they two methods, or one?*

We now have two procedures that use the same sample, the same standard error and the same critical value. It would be strange if they were unrelated, and they are not.

Start by computing the interval for our students, exactly as in the previous unit.

```
xbar + c(-1, 1) * qnorm(0.975) * (8 / sqrt(36))
```

```
#> [1] 165.4 170.6
```

The 95% interval is $[165.39,\ 170.61]$. The claimed value of 165 lies **outside** it — the same verdict the test reached, and not a coincidence.

The reason is visible in the algebra. We fail to reject H_0 when the test statistic is small enough, that is when

$$\left| \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} \right| \le z_{\alpha/2}$$

Multiplying through by $\sigma/\sqrt{n}$ and rearranging for μ_0 gives

$$\bar{x} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}} \ \le\ \mu_0 \ \le\ \bar{x} + z_{\alpha/2}\frac{\sigma}{\sqrt{n}}$$

which is the confidence interval. The condition for *not rejecting a claim* and the formula for the *interval* are the same inequality, solved for different things.

Testing every possible claim at once

That equivalence can be checked rather than taken on trust. Rather than testing the single claim $\mu_0 = 165$, test every value on a fine grid and collect the ones that survive.

```
se   <- 8 / sqrt(36)
grid <- seq(160, 176, by = 0.001)

not_rejected <- grid[abs(xbar - grid) / se <= qnorm(0.975)]
```

```
surviving <- range(not_rejected)
interval  <- xbar + c(-1, 1) * qnorm(0.975) * se

rbind(claims_that_survive_the_test = surviving,
      confidence_interval          = interval,
      difference                   = surviving - interval)
```

```
#>                                   [,1]        [,2]
#> claims_that_survive_the_test 165.3870000 170.6130000
#> confidence_interval          165.3867147 170.6132853
#> difference                     0.0002853  -0.0002853
```

The two agree to within a thousandth of a centimetre, which is the spacing of the grid — test a finer grid and the discrepancy shrinks with it. The set of claims a test would not reject *is* the confidence interval. Nothing was assumed to make this happen; it follows from the two being one inequality solved for different unknowns.

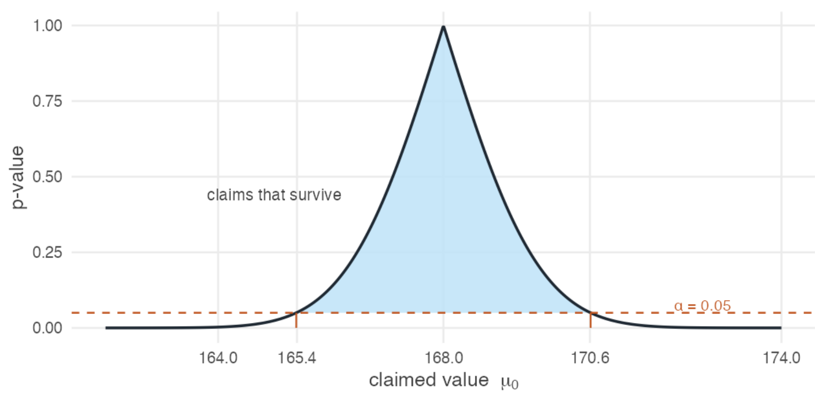

Figure 4.3: The p-value for every possible claimed value of mu. Claims above the 0.05 line survive the test, and the range of surviving claims is exactly the 95% confidence interval.

The curve peaks at $\mu_0 = \bar{x} = 168$, where the claim matches the data exactly and the p-value is 1. It falls away in both directions, crossing 0.05 precisely at the two endpoints of the interval.

REMEMBER THIS

A $(1 - \alpha)$ confidence interval is exactly the set of values μ_0 that a two-tailed test at level α would **fail** to reject.

Test and interval are one object viewed from two sides. If the interval

> misses the claimed value, the test rejects it. Always.

So why keep both?

Because they carry different information, and the figure above shows what each one throws away.

A confidence interval answers infinitely many tests at once, and reports the answer in the units of the problem. Handed $[165.39,\ 170.61]$, a reader can evaluate any claim about average height — 165, 166, 172 — without recomputing anything. A test answers one question, chosen in advance.

A test, however, reports *how far* rather than merely *in or out*. Both 165 and 160 fall outside our interval, and the interval says the same thing about both: rejected. The p-values do not agree — 0.024 for 165, and effectively zero for 160. One claim is marginally implausible and the other is not remotely survivable, and the interval alone does not distinguish them.

> **THE IDEA IN PLAIN WORDS**
>
> Report the interval. It contains the test, states its answer in the units the reader cares about, and does not tempt anyone into treating a single threshold as a verdict.
> Report the p-value alongside it when one specific claim is genuinely at issue, since that is where the interval is silent about degree.

Everything Section 3.8 said about reading a confidence interval applies unchanged here. In particular, the 95% still describes the procedure and not this interval, and an interval that excludes a claimed value has not proved the claim false — it has reported that the data are unlikely under it.

4.7 When σ is unknown: the t-test

Everything so far assumed we knew the population standard deviation σ. We almost never do. Replacing it with the sample standard deviation s adds a second source of uncertainty, and the statistic

$$t_{calc} = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

no longer follows a normal distribution. It follows Student's t with $n - 1$ **degrees of freedom** — a distribution with fatter tails, which demands more evidence before it will let you reject.

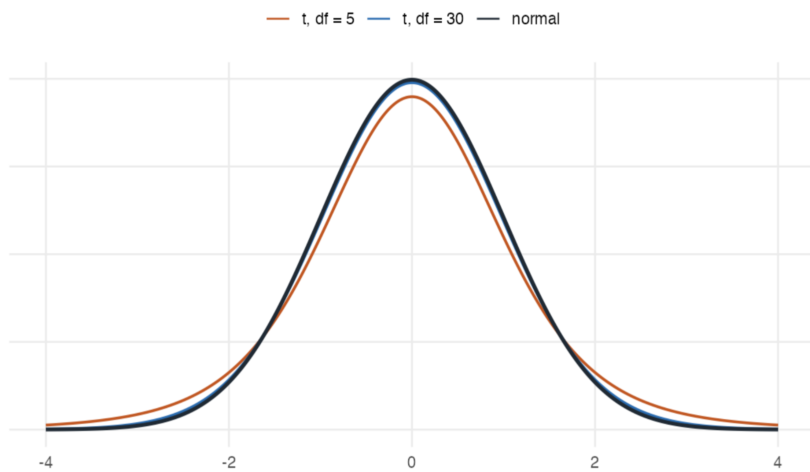

Figure 4.4: Student's t against the standard normal. With few degrees of freedom the tails are noticeably heavier, so critical values sit further out.

Settling the question with the actual class

We opened this chapter with a made-up sample of 36 students. We can now do the real thing. `stats-class.csv` holds the survey this class filled in — names, ages, heights, gender:

```r
students <- read.csv("data/stats-class.csv")

str(students)
```

```
#> 'data.frame':	33 obs. of  4 variables:
#>  $ student_id: chr  "S01" "S02" "S03" "S04" ...
#>  $ age       : int  31 19 18 18 19 18 18 21 18 19 ...
#>  $ height    : int  178 161 181 164 170 175 176 186 161 170 ...
#>  $ gender    : chr  "M" "F" "M" "F" ...
```

Thirty-three rows, but look at the last one. One student left age and height blank, and another did not give a gender:

```r
students[!complete.cases(students), ]
```

```
#>    student_id age height gender
#> 8         S08  21    186   <NA>
#> 33        S33  NA     NA      F
```

WATCH OUT

A blank is not always an `NA`. In a numeric column an empty field is read as `NA`, but in a text column it is read as the empty string `""` — and `""` is a perfectly ordinary value as far as `is.na()`, `complete.cases()` and `na.rm` are concerned. A gap in a text column can therefore hide in plain sight. If a dataset you did not prepare yourself has blanks, say so explicitly when you read it:

```
read.csv("somefile.csv", na.strings = c("", "NA"))
```

The files in this book are already written so that plain `read.csv()` does the right thing.

WATCH OUT

`mean()` returns `NA` if even one value is missing — it refuses to guess what you meant. Pass `na.rm = TRUE` to drop the gaps, and always know how many you dropped.

```
mean(students$height)                  # NA — one height is missing
```

```
#> [1] NA
```

```
mean(students$height, na.rm = TRUE)    # 32 students
```

```
#> [1] 169.5
```

Now the test. We do **not** know the population standard deviation of student heights, so this is a t-test, not a Z-test:

```
heights_real <- students$height[!is.na(students$height)]

c(n = length(heights_real),
  mean = mean(heights_real),
  sd = sd(heights_real))
```

```
#>      n   mean     sd
#>  32.00 169.50   8.71
```

```
t.test(heights_real, mu = 165)
```

```
#>
```

```
#>  One Sample t-test
#>
#> data:  heights_real
#> t = 2.9, df = 31, p-value = 0.006
#> alternative hypothesis: true mean is not equal to 165
#> 95 percent confidence interval:
#>  166.4 172.6
#> sample estimates:
#> mean of x
#>     169.5
```

$t = 2.92$ on 31 degrees of freedom, with a p-value of 0.0064. We **reject** the claim that the average is 165 cm. The 95% interval, $[166.4,\ 172.6]$, excludes 165 — the same verdict, as always.

> **THE IDEA IN PLAIN WORDS**
>
> Notice how much narrower this is than the textbook version of the problem. The opening example assumed we somehow knew $\sigma = 8$. Here we had to estimate it from 32 people, and the t distribution charges us for that uncertainty with a wider interval and a higher bar to clear.
> Also notice what the data is not: 32 students in one classroom are not a random sample of all APU students. The arithmetic is correct; whether it answers the question about *the university* is a separate matter, and a harder one.

Worked example

The dean estimates that full-time faculty teach 11.0 classroom hours per week. You sample eight faculty members. Can you reject the dean's claim at $\alpha = 0.10$? At $\alpha = 0.01$?

```r
hours <- c(11.8, 8.6, 10.5, 7.9, 6.4, 10.4, 10.0, 8.2)

c(n = length(hours), mean = mean(hours), sd = sd(hours))
```

```
#>     n  mean    sd
#> 8.000 9.225 1.749
```

On paper. With $\bar{x} = 9.225$, $s = 1.749$, $n = 8$ and $df = 7$:

$$t_{calc} = \frac{9.225 - 11}{1.749/\sqrt{8}} = \frac{-1.775}{0.618} = -2.87$$

The critical values are $t_{0.05,7} = 1.895$ and $t_{0.005,7} = 3.499$.

- At $\alpha = 0.10$: $2.87 > 1.895 \Rightarrow$ **reject** H_0.
- At $\alpha = 0.01$: $2.87 < 3.499 \Rightarrow$ **fail to reject** H_0.

In R. Both the pieces and the packaged version:

```
t_calc <- (mean(hours) - 11) / (sd(hours) / sqrt(length(hours)))
t_crit <- qt(c(0.95, 0.995), df = 7)     # 10% and 1%, two-tailed

c(t_calc = t_calc, crit_10pct = t_crit[1], crit_1pct = t_crit[2])
```

```
#>     t_calc crit_10pct  crit_1pct
#>     -2.870      1.895      3.499
```

```
t.test(hours, mu = 11, conf.level = 0.90)
```

```
#>
#>  One Sample t-test
#>
#> data:  hours
#> t = -2.9, df = 7, p-value = 0.02
#> alternative hypothesis: true mean is not equal to 11
#> 90 percent confidence interval:
#>   8.053 10.397
#> sample estimates:
#> mean of x
#>     9.225
```

The p-value is 0.024 — below 0.10 but above 0.01, exactly matching the two different verdicts we reached by hand.

> **THE IDEA IN PLAIN WORDS**
>
> The same data can be "significant" at one level and not at another. Nothing has changed about the world between those two lines — only how much risk of a false alarm we were willing to tolerate. This is why reporting the p-value itself is far more informative than reporting a bare "significant / not significant."

4.8 One-tailed tests

Sometimes only one direction matters. A firm claims its new curing technique gets nicotine content *below* 1.5 mg; we only care if the true mean is lower.

$$H_0 : \mu \geq 1.5 \qquad H_A : \mu < 1.5$$

All the probability we are willing to risk now goes into a single tail, so the critical value moves *inward*: $-Z_{0.05} = -1.645$ rather than ± 1.96.

```
qnorm(0.05)           # left-tailed critical value at 5%
```

```
#> [1] -1.645
```

```
qnorm(0.95)           # right-tailed critical value at 5%
```

```
#> [1] 1.645
```

In R, add the `alternative` argument:

```
nicotine <- make_sample(n = 20, mean = 1.42, sd = 0.7)

z.test(nicotine, mu = 1.5, sigma.x = 0.7, alternative = "less",
       conf.level = 0.95)
```

```
#>
#>  One-sample z-Test
#>
#> data:  nicotine
#> z = -0.51, p-value = 0.3
#> alternative hypothesis: true mean is less than 1.5
#> 95 percent confidence interval:
#>     NA 1.677
#> sample estimates:
#> mean of x
#>      1.42
```

$Z_{calc} = -0.51$, which is not below -1.645, so we fail to reject. The firm's claim is not supported — a sample mean of 1.42 would arise about 30% of the time even if the true mean were 1.5.

The one-sided interval looks backwards

This trips up nearly everyone. For the left-tailed test above, the confidence interval is

$$\left(-\infty,\ \bar{x} + Z_\alpha \frac{\sigma}{\sqrt{n}}\right)$$

An *upper* bound — for a *left*-tailed test. Why?

WHERE THIS COMES FROM

Start from the rejection rule and solve for μ_0 rather than for $\bar{x}$:

$$\frac{\bar{x}-\mu_0}{\sigma/\sqrt{n}} < -Z_\alpha$$

$$\bar{x}-\mu_0 < -Z_\alpha \frac{\sigma}{\sqrt{n}}$$

$$\mu_0 > \bar{x} + Z_\alpha \frac{\sigma}{\sqrt{n}}$$

The inequality flipped when μ_0 moved across. So we reject exactly when μ_0 is **too large**, and the values we *fail* to reject are those with $\mu \leq \bar{x} + Z_\alpha\,\sigma/\sqrt{n}$.

THE IDEA IN PLAIN WORDS

The rejection region is defined in Z**-space**; the confidence interval lives in **parameter space**. Solving the inequality for μ reverses the direction. Put plainly: a left-tailed test rejects when the sample mean comes out too small. So the values of μ it rules out are the ones that are too *large* relative to what we saw. That leaves an upper bound.

4.9 Reference tables

Table 4.3: One-sample tests at a glance. Use Z and `z.test()` when σ is known; use t with $df = n - 1$ and `t.test()` when it is not.

Test	Hypotheses	Reject when	R argument
Two-tailed	$H_0 : \mu = \mu_0$ vs $H_A : \mu \neq \mu_0$	$\lvert Z_{calc}\rvert > Z_{\alpha/2}$	(default)
Left-tailed	$H_0 : \mu \geq \mu_0$ vs $H_A : \mu < \mu_0$	$Z_{calc} < -Z_\alpha$	`alternative = "less"`
Right-tailed	$H_0 : \mu \leq \mu_0$ vs $H_A : \mu > \mu_0$	$Z_{calc} > Z_\alpha$	`alternative = "greater"`

Table 4.4: Standard normal critical values. Every figure below is `qnorm()` evaluated inline, not copied from a printed table.

α	Two-tailed, $Z_{\alpha/2}$	One-tailed, Z_α
10%	1.645	1.282
5%	1.96	1.645
1%	2.576	2.326

4.10 Summary

REMEMBER THIS

1. A gap between a sample and a claim proves nothing on its own — sampling noise guarantees some gap. The test asks whether the gap is *bigger than noise*.
2. $Z_{calc} = \dfrac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$ when σ is known; $t_{calc} = \dfrac{\bar{x} - \mu_0}{s/\sqrt{n}}$ with $df = n-1$ when it is not.
3. Reject when the statistic beats the critical value, or equivalently when the p-value falls below α. The two rules never disagree.
4. A confidence interval is the set of null values a test would *not* reject. Interval and test are the same statement.
5. A p-value is $P(\text{data} \mid H_0)$, never $P(H_0 \mid \text{data})$.
6. The equals sign always lives in H_0. "Can you reject the claim?" puts the claim in H_0; "is there evidence supporting the claim?" puts it in H_A.
7. `qnorm()` and `qt()` replace every printed table you own.

4.11 Practice problems

YOUR TURN

Work each one on paper first, then check yourself in R. The solutions are one click away, which makes them worth nothing unless you try first. These are for learning, so the workings are shown in full. The question bank that follows is for testing yourself, and has no answers.

11.1 A consumer research organisation states that the mean caffeine content per 12-ounce bottle of a caffeinated soft drink is 37.7 mg. A random sample of 36 bottles has a mean of 35.28 mg. The population standard deviation is 10.8 mg. At $\alpha = 0.01$, can you reject the organisation's claim?

SOLUTION

$H_0 : \mu = 37.7$ against $H_A : \mu \neq 37.7$. Two-tailed at $\alpha = 0.01$, so the critical value is $Z_{0.005} = 2.58$.

$$Z_{calc} = \frac{35.28 - 37.7}{10.8/\sqrt{36}} = \frac{-2.42}{1.8} = -1.34$$

```r
caffeine <- make_sample(n = 36, mean = 35.28, sd = 10.8)
z.test(caffeine, mu = 37.7, sigma.x = 10.8, conf.level = 0.99)
```

```
#>
#>  One-sample z-Test
#>
#> data:  caffeine
#> z = -1.3, p-value = 0.2
#> alternative hypothesis: true mean is not equal to 37.7
#> 99 percent confidence interval:
#>  30.64 39.92
#> sample estimates:
#> mean of x
#>     35.28
```

$|-1.34| < 2.58$ and the p-value of 0.179 is far above 0.01, so we **fail to reject**. If the true mean really were 37.7 mg, a sample mean this far away would turn up about 18% of the time — thoroughly unremarkable.

11.2 Historical data show household water use is normally distributed with mean 360 gallons and standard deviation 40 gallons per day. A sample of 200 households averages 374 gallons. Are these data consistent with the historical distribution at the 5% level? What is the p-value?

SOLUTION

$$Z_{calc} = \frac{374 - 360}{40/\sqrt{200}} = \frac{14}{2.828} = 4.95$$

```r
water <- make_sample(n = 200, mean = 374, sd = 40)
z.test(water, mu = 360, sigma.x = 40, conf.level = 0.95)
```

```
#>
#>  One-sample z-Test
#>
#> data:  water
```

```
#> z = 4.9, p-value = 0.0000007
#> alternative hypothesis: true mean is not equal to 360
#> 95 percent confidence interval:
#>  368.5 379.5
#> sample estimates:
#> mean of x
#>       374
```

$4.95 \gg 1.96$, and the p-value is about 0 — under one in a million. **Reject** H_0. Water use has moved away from the historical mean.

11.3 A study claims the average monthly salary of full professors in India is ₹87,800. Students at one university sample ten salaries (in thousands):

```
91.0, 79.8, 102.0, 93.5, 82.0, 88.6, 90.0, 98.6, 101.0, 84.0
```

They believe their professors earn *more*. Test at the 10% and 5% levels.

SOLUTION

The word "more" makes this **right-tailed**: $H_0 : \mu \leq 87.8$ against $H_A : \mu > 87.8$. Since σ is unknown and $n = 10$, use a t-test with $df = 9$.

```
salary <- c(91.0, 79.8, 102.0, 93.5, 82.0, 88.6, 90.0, 98.6, 101.0,
 ↪  84.0)

c(mean = mean(salary), sd = sd(salary))
```

```
#>   mean     sd
#> 91.050  7.797
```

```
t.test(salary, mu = 87.8, alternative = "greater")
```

```
#>
#>  One Sample t-test
#>
#> data:  salary
#> t = 1.3, df = 9, p-value = 0.1
#> alternative hypothesis: true mean is greater than 87.8
#> 95 percent confidence interval:
#>  86.53   Inf
#> sample estimates:
#> mean of x
#>     91.05
```

$t_{calc} = 1.318$, against critical values $t_{0.10,9} = 1.383$ and $t_{0.05,9} = 1.833$. The statistic falls short of both. **Fail to reject** at either level — there is not enough evidence that these professors are paid above the national figure.

11.4 A Department of Transportation claims mean wait time is *at most* 6 minutes. A sample of 34 locations has mean 10.3 minutes, standard deviation 8.0. Test at 5% and 1%.

SOLUTION

"At most" makes this right-tailed: $H_0 : \mu \leq 6$ against $H_A : \mu > 6$, t-test with $df = 33$.

```
waits <- make_sample(n = 34, mean = 10.3, sd = 8.0)
t.test(waits, mu = 6, alternative = "greater")
```

```
#>
#>  One Sample t-test
#>
#> data:  waits
#> t = 3.1, df = 33, p-value = 0.002
#> alternative hypothesis: true mean is greater than 6
#> 95 percent confidence interval:
#>  7.978   Inf
#> sample estimates:
#> mean of x
#>      10.3
```

$t_{calc} = 3.14$, well beyond $t_{0.05,33} = 1.692$ and $t_{0.01,33} = 2.445$. **Reject at both levels** — wait times are longer than claimed.

11.5 `marriage-ages.csv` records the ages of both partners in ten couples. Is there evidence that one partner tends to be older? Test at the 5% level.

Hint: the two columns are not independent samples — they are ten couples. What is the one number per couple that captures the question?

SOLUTION

The two columns are **paired**: each row is one couple. Treating them as two independent samples throws away that pairing and inflates the standard error. The right move is to reduce each couple to a single number — the age gap — and run a **one-sample** test on it against zero.

```
couples <- read.csv("data/marriage-ages.csv")

gap <- couples$partner1 - couples$partner2
gap
```

```
#>  [1] 1 5 4 1 1 4 0 8 1 0
```

```r
c(mean = mean(gap), sd = sd(gap), n = length(gap))
```

```
#>   mean     sd      n
#>  2.500  2.635 10.000
```

```r
t.test(gap, mu = 0)
```

```
#>
#>  One Sample t-test
#>
#> data:  gap
#> t = 3, df = 9, p-value = 0.01
#> alternative hypothesis: true mean is not equal to 0
#> 95 percent confidence interval:
#>  0.6149 4.3851
#> sample estimates:
#> mean of x
#>       2.5
```

The mean gap is 2.5 years, $t = 3.0$ on 9 degrees of freedom, $p = 0.015$. Since $0.015 < 0.05$, we **reject** H_0: partner 1 is reliably older.
R will do the pairing for you if you ask:

```r
t.test(couples$partner1, couples$partner2, paired = TRUE)
```

```
#>
#>  Paired t-test
#>
#> data:  couples$partner1 and couples$partner2
#> t = 3, df = 9, p-value = 0.01
#> alternative hypothesis: true mean difference is not equal to 0
#> 95 percent confidence interval:
#>  0.6149 4.3851
#> sample estimates:
#> mean difference
#>             2.5
```

Identical output — `paired = TRUE` computes exactly the differences we formed by hand. Compare what happens if you forget:

```r
t.test(couples$partner1, couples$partner2, paired = FALSE)$p.value
```

```
#> [1] 0.607
```

The same ten couples, a p-value of 0.607 instead of 0.015, and the opposite conclusion. **Pairing is not a detail.**

11.6 *(No computation.)* A classmate reports "$p = 0.03$, so there is a 3% chance the null hypothesis is true." Explain what is wrong, and state what $p = 0.03$ does mean.

SOLUTION

The p-value is computed **assuming the null is true**. It cannot then also be the probability that the null is true — that would be circular. Formally, your classmate has swapped $P(\text{data} \mid H_0)$ for $P(H_0 \mid \text{data})$, which are different quantities entirely.
The correct reading: *if* H_0 were true, samples at least as extreme as this one would arise 3% of the time. Since that is unusual, we treat it as evidence against H_0 — but the statement is about how surprising the data are, not about how probable the hypothesis is.

4.12 Question bank

WATCH OUT

No solutions are given for this section, by design. These questions are drawn from past quizzes and exams. If you can work them without checking an answer key, you are ready; if you cannot, the section to reread is named beside each one.
They run from easiest to hardest. Do them in order.

Multiple choice

QB 11.1 The level of significance, α, is the risk of *(select all that apply)*

a. Rejecting the null hypothesis when it is true
b. Rejecting the null hypothesis when the alternative hypothesis is true
c. Rejecting the alternative hypothesis when the null hypothesis is true
d. Rejecting the alternative hypothesis when it is true

See Section 4.3.

QB 11.2 An economist sets the alternative hypothesis that the average CGPA of undergraduate students is less than 8. What type of test should be used, and what is the critical value at $\alpha = 5\%$?

a. One-tailed test and $+1.96$
b. One-tailed test and -1.96

c. One-tailed test and $+1.645$
d. One-tailed test and -1.645

See Section 4.8.

QB 11.3 The margin of error for a confidence interval is $\pm Z_{\alpha/2}\,\sigma/\sqrt{n}$. Which statements are correct? *(select all that apply)*

a. Increasing the sample size will decrease the margin of error
b. The margin of error decreases as α rises
c. The larger the margin of error, the less confident one is about the estimate
d. The margin of error increases with the population standard deviation σ

See Section 4.6.

QB 11.4 A 95% confidence interval for mean student height is $(165.41,\ 170.95)$. You now compute a 99% interval from the same data. Which statements are likely correct? *(select all that apply)*

a. The 90%, 95% and 99% intervals are the same
b. The 99% interval is $(164.54,\ 171.83)$
c. The 99% interval is $(165.86,\ 170.51)$
d. The answer cannot be determined from the information given

See Section 4.6. Ask yourself which way an interval must move when you demand more confidence.

QB 11.5 A company tests the null hypothesis that the average time to accept a job offer is at most 7 days. The test concludes that the average exceeds 7 days. It is later learned that the true mean is exactly 7 days. What error occurred?

a. Type II error
b. Type I error
c. A correct decision was made
d. Cannot be determined from the information provided

Short reasoning

QB 11.6 "A high p-value is evidence that the null hypothesis is true." Do you agree? Why or why not?

See Section 4.5.

QB 11.7 Two economists, Amit and Anand, collect data on Indian agricultural wages to test the effect of MNREGA. Amit concludes that wages after the policy are higher than before, when in fact wages have not changed. Anand concludes that wages before and after are the same — also incorrectly.

It is claimed that Amit has committed a Type I error and Anand a Type II error. Do you agree? Why or why not?

Numerical

QB 11.8 A researcher wants to estimate the average weekly study hours of university students. The population standard deviation is 6 hours. What sample size is needed to estimate the mean with a margin of error of 1 hour at 90% confidence?

a. $n = 98$
b. $n = 64$
c. $n = 121$
d. $n = 82$

QB 11.9 The numbers of people taking the morning university shuttle on 16 randomly chosen days are:

```
27, 46, 35, 33, 29, 45, 28, 24, 30, 40, 38, 35, 41, 27, 38, 29
```

The administration claims the daily average is 30. Taking $\sigma = 7$:

a. Construct a 99% confidence interval for the mean number of riders.
b. Use that interval to test the administration's claim.
c. Would your conclusion change at the 95% level? Explain *why* before you compute it.

Part (c) is the one that separates understanding from arithmetic.

4.13 Looking ahead

Every test in this unit compared a sample against a *claimed* number — a figure from a government report, a manufacturer's specification, a previous year's estimate. That number had to come from somewhere outside the data.

Most questions worth asking do not arrive in that form.

Did the households that received the transfer end up better off than those that did not? Do girls in the scheme attend school more than girls outside it? Did the districts that got the road see wages rise more than the districts that did not? In each case there is no external claim to test against. There are two groups, each with its own sample mean, and the question is whether the difference between them is larger than sampling variation alone would produce.

That is the subject of the next unit. It is also where the difficulty changes character. Deciding whether two groups differ is a statistical problem, and we will solve it. Deciding whether one group differs *because* of the treatment is a question about how the groups came to be formed — and no test statistic can answer it.